The heat generator on the retained inverse-fibre cover

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Abstract

We transport the actual de Bruijn–Newman initial function through the retained cover $s = -r^2/2$. We give its connection, square, branch commutators, analytic heat realization, and the exact evolution of its arithmetic relation. We also compute the full Euclidean spatial diffusion of its actual scalar observable and its transported covector. The original inverse-fibre coordinates and all analytic-unit terms remain present. Complete proofs specify the relations between branch escape, arithmetic zeros, and the transported operators. The construction does not exhibit a counterexample to the Riemann hypothesis or to the classical Euclidean Navier–Stokes problem. The source's exact fourth-order interacting electric-correlation coefficient is now propagated into this arithmetic heat flow through an explicitly invertible Fourier operator. Its full source proof, spectral derivative term, coupling scales, and spatial pullbacks are included. The complete arithmetic quotient continuation proves exact heat transport, closure and multiplicity-jet results in its stated topologies, and Mellin, principal-part and compact-test morphisms. The common-representative map identifies its heat dynamics with the original Fourier dynamics. The complete second Euclidean diffusion has an explicit jet residue. We prove that its divergence along the original escaping trajectory survives in the actual fourth coefficient at sufficiently negative Newman times, with a full error bound. The original three special points of the supplied period construction also give an exact real three-torus quotient. Their retained twists and cusp direction generate a three-dimensional forced fluid with all mixed interactions and localization terms calculated explicitly. For a supplied incompressible flow we also construct three profiles in the original polynomial target and recover its velocity from their arithmetic temporal defect by an explicit inverse integral and norm identity. The complete force-dependent acceleration, material diffusion and translated zero divisors are calculated.

Contents

1 Original coordinates and the retained cover

The original source coordinate triple is (x, y, w) and its target is (a, b, c). Its polynomial is

$$
\begin{aligned}
F_1 &= (1+xy)^3 w + y^2(1+xy)(4+3xy),\\
F_2 &= y + 3x(1+xy)^2 w + 3xy^2(4+3xy),\\
F_3 &= 2x - 3x^2 y - x^3 w.
\end{aligned} \tag{1}
$$

All coordinate spaces here are complex unless a real locus is stated. The original $x \neq 0$ inverse chart is

$$
\begin{aligned}
P_{a,b,c}(r) = cr^3 - 2r^2 + br - 2a, \qquad \alpha = \tfrac{1}{2}P_r(r),\\
(x, y, w) = (\alpha^{-1}, r - \alpha, 5\alpha^2 - 3r\alpha - c\alpha^3), \qquad \alpha \neq 0.
\end{aligned} \tag{2}
$$

Here is the complete verification needed for this chart. At a point with $x \neq 0$, set $\alpha = 1/x$ and $r = y + 1/x$; solving $F_3 = c$ gives the third coordinate in (2). Substitution into the other original components gives

$$
F_2 = 4r + 2\alpha - 3cr^2, \qquad F_1 = r^2 + r\alpha - cr^3.
$$

The first equation equals b precisely when $2\alpha = P_r(r)$; after that substitution $2(F_1 - a) = P(r)$. This proves both directions and uniqueness of the root label. A multiple root gives no finite point in this chart. Directly, $F(0, y, w) = (w + 4y^2, y, 0)$, so $c = 0$ has the additional inverse point $(0, b, a - 4b^2)$ and $c \neq 0$ has none.

On $b = c = 0$ the root cover is $a = -r^2$, with $\alpha = -2r$. The arithmetic coordinate is the specific further map

$$a = 2s, \qquad s = -r^2/2. \tag{3}$$

Its unramified domain is $\mathbb{C}_r^* = \mathbb{C}_r \setminus \{0\}$ over $\mathbb{C}_s^*$. The point $r = 0$ is the ramification point of the completed cover and is excluded by $\alpha \neq 0$ from the original inverse chart. These are explicit restrictions of the original map.

The constant-coefficient heat evolution in the original root variable has the following exact map to the arithmetic base, retaining both sheets. For $V \subset \mathbb{C}_s^*$ and $\widetilde{V} = \{r : -r^2/2 \in V\}$ put $T_V : \mathcal{O}(V)^2 \to \mathcal{O}(\widetilde{V})$, $T_V(q_0, q_1)(r) = q_0(-r^2/2) + rq_1(-r^2/2)$. Its inverse is explicitly the even part and the odd part divided by r; their invariance under $r \mapsto -r$ makes them holomorphic on V. Thus the formulas give an isomorphism in both directions. In the original basis $(1, r)$ put

$$R = \begin{pmatrix} 0 & -2s \\ 1 & 0 \end{pmatrix}, \qquad D = \partial_s + \begin{pmatrix} 0 & 0 \\ 0 & 1/(2s) \end{pmatrix}.$$

The identities $r^2 = -2s$ and $\partial_s r = -1/r = r/(2s)$ give $\partial_r T_V = T_V(-RD)$ by direct differentiation of both components. Also $DR - RD = -R^{-1}$, since differentiation of the factor r gives $-1/r$. Applying the first identity twice proves the full typed operator morphism

$$T_V^{-1}\partial_r^2 T_V = RDRD = R^2D^2 - D = -2sD^2 - D.$$

The arithmetic Newman generator has the corresponding pullback $T_V(-D^2/4)T_V^{-1}$, since D transports ∂_s. This proves the exact relationship between both generators, including the first derivative and both sheet components. The original polynomial has component vector $(4s - 2a, b - 2cs)$ under this map. Consequently its full root heat evolution is

$$P(\tau, r) = P_{a_0+2\tau, b_0+6c\tau, c}(r), \qquad P_\tau = 6cr - 4 = P_{rr}.$$

Here τ is forward heat time. The explicit reversal $\widetilde{P}(t, r) = P(-t, r)$ satisfies $\widetilde{P}_t = -\widetilde{P}_{rr}$. This is the backward sign convention; the arithmetic Newman generator below acts in its specified s coordinate. The arithmetic pullback below transports differentiation in $s = a/2$, which must also carry the derivative of r with respect to s. Both operators will therefore be displayed with their exact coordinates. In the displayed component vector, $(-2sD^2 - D)$ gives $(-4, 6c)$: the first component is $-\partial_s(4s - 2a) = -4$, while for the second $D(b - 2cs) = b/(2s) - 3c$ and $D^2(b - 2cs) = -b/(4s^2) - 3c/(2s)$. Their combination is exactly $6c$. These are precisely the parameter derivatives $a_\tau = 2$, $b_\tau = 6c$, $c_\tau = 0$ in the original polynomial, verifying the whole morphism on the original family.

For an open set $V \subset \mathbb{C}_s^*$, set $\widetilde{V} = \{r : -r^2/2 \in V\}$. Every holomorphic function on $\widetilde{V}$ has a unique decomposition

$$f(r) = q_0(s) + rq_1(s), \qquad q_0, q_1 \in \mathcal{O}(V). \tag{4}$$

Indeed its even part is $(f(r) + f(-r))/2$ and its odd coefficient is $(f(r) - f(-r))/(2r)$. Both are invariant under $r \mapsto -r$ and therefore define holomorphic functions of s on local inverse charts, agreeing on their overlaps. These formulas also prove uniqueness. They define an isomorphism $T_V : \mathcal{O}(V)^2 \to \mathcal{O}(\widetilde{V})$. The same decomposition holds over V containing zero: power series show that the odd numerator divided by r extends holomorphically and that each even power series is a power series in $s = -r^2/2$. The derivative operator below, however, is first defined only over $V \subset \mathbb{C}^*$.

2 Connection, square, and branch coupling

Write a section as the column $q = (q_0, q_1)^\mathsf{T}$ in the ordered basis $(1, r)$. On the unramified cover, $ds/dr = -r$, so differentiation in the original spectral coordinate is $\partial_s^{\text{lift}} = -r^{-1}\partial_r$. In particular $\partial_s r = -1/r = r/(2s)$. Consequently

$$D = \partial_s + A(s), \quad A(s) = \begin{pmatrix} 0 & 0 \\ 0 & 1/(2s) \end{pmatrix}, \qquad T_V D = \partial_s^{\text{lift}} T_V. \tag{5}$$

This is a connection: for $g \in \mathcal{O}(V)$, $D(gq) = g'q + gDq$. Its curvature in this one complex coordinate is zero; the nontrivial information is its ramification and monodromy.

Proposition 2.1 (The full second derivative and its commutators)**.** *On $\mathcal{O}(V)^2$ one has*

$$D^2 = \begin{pmatrix} \partial_s^2 & 0 \\ 0 & \partial_s^2 + s^{-1}\partial_s - 1/(4s^2) \end{pmatrix}, \tag{6}$$

$$R = \begin{pmatrix} 0 & -2s \\ 1 & 0 \end{pmatrix}, \quad R^2 = -2sI, \quad [D, R] = -R^{-1}, \tag{7}$$

$$[D^2, R] = -2R^{-1}D - R^{-3}. \tag{8}$$

Here R is multiplication by the original root r; commutators include differentiation of the matrix coefficients.

Proof. Expanding $(\partial_s + A)^2$ gives $\partial_s^2 + 2A\partial_s + A' + A^2$. Its lower diagonal constant is $-1/(2s^2) + 1/(4s^2) = -1/(4s^2)$, proving (6). Multiplication $r(q_0 + rq_1) = -2sq_1 + rq_0$ proves R and R^2. Matrix multiplication then gives

$$R' + AR - RA = \begin{pmatrix} 0 & -1 \\ 1/(2s) & 0 \end{pmatrix} = -R^{-1}.$$

Differentiating $RR^{-1} = I$ covariantly gives $[D, R^{-1}] = -R^{-1}[D, R]R^{-1} = R^{-3}$. Finally $[D^2, R] = D[D, R] + [D, R]D = -2R^{-1}D - R^{-3}$. This proves every term, including the zeroth-order branch term. ☐

The sheet involution is $J = \operatorname{diag}(1, -1)$; it commutes with D, D^2 and anticommutes with R. The trace is $(q_0, q_1) \mapsto 2q_0$, with kernel the odd summand, and its half has the section $q_0 \mapsto (q_0, 0)$. D preserves the two summands on the punctured base, whereas R exchanges them and couples the odd coefficient into $-2sq_1$. This describes exactly the information lost by tracing.

On a simply connected $V \subset \mathbb{C}^*$ choose a holomorphic root $r(s)$ satisfying $r(s)^2 = -2s$. The invertible matrix $G = \operatorname{diag}(1, r(s))$ obeys $D = G^{-1}\partial_s G$. Horizontal coefficients satisfy $q_0' = 0$ and $q_1' + q_1/(2s) = 0$, hence $(q_0, q_1) = (c_0, c_1/r(s))$. Continuation once around zero changes the sign of the latter coefficient. Thus the monodromy is exactly $\operatorname{diag}(1, -1)$; no single-valued global choice of $r(s)$ on $\mathbb{C}^*$ is implied by this local gauge formula.

3 The actual arithmetic heat function

We use the convention of [1, Section 1, equations (1)–(4)]:

$$\begin{aligned} \Phi(u) &= \sum_{n \ge 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ H_t(z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du, \\ Z_t(s) &= 8H_t(2s), \qquad Z_0(s) = \Xi(s) = \xi(1/2 + is), \\ \xi(\sigma) &= \tfrac{1}{2}\sigma(\sigma - 1)\pi^{-\sigma/2}\Gamma(\sigma/2)\zeta(\sigma). \end{aligned} \tag{9}$$

The Mellin and spectral coordinates are related by $\sigma = 1/2 + is$, with inverse $s = -i(\sigma - 1/2)$. They are not silently identified.

We rederive the integral identity and its analytic bounds. Set

$$h(x) = \frac{\pi}{2}x^2(2\pi x^2 - 3)e^{-\pi x^2}, \quad S(\lambda) = \sum_{n\geq 1} h(n\lambda), \quad k(v) = e^{v/2}S(e^v).$$

For the Fourier convention $\widehat{h}(q) = \int_{\mathbb{R}} h(x)e^{-2\pi i x q}dx$, the Gaussian transform and its second and fourth derivatives give

$$\widehat{x^2e^{-\pi x^2}} = (1/(2\pi) - q^2)e^{-\pi q^2},$$
$$\widehat{x^4e^{-\pi x^2}} = (q^4 - 3q^2/\pi + 3/(4\pi^2))e^{-\pi q^2}.$$

Multiplying by $-3\pi/2$ and π^2 respectively gives $\widehat{h} = h$. Since $h(0) = 0$, Poisson summation gives $S(\lambda) = \lambda^{-1}S(1/\lambda)$ and $k(v) = k(-v)$. For $\lambda \geq 1$, $|S(\lambda)| \leq C\lambda^4e^{-\pi\lambda^2}$, because the remaining series is bounded by a constant times $\sum_{n\geq 1} n^4e^{-\pi(n^2-1)} < \infty$. The symmetry gives the corresponding bound as $\lambda \downarrow 0$. Each fixed number of v derivatives introduces only finitely many additional powers of ne^v. It follows, with constants depending on the derivative order, that

$$|k^{(j)}(v)| \leq C_je^{C_j|v|}e^{-\pi e^{2|v|}}. \tag{10}$$

Termwise differentiation on compact sets is justified by the same summable bounds. On the negative half-line differentiate the proved evenness identity. This proves (10) on both ends.

For $\Re\sigma > 1$, absolute integrability permits summation inside the Mellin integral. The substitution $q = \pi x^2$ gives

$$\begin{aligned}\int_0^\infty h(x)x^{\sigma-1}dx &= \pi^{-\sigma/2}\left[\tfrac{1}{2}\Gamma(\sigma/2+2) - \tfrac{3}{4}\Gamma(\sigma/2+1)\right]\\ &= \frac{\sigma(\sigma-1)}{8}\pi^{-\sigma/2}\Gamma(\sigma/2),\\ \int_0^\infty S(\lambda)\lambda^{\sigma-1}d\lambda &= \xi(\sigma)/4.\end{aligned}$$

The two-ended bounds make the last integral entire in σ, so the equality extends by analytic continuation. Substituting $\lambda = e^v$ and $\sigma = 1/2 + is$ proves

$$Z_t(s) = 4\int_{\mathbb{R}} e^{tv^2/4}k(v)e^{isv}\,dv. \tag{11}$$

Indeed direct expansion also gives $\Phi(u) = 2k(2u)$; the substitution $v = 2u$ and the evenness of k carry the stated H_t integral to (11), retaining the factors $2, 4, 8$. The bound (10) dominates every fixed derivative uniformly on compact complex (t, s) sets. Thus Z is jointly entire and differentiation under its integral proves exactly

$$\partial_tZ_t = -\tfrac{1}{4}\partial_s^2Z_t. \tag{12}$$

This supplies the actual initial datum and the actual time orientation.

4 An explicit domain for the heat evolution

The differential expression alone does not specify a heat evolution on every entire-function space. We now give a precise invariant domain containing the actual kernel.

Definition 4.1. Let $\mathscr{E}$ consist of $\phi \in C^\infty(\mathbb{R}, \mathbb{C})$ for which

$$p_{A,B,m,j}(\phi) = \int_{\mathbb{R}} |\phi^{(j)}(v)|(1+|v|)^m e^{Av^2+B|v|}\, dv < \infty$$

for every nonnegative integer A, B, m, j. Give it these seminorms. Define

$$T\phi(s) = 4\int_{\mathbb{R}} \phi(v)e^{isv}\, dv, \qquad \mathscr{A} = T\mathscr{E},$$

and equip $\mathscr{A}$ with the transported topology.

The map T is injective. To give the needed uniqueness argument, let $T\phi$ vanish on the real line and let $g_\epsilon(v) = (4\pi\epsilon)^{-1/2}e^{-v^2/(4\epsilon)}$. Its Gaussian integral representation is $g_\epsilon(v) = (2\pi)^{-1}\int_{\mathbb{R}} e^{-\epsilon s^2}e^{isv}ds$. Fubini is valid because $\phi \in L^1$ and the Gaussian is integrable; it expresses $\phi * g_\epsilon$ as a Gaussian-weighted integral of the vanishing Fourier transform. Thus $\phi * g_\epsilon = 0$. As $\epsilon \downarrow 0$, these convolutions converge to ϕ in L^1: translations are continuous in L^1, first for continuous compactly supported functions by uniform continuity, then for all L^1 functions by approximation; splitting the Gaussian integral into a small translation ball and its vanishing tail proves the claim. Consequently $\phi = 0$ almost everywhere and, being continuous, everywhere. This also justifies the transported topology without an unspecified choice of representative.

Proposition 4.2 (Heat evolution with its inverse)**.** *For every complex t, multiplication $E_t\phi(v) = e^{tv^2/4}\phi(v)$ is a continuous automorphism of $\mathscr{E}$. The operators $U_t = TE_tT^{-1}$ on $\mathscr{A}$ satisfy*

$$\begin{gathered} U_tU_u = U_{t+u}, \qquad U_t^{-1} = U_{-t}, \qquad \partial_t U_t = -\tfrac{1}{4}\partial_s^2 U_t, \\ U_t\Xi = Z_t, \qquad U_t\partial_s = \partial_s U_t, \qquad U_tM_sU_{-t} = M_s - \tfrac{t}{2}\partial_s. \end{gathered} \tag{13}$$

Here M_s is multiplication by the unchanged spectral coordinate.

Proof. Every v derivative of $e^{tv^2/4}\phi$ is a finite sum of $e^{tv^2/4}$ times a polynomial in v times a derivative of ϕ. Each target seminorm is bounded by finitely many source seminorms with larger A, m; on compact t sets these bounds are uniform. The inverse multiplier is $e^{-tv^2/4}$. Differentiation in t inserts $v^2/4$, while two s derivatives in the transform insert $-v^2$, proving the generator and the group identities. The kernel bound proves $k \in \mathscr{E}$, so $U_t\Xi = Z_t$ follows from (11).

Integration by parts gives $M_sT\phi = T(i\phi')$. For completeness the boundary terms vanish: each derivative of $e^{isv}\phi$ is integrable on a horizontal real-v line and the function itself is integrable, so its limits at both ends exist and are zero. The same argument applies after exponential weights. Thus multiplication by s preserves $\mathscr{A}$; differentiation does also, since $\partial_sT\phi = T(iv\phi)$. Finally

$$e^{tv^2/4}i\partial_v(e^{-tv^2/4}\phi) = i\phi' - \tfrac{t}{2}iv\phi.$$

Applying T proves the last identity in (13). All displayed operators have the domains just specified. ☐

5 Transport of the moving arithmetic relation

Put $B_t = M_s - (t/2)\partial_s$. For every polynomial p, the automorphism calculation proves

$$U_t(p(s)\Xi(s)) = p(B_t)Z_t(s). \tag{14}$$

The right side is the polynomial in the actual differential operator, with composition order retained. For example the first two nonconstant relations are

$$B_t Z_t = sZ_t - \tfrac{t}{2} Z_t',$$
$$B_t^2 Z_t = (s^2 - t/2)Z_t - tsZ_t' + \tfrac{t^2}{4} Z_t''.$$

These follow from $\partial_s(sq) = q + s\partial_s q$. In particular, an evolving polynomial multiple is not obtained by silently leaving multiplication by s fixed.

One can take a closure in the topology that has actually been defined: let $N_0 = \overline{\mathbb{C}[s]\Xi}^{\mathscr{A}}$ and $N_t = U_t N_0$. Continuity and invertibility give $N_t = \overline{\{p(B_t)Z_t : p \in \mathbb{C}[s]\}}^{\mathscr{A}}$. M_s preserves N_0, since it is continuous and preserves its dense generating subspace; hence B_t preserves N_t. The map $[f] \mapsto [U_t f]$ is a well-defined continuous isomorphism $\mathscr{A}/N_0 \to \mathscr{A}/N_t$ with inverse induced by U_{-t} and intertwines the induced M_s with the induced B_t. The exact map from these explicit spaces to the original Connes quotient is constructed immediately below, with both original topologies retained.

5.1 A coordinate-level correspondence with the actual Connes quotient

We now construct the comparison with the actual object of [2, Section 3.6, Proposition 3.6]. Write $\mathcal{H} = \mathcal{H}_{\le 1}$ for its ring of entire functions of Hadamard order at most one, retaining its stated Hadamard topology. Write

$$N_{\mathrm{C}} = \overline{\mathbb{C}[s]\Xi}^{\mathcal{H}}, \qquad Q_{\mathrm{C}} = \mathcal{H}/N_{\mathrm{C}}, \qquad Q_t = \mathscr{A}/N_t.$$

Here s is the same coordinate in both quotients. The closure defining N_{C} is precisely the closure of the generators in that proposition; the following calculation proves the equality of their algebraic spans, including the transform conventions. The comparison constructed below uses the actual topology on $\mathcal{H}$, through a subspace of a product; it does not substitute the topology of $\mathscr{A}$ for it.

The original summation, Mellin transform, and their coordinates. Let $\mathcal{G}_0$ be the vector space of the finite sums

$$f(x) = e^{-\pi x^2} \sum_{j=0}^{n} c_j \pi^j x^{2j}, \qquad c_0 = 0, \qquad \sum_{j=0}^{n} c_j \frac{(2j)!}{4^j j!} = 0. \tag{15}$$

The two constraints say exactly $f(0) = \widehat{f}(0) = 0$ for $\widehat{f}(y) = \int_{\mathbb{R}} f(x)e^{-2\pi i x y}dx$. Indeed the Gaussian moment is $\int_{\mathbb{R}} \pi^j x^{2j} e^{-\pi x^2} dx = (2j)!/(4^j j!)$. The original maps of [2, equations (3.3), (3.4), and Section 3.6] are

$$(\mathcal{E}_{\mathrm{C}} f)(\lambda) = \lambda^{1/2} \sum_{n\ge 1} f(n\lambda), \qquad \lambda > 0,$$
$$(\mathcal{F}_\mu g)(s) = \int_0^\infty g(\lambda)\lambda^{-is}\frac{d\lambda}{\lambda}. \tag{16}$$

Both signs and the measure are retained. With $\lambda = e^v$, define

$$k_f(v) = e^{v/2} \sum_{n\ge 1} f(ne^v), \qquad \iota f(v) = \frac{1}{4} k_f(-v). \tag{17}$$

Then there is an exact typed map

$$\iota : \mathcal{G}_0 \longrightarrow \mathscr{E}, \qquad T\iota f(s) = \int_{\mathbb{R}} k_f(v) e^{-isv} dv = \mathcal{F}_\mu \mathcal{E}_{\mathrm{C}} f(s). \tag{18}$$

The reflection and the factor $1/4$ in (17) are necessary for the defined T, and have not been suppressed.

Here is the convergence proof. For $v \geq 0$, every derivative of k_f is bounded by $C_j e^{C_j v} e^{-\pi e^{2v}}$, by differentiating the finite polynomial times Gaussian and summing the resulting convergent series in n. The Fourier transform of a polynomial times this Gaussian is again a polynomial times the same Gaussian. Poisson summation and the two zero constraints give

$$\sum_{n\geq 1} f(n\lambda) = \lambda^{-1} \sum_{n\geq 1} \widehat{f}(n/\lambda), \qquad k_f(v) = k_{\widehat{f}}(-v).$$

The positive-half-line estimate for $\widehat{f}$ therefore proves the estimate on the negative half-line as well, for every derivative. Every defining seminorm of $\mathscr{E}$ is finite. These bounds justify the change of variable, reflection, and all entire continuations in (18).

Put $z = 1/2 - is$, whose inverse is $s = i(z - 1/2)$, and put

$$P_f(z) = \sum_{j=0}^{n} c_j (z/2)_j, \qquad (z/2)_j = \prod_{k=0}^{j-1} (z/2 + k), \qquad (z/2)_0 = 1.$$

For $\Re z > 1$, the absolutely convergent Mellin integral gives

$$\begin{aligned} \mathcal{F}_\mu \mathcal{E}_{\mathrm{C}} f(s) &= \zeta(z) \int_0^\infty f(x) x^{z-1} dx \\ &= \frac{1}{2} \pi^{-z/2} \Gamma(z/2) \zeta(z) P_f(z). \end{aligned} \tag{19}$$

The two constraints in (15) are respectively $P_f(0) = 0$ and $P_f(1) = 0$, since $(1/2)_j = (2j)!/(4^j j!)$. There is consequently a unique polynomial

$$q_f(s) = \left. \frac{P_f(z)}{z(z-1)} \right|_{z=1/2-is}, \qquad T\iota f(s) = q_f(s)\Xi(s). \tag{20}$$

For the last equality, the right side of (19) is $q_f(s)\xi(1/2 - is) = q_f(s)\Xi(-s)$, and the proved evenness $\Xi(-s) = \Xi(s)$ supplies the final sign conversion. Entire continuation then proves the identity on all $\mathbb{C}_s$.

This map is onto every polynomial multiple, with an explicit inverse. For $q \in \mathbb{C}[s]$, expand the unchanged polynomial

$$z(z-1) q(i(z - 1/2)) = \sum_{j=0}^{\deg q + 2} c_j (z/2)_j. \tag{21}$$

There is a unique such expansion because the jth basis polynomial has degree j and leading coefficient 2^{-j}. Evaluation at 0 and 1 proves the two required constraints, giving $f \in \mathcal{G}_0$. This proves both injectivity and surjectivity of $f \mapsto q_f$ and of $f \mapsto q_f \Xi$, since Ξ is not zero identically. Here is the full basis argument connecting to the source's Hermite presentation. Its functions are

$$h_n(x) = \frac{2^{1/4}}{\sqrt{2^n n!}} H_n(\sqrt{2\pi} x) e^{-\pi x^2}, \qquad H_n(y) = n! \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j (2y)^{n-2j}}{j!(n-2j)!}.$$

Their even members are a triangular basis of all even polynomial Gaussians, since each has nonzero leading coefficient in its successive even degree. Put $L = \sqrt{\pi} x - (2\sqrt{\pi})^{-1} \partial_x$. Differentiating the displayed polynomial proves $h_n = L^n h_0 / \sqrt{n!}$. The Gaussian transform gives $\widehat{h}_0 = h_0$, while integration by parts gives $\mathcal{F}_{\mathbb{R}} L = -iL\mathcal{F}_{\mathbb{R}}$. Thus $\widehat{h}_{2n} = (-1)^n h_{2n}$, with both signs proved in the original Fourier convention. The finite polynomial formula also gives $h_{2n}(0) = (-1)^n 2^{1/4-n}\sqrt{(2n)!}/n! \neq 0$. Split an even polynomial Gaussian into the spans of the $h_{4\ell}$ and the $h_{4\ell+2}$. The two constraints $f(0) = \widehat{f}(0) = 0$ are exactly vanishing at zero of each of those two components. In the first span subtracting its constant-basis coordinate gives $h_{4\ell} - h_{4\ell}(0)h_0/h_0(0)$; in the second it gives $-h_{4\ell+2} + h_{4\ell+2}(0)h_2/h_2(0)$, with the displayed minus sign retained. These are precisely the source's ψ_ℓ^+, ψ_ℓ^-, for $\ell \geq 1$. They span the two constraint kernels by solving for their respective h_0, h_2 coefficients. Thus their union spans $\mathcal{G}_0$, and (20) proves the required equality of spans. For the particular original kernel $h(x) = \pi^2 x^4 e^{-\pi x^2} - (3\pi/2)x^2 e^{-\pi x^2}$, $P_h(z) = z(z-1)/4$, so $q_h = 1/4$, $\iota h = k/4$, and

$$\mathcal{F}_\mu \mathcal{E}_{\mathrm{C}} h = \Xi/4, \qquad T\iota(4h) = \Xi. \tag{22}$$

The factor 4 in the actual arithmetic function is thereby accounted for.

The original scaling differential operator is $S = -i(x\partial_x + 1/2) : \mathcal{G}_0 \to \mathcal{G}_0$. It preserves the two constraints: its value at zero is $-if(0)/2$, and integration by parts gives $\int Sf = (i/2)\int f$. Termwise differentiation proves $\mathcal{E}_{\mathrm{C}} Sf = -i\lambda\partial_\lambda \mathcal{E}_{\mathrm{C}} f$. Integration by parts, with the proved two-ended bounds, then gives

$$T\iota Sf = M_s T\iota f, \qquad \iota Sf = i\partial_v \iota f. \tag{23}$$

The second identity also follows from differentiating $k_f(-v)/4$; it records the sign produced by the reflection. Thus the comparison intertwines the actual scaling operator and multiplication by the same s.

The correspondence between the two completed quotients. Define a space of common representatives, with its topology specified by the displayed embedding into the actual product:

$$C = \mathscr{A} \cap \mathcal{H}, \qquad C \longrightarrow \mathscr{A} \times \mathcal{H}, \quad f \longmapsto (f, f). \tag{24}$$

Give C the subspace topology of this embedding and put

$$K = N_0 \cap N_{\mathrm{C}}, \qquad \mathscr{R} = C/K. \tag{25}$$

Both intersections mean equality of entire functions in the unchanged s coordinate. In particular $K \subset C$. Since the two quotient maps are continuous, the following is a continuous injective linear map, with its complete coordinate formula:

$$j : \mathscr{R} \longrightarrow Q_0 \times Q_{\mathrm{C}}, \qquad j([f]_K) = ([f]_{N_0}, [f]_{N_{\mathrm{C}}}). \tag{26}$$

Indeed the map on C is continuous by the product-induced topology; its kernel consists exactly of representatives belonging to both N_0 and N_{C}, which is K. The definition of the quotient topology therefore proves continuity and injectivity, without asserting that this injection is a topological embedding into the product.

Writing π_0, π_{C} for its two coordinate projections, their images and kernels are exactly

$$\begin{aligned} \operatorname{im}\pi_0 &= \{[f]_{N_0} : f \in C\}, & \ker\pi_0 &= (N_0 \cap \mathcal{H})/K, \\ \operatorname{im}\pi_{\mathrm{C}} &= \{[f]_{N_{\mathrm{C}}} : f \in C\}, & \ker\pi_{\mathrm{C}} &= (\mathscr{A} \cap N_{\mathrm{C}})/K. \end{aligned} \tag{27}$$

For example $\pi_0([f]_K) = 0$ means precisely $f \in N_0$, while $f \in C$ already supplies $f \in \mathcal{H}$; this proves the first kernel identity in both directions. The other identity follows by the same direct test in the second coordinate. Formula (27) gives an explicit representative rule for the correspondence even when either coordinate has several possible lifts.

The first image in (27) is dense in Q_0. To prove this, take $\phi \in \mathscr{E}$, and choose a fixed smooth cutoff χ equal to one on $[-1, 1]$ and zero outside $[-2, 2]$. Set $\phi_R(v) = \chi(v/R)\phi(v)$

for $R \geq 1$. Leibniz's formula bounds every seminorm of $\phi - \phi_R$ by a finite sum of weighted integrals of derivatives of ϕ over $|v| \geq R$, multiplied by bounded constants and factors R^{-k}. Those integrals tend to zero by their absolute integrability. Thus $\phi_R \to \phi$ in $\mathscr{E}$. Moreover

$$|T\phi_R(s)| \leq 4\|\phi_R\|_{L^1} e^{2R|\Im s|},$$

so $T\phi_R$ is of order at most one and belongs to C. Consequently C is dense in $\mathscr{A}$; taking images under the continuous surjective quotient map proves the assertion. All generators (20) also lie in C by their proved $\mathscr{E}$ representatives and their order at most one.

Finally this same correspondence transports exactly through every complex Newman time. For $t \in \mathbb{C}$, define

$$j_t : \mathscr{R} \longrightarrow Q_t \times Q_{\mathrm{C}}, \qquad j_t([f]_K) = ([U_t f]_{N_t}, [f]_{N_{\mathrm{C}}}). \tag{28}$$

It is continuous and injective: U_t is a continuous automorphism and $U_t f \in N_t$ is equivalent to $f \in N_0$. Its two kernels as coordinate projections are the same as in (27), and its first image is dense in Q_t. These assertions follow by applying the proved isomorphism $Q_0 \to Q_t$ to the first coordinate. The space C is stable under M_s: it is stable in $\mathscr{A}$ by integration by parts and in $\mathcal{H}$ because the latter is the specified topological ring. Each closed relation subspace is stable under M_s, by continuity and invariance of $\mathbb{C}[s]\Xi$. Thus $[f]_K \mapsto [sf]_K$ is defined, continuous, and obeys

$$j_t([sf]_K) = ([B_t U_t f]_{N_t}, [sf]_{N_{\mathrm{C}}}), \qquad B_t = M_s - (t/2)\partial_s. \tag{29}$$

This is the exact operator correspondence: the first coordinate has the heat-transported multiplication operator, while the second has the original Connes multiplication operator. It follows directly from $U_t M_s = B_t U_t$ and retains the full derivative term. No inclusion of all of $\mathscr{A}$ into $\mathcal{H}$, no equality of their closures, and no identification of their spectra is needed for the proved maps (26)–(29).

5.2 Complete local transport at the moving divisor

There is a second exact description, local at the moving divisor. Let $Z = Z_t(s)$ and $\mathcal{P} = \partial_t + \frac{1}{4}\partial_s^2$. For every holomorphic $g(t,s)$ the complete product calculation is

$$\mathcal{P}(Zg) = Z\mathcal{P}g + \tfrac{1}{2} Z_s g_s. \tag{30}$$

The term $\mathcal{P}Z$ is zero by (12); the mixed product term is retained. On the complement of $Z = 0$, set $b = Z_s/Z$. Direct differentiation of f/Z gives the exact conjugation

$$Z\mathcal{P}Z^{-1} = \partial_t + \tfrac{1}{4}\partial_s^2 - \tfrac{1}{2} b\partial_s + \tfrac{1}{2} b^2. \tag{31}$$

To check the potential term before canceling any expression, it is $\frac{1}{2}(Z_s/Z)^2 - \frac{1}{4}Z_{ss}/Z - Z_t/Z$; the last two terms cancel by the established heat equation. Thus the operator in (31) sends Zg exactly to $Z\mathcal{P}g$. Its coefficients are meromorphic at zeros of Z, but its restriction to the ideal of holomorphic multiples of Z has the displayed holomorphic value. Division by Z on this ideal is well-defined and unique by analytic continuation. There is no claim that division is holomorphic on arbitrary sections at a zero.

The identical calculation on the two-component module replaces ∂_s by D:

$$Z(\partial_t + \tfrac{1}{4}D^2)Z^{-1} = \partial_t + \tfrac{1}{4}D^2 - \tfrac{1}{2}bD + \tfrac{1}{2}b^2 I. \tag{32}$$

Indeed $D(Zq) = Z_s q + ZDq$, so the product expansion is exactly the scalar one, retaining $A(s)$ in D. This gives the full moving arithmetic relation on the retained cover without assuming a derivative acts on a frozen quotient.

6 The exact branch-sensitive heat defect

The actual pulled arithmetic solution is $q(t,s) = (Z_t(s), 0)$. Equation (6) proves $(\partial_t + \frac{1}{4}D^2)q = 0$. Its function on the cover is $f(t,r) = Z_t(-r^2/2)$. By differentiating $s = -r^2/2$ twice, the scalar generator is

$$L_r = -\frac{1}{4r^2}\partial_r^2 + \frac{1}{4r^3}\partial_r, \qquad \partial_t f = L_r f \quad (r \neq 0). \tag{33}$$

This retains the original spectral diffusion; it does not substitute the constant-coefficient r-heat equation of the fibre polynomial.

The branch observable rZ_t is the column $Rq = (0, Z_t)$. From the already proved commutator its complete defect is

$$\begin{aligned}(\partial_t + \tfrac{1}{4}D^2)(Rq) &= -\tfrac{1}{2}R^{-1}Dq - \tfrac{1}{4}R^{-3}q \\ &= \begin{pmatrix} 0 \\ Z_t'/(4s) - Z_t/(16s^2) \end{pmatrix}.\end{aligned} \tag{34}$$

Equivalently, as a function on the cover it is $-Z_t'/(2r) - Z_t/(4r^3)$. These terms are the exact dynamics of the retained odd observable, not a reason to discard that observable or declare it unrelated. For real t, $Z_t(0) > 0$ (proved from the integral in (45)), so its leading branch pole is nonzero. The original arithmetic solution itself remains entire at the branch point.

7 Keeping the other original fibre coefficients

The preceding branch at $s = 0$ belongs to the particular slice $b = c = 0$. We now retain the full source family, with b, c fixed and $a = 2s$. Its cover is defined by

$$cr^3 - 2r^2 + br - 4s = 0, \qquad s = \frac{cr^3 - 2r^2 + br}{4}. \tag{35}$$

On its original inverse chart $P_r = 3cr^2 - 4r + b = 2\alpha \neq 0$, implicit differentiation gives

$$\partial_s^{\text{lift}} = \frac{4}{P_r}\partial_r, \qquad L_{b,c} = -\frac{4}{P_r^2}\partial_r^2 + \frac{4P_{rr}}{P_r^3}\partial_r. \tag{36}$$

To check the second identity, apply the first operator twice: $16P_r^{-2}\partial_r^2 - 16P_{rr}P_r^{-3}\partial_r$, then multiply by $-1/4$. Thus the actual function $Z_t((cr^3 - 2r^2 + br)/4)$ satisfies this exact transported equation on $P_r \neq 0$. The branch equations, with no coefficient removed, are

$$b = 4r_0 - 3cr_0^2, \qquad s_0 = \frac{r_0^2 - cr_0^3}{2}. \tag{37}$$

They follow by solving $P_r = 0$ and substituting in (35).

For the two-sheet subfamily $c = 0$, retain arbitrary $b \in \mathbb{C}$ and put $\Delta = b^2 - 32s$. The algebraic relation is $r^2 - (b/2)r + 2s = 0$. In the unchanged ordered basis $(1, r)$,

$$R_b = \begin{pmatrix} 0 & -2s \\ 1 & b/2 \end{pmatrix}, \qquad D_b = \partial_s + A_b, \quad A_b = \begin{pmatrix} 0 & 4b/\Delta \\ 0 & -16/\Delta \end{pmatrix}. \tag{38}$$

These formulas hold on $\Delta \neq 0$. Indeed $(b - 4r)^2 = \Delta$ in the algebra, hence $\partial_s r = 4/(b - 4r) = 4(b - 4r)/\Delta$. This gives both coefficient entries of A_b. Multiplication by r gives R_b. Direct differentiation and multiplication then prove

$$\begin{aligned} R_b^2 - \tfrac{b}{2}R_b + 2sI &= 0, \\ [D_b, R_b] &= 4(bI - 4R_b)/\Delta, \\ D_b^2 &= \partial_s^2 + 2A_b\partial_s + (16/\Delta)A_b. \end{aligned} \tag{39}$$

For the last equality, $A_b' = (32/\Delta)A_b$ and $A_b^2 = (-16/\Delta)A_b$. At $b = 0$ these identities recover (5)–(6) exactly.

The other root is $b/2 - r$. Thus the sheet involution and trace in this original basis are respectively

$$J_b = \begin{pmatrix} 1 & b/2 \\ 0 & -1 \end{pmatrix}, \qquad \operatorname{Tr}_b(q_0, q_1) = 2q_0 + (b/2)q_1.$$

$J_b^2 = I$, and J_b commutes with D_b because b is fixed and $A_b J_b = J_b A_b$. Also $\operatorname{Tr}_b D_b q = \partial_s \operatorname{Tr}_b q$, as the two added terms are $8bq_1/\Delta - 8bq_1/\Delta = 0$. The trace kernel is exactly the set of pairs $(-bq_1/4, q_1)$; the half-trace has the section $q \mapsto (q, 0)$. This explicitly exhibits the branch information and its coupling when the original linear coefficient is present.

The branch point is $r_0 = b/4$, $s_0 = b^2/32$. The exact coordinate change $\eta = r - b/4$, with inverse $r = \eta + b/4$, gives

$$s = s_0 - \eta^2/2, \qquad Z_t(s) = Z_t(s_0 - \eta^2/2). \tag{40}$$

The arithmetic function and its time origin have not changed. The displayed translation applies to the cover only. For any specified $s_0 \in \mathbb{C}$ there is a coefficient b with $b^2 = 32s_0$, so every base point occurs as a branch point of an original $c = 0$ fibre family. If $Z_t(s_0) = 0$ has multiplicity m, its factorization $Z_t(s) = (s - s_0)^m A(s)$, $A(s_0) \neq 0$, pulls back to $(-1/2)^m \eta^{2m} A(s_0 - \eta^2/2)$ and has multiplicity $2m$. This proves the exact contact with any actual arithmetic zero and keeps its analytic unit.

The original inverse point there has

$$\alpha = b/2 - 2r = -2\eta, \quad x = -1/(2\eta), \quad y = b/4 + 3\eta, \quad w = 26\eta^2 + (3b/2)\eta.$$

These follow directly from (2). At $\eta = 0$ its two chart branches escape, while the separate point $(0, b, 2s_0 - 4b^2)$ remains in the finite fibre. Thus the zero-free origin of the $b = 0$ arithmetic pullback does not remove the branch mechanism from the entire original coefficient family. Conversely, choosing a coefficient to place a branch at a specified base point does not prove that that point is an arithmetic zero, or determine whether a zero lies off the real s axis. The formulas state exactly what each operation retains.

8 The actual arithmetic zero divisor on the retained inverse cover

This section uses the arithmetic spectral coordinate s, the Newman time t, and the original inverse-root coordinate r. In particular, the letter s here is not the completed-zeta argument ρ. The complete coordinate dictionary is

$$a = 2s = -r^2, \qquad z = 2s = -r^2, \qquad \rho = \tfrac{1}{2} + is = \tfrac{1}{2} - \tfrac{i}{2}r^2. \tag{41}$$

The maps $s \mapsto a$ and $s \mapsto z$ have inverses $a \mapsto a/2$ and $z \mapsto z/2$; $s \mapsto \rho$ has inverse $\rho \mapsto -i(\rho - 1/2)$. The last root map is a two-sheet map, whose two local inverses on a simply connected open subset of $\mathbb{C}_s \setminus \{0\}$ are $r = \sqrt{-2s}$ and $r = -\sqrt{-2s}$. No single inverse on all of $\mathbb{C}_s \setminus \{0\}$ is asserted.

The primary analytic conventions below are those of Brad Rodgers and Terence Tao[1, Section 1, equations (1)–(4)] and D. H. J. Polymath[4, Section 3, Proposition 3.1]. The local calculations below prove their transport in the coordinates (41), including the analytic factors in zero motion.

8.1 The specified datum, its moments, and the branch point

Keep the exact kernel and constants

$$\begin{aligned}
\Phi(u) &= \sum_{n=1}^{\infty}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u})e^{-\pi n^2 e^{4u}},\\
H_t(z) &= \int_0^{\infty} e^{tu^2}\Phi(u)\cos(zu)\,du,\\
Z(t,s) &= 8H_t(2s), \qquad f(t,r) = Z(t,-r^2/2) = 8H_t(-r^2).
\end{aligned} \tag{42}$$

Thus $Z(0,s) = \Xi(s) = \xi(1/2+is)$ by the cited Fourier identity, where $\xi(\rho) = \rho(\rho-1)\pi^{-\rho/2}\Gamma(\rho/2)\zeta(\rho)/2$, with its entire continuation at the removable exceptional points. This identity fixes the initial datum; the inverse cubic below is never substituted for it.

For $u \ge 0$ each summand of Φ is positive, since its polynomial factor is

$$\pi n^2 e^{5u}(2\pi n^2 e^{4u} - 3) > 0.$$

For a completely explicit integrable majorant put $C_\Phi = \sum_{n\ge 1} n^4 e^{-\pi(n^2-1)} < \infty$. The finiteness follows by comparison with $\sum_{n\ge1} n^4 e^{-\pi(n-1)}$. With $\lambda = \pi e^{4u} \ge \pi$, one has

$$0 < \Phi(u) \le 2\pi^2 e^{9u}\sum_{n\ge1} n^4 e^{-\lambda n^2} \le 2\pi^2 C_\Phi e^{9u-\pi e^{4u}}.$$

On $|t| \le T$, $|s| \le S$, every derivative of the integrand in (42) is bounded by a constant times $u^N\exp(Tu^2 + (9+2S)u - \pi e^{4u})$, for an integer $N \ge 0$. This is integrable: $\pi e^{4u}/2$ eventually exceeds $Tu^2 + (9+2S)u + N\log(1+u)$, and $\exp(-\pi e^{4u}/2)$ is integrable. The same argument applies on compact r sets, with $2S$ replaced by a bound for $|r|^2$. Differentiation under the integral and locally uniform holomorphic integration therefore show that Z is entire on $\mathbb{C}_t \times \mathbb{C}_s$, and f is entire on $\mathbb{C}_t \times \mathbb{C}_r$. They give the exact equations

$$Z_t = -\tfrac14 Z_{ss}, \qquad f_t = \mathcal{L}f, \qquad \mathcal{L} = -\frac{1}{4r^2}\partial_r^2 + \frac{1}{4r^3}\partial_r \quad (r \ne 0). \tag{43}$$

Indeed $f_r = -rZ_s$ and $f_{rr} = -Z_s + r^2 Z_{ss}$, so the two terms in $\mathcal{L}f$ are $Z_s/(4r^2) - Z_{ss}/4$ and $-Z_s/(4r^2)$, respectively. Their sum is $-Z_{ss}/4$.

Define the actual moments, with their original measure,

$$M_k(t) = \int_0^{\infty} u^{2k} e^{tu^2}\Phi(u)\,du \quad (k = 0,1,2,\ldots).$$

The majorant just proved gives $M_k'(t) = M_{k+1}(t)$ and

$$\begin{aligned}
&\partial_s^{2k} Z(t,0) = 8(-1)^k 2^{2k} M_k(t), \qquad \partial_s^{2k+1} Z(t,0) = 0,\\
&f(t,r) = 8\sum_{k=0}^{\infty}\frac{(-1)^k M_k(t)}{(2k)!} r^{4k}.
\end{aligned} \tag{44}$$

The series converges locally uniformly in both variables, by expansion of the cosine and the same compact-set majorant. In particular

$$\begin{aligned}
\Xi(0) &= 8M_0(0) = -\frac{\Gamma(1/4)\zeta(1/2)}{8\pi^{1/4}} > 0,\\
\Xi'(0) &= 0, \qquad \Xi''(0) = -32M_1(0) < 0,\\
f(t,r) &= 8M_0(t) - 4M_1(t)r^4 + \tfrac13 M_2(t) r^8 + O(r^{12}).
\end{aligned}$$

The equality with $\Gamma(1/4)\zeta(1/2)$ follows by evaluating the specified completed-zeta identity, while both strict signs follow directly from the positive integrals. Neither sign requires a numerical evaluation of $\zeta(1/2)$.

For every real t and v there is the stronger exact assertion, also recorded in the first paragraph of Polymath's Section 3[4],

$$Z(t, iv) = 8 \int_0^\infty e^{tu^2} \Phi(u) \cosh(2vu)\, du > 0. \tag{45}$$

In particular the actual arithmetic zero divisor avoids $s = 0$ for every real Newman time. The completed root cover has a branch point there, but no arithmetic zero reaches that branch point at any real time. Evenness of the cosine also proves, for all complex t, s, r,

$$Z(t, -s) = Z(t, s), \qquad f(t, -r) = f(t, r), \qquad f(t, ir) = f(t, r). \tag{46}$$

For real t complex conjugation gives $\overline{Z(t, s)} = Z(t, \overline{s})$ and $\overline{f(t, r)} = f(t, \overline{r})$, because the kernel is real.

8.2 Exact function maps and the removable and nonremovable terms

Write $\mathcal{O}(\mathbb{C})$ for entire functions with the topology of uniform convergence on compact sets. The pullback

$$\pi^* : \mathcal{O}(\mathbb{C}_s) \longrightarrow \mathcal{O}(\mathbb{C}_r), \qquad q \longmapsto q(-r^2/2)$$

is injective, since every $s \in \mathbb{C}$ has a root of $r^2 = -2s$. Its image is exactly the even entire functions. To prove the latter statement without a choice of square root, expand an even entire function as $g(r) = \sum_{n \geq 0} a_{2n} r^{2n}$ and define $q(s) = \sum_{n \geq 0} (-2)^n a_{2n} s^n$. This series converges for every s: choose $R > \sqrt{2|s|}$ and apply the Cauchy bound $|a_{2n}| \leq \max_{|r|=R} |g(r)| R^{-2n}$. It gives $q(-r^2/2) = g(r)$, and proves uniqueness by comparing coefficients. It also proves continuity of the inverse on compact sets. Every entire cover function has the unique decomposition

$$g(r) = q_0(-r^2/2) + r q_1(-r^2/2).$$

Here q_0 is recovered from $(g(r) + g(-r))/2$ and q_1 from $(g(r) - g(-r))/(2r)$, with the latter expression extended at $r = 0$ by its power series. The sheet involution $\iota g(r) = g(-r)$ has trace map $g \mapsto g + \iota g$ with kernel exactly the odd entire functions. These are maps of actual holomorphic functions, before any arithmetic quotient is formed.

For every integer $n \geq 0$ the operator in (43) gives

$$\mathcal{L}(r^n) = \frac{n(2-n)}{4} r^{n-4}. \tag{47}$$

Consequently, for $g(r) = \sum_{n \geq 0} a_n r^n$, its Laurent expansion is

$$\mathcal{L}g = \frac{a_1}{4r^3} - \frac{3a_3}{4r} + \sum_{n \geq 4} \frac{n(2-n)}{4} a_n r^{n-4}.$$

The terms $n = 0, 2$ are zero. Thus $\mathcal{L}g$ extends to an entire function exactly when $a_1 = a_3 = 0$. Its value at zero in that case is $-2a_4 = -g^{(4)}(0)/12$. In particular $\mathcal{L}f(t, 0) = 8M_1(t) = f_t(t, 0)$: the singular coordinate coefficients cancel on the actual arithmetic datum.

The largest subspace of $\mathcal{O}(\mathbb{C}_r)$ on which every iterate $\mathcal{L}^j$ remains entire is the even subspace. For even monomials successive applications of (47) reach r^0 or r^2, which are annihilated, without producing a pole. Equivalently, $\mathcal{L}^j \pi^* q = (-1/4)^j \pi^* q^{(2j)}$. Conversely, let a_n be the first nonzero odd coefficient of g. When $n = 4j + 1$, applying $\mathcal{L}^{j+1}$ produces a nonzero multiple of r^{-3} from this term; when $n = 4j + 3$ it produces a nonzero multiple of r^{-1}.

Every factor $(n-4\ell)(2-n+4\ell)/4$ is nonzero for these odd exponents. Distinct initial powers retain distinct exponents after the same number of applications, so no cancellation with a higher power is possible. Even powers cannot contribute a pole. This proves the assertion and maximality. It asserts stability of all differential iterates, not existence of a heat evolution for every even entire datum.

There is an equivalent spacetime statement. A function g holomorphic on a product of a time disk with a full disk about $r=0$, satisfying $g_t=\mathcal{L}g$ for $r\neq 0$, is even in r. Indeed successive time differentiation gives $\mathcal{L}^j g=\partial_t^j g$ on the punctured disk, and the right side is holomorphic at zero for every j. The preceding first-odd-coefficient argument, which is local in r, applies at every time. The equation therefore descends there through the proved even-function inverse to $q_t=-q_{ss}/4$. On a connected punctured plane the kernel of $\mathcal{L}$ consists exactly of $A+Br^2$: multiplying the equation by $4r^3$ gives $rg_{rr}=g_r$, hence $(g_r/r)'=0$ and then $g=A+Br^2$.

8.3 The original cubic, all finite fibres, and the zero divisor

Keep the original map in spatial coordinates (x,y,w):

$$\begin{aligned}F_1&=(1+xy)^3w+y^2(1+xy)(4+3xy),\\F_2&=y+3x(1+xy)^2w+3xy^2(4+3xy),\\F_3&=2x-3x^2y-x^3w.\end{aligned}$$

Its retained polynomial is

$$P_{a,b,c}(r)=cr^3-2r^2+br-2a,\qquad \alpha=\tfrac12 P'_{a,b,c}(r).$$

For $x\neq 0$ the inverse chart is

$$x=\alpha^{-1},\qquad y=r-\alpha,\qquad w=5\alpha^2-3r\alpha-c\alpha^3,\qquad P_{a,b,c}(r)=0,\quad P'_{a,b,c}(r)\neq 0. \tag{48}$$

Here is the needed exact verification. Set $r=y+1/x$ and $\alpha=1/x$. Solving $F_3=c$ gives the displayed expression for w. Substitution in the other components gives $F_2=4r+2\alpha-3cr^2$ and $F_1=r^2+r\alpha-cr^3$. The equality $F_2=b$ is $2\alpha=P'(r)$, and after its substitution $2(F_1-a)=P(r)$. These identities prove both directions and recover r from the inverse image. In particular a multiple root $P'=0$ has no finite point in this chart. On $x=0$, direct substitution gives $F(0,y,w)=(w+4y^2,y,0)$; there is the additional point $(0,b,a-4b^2)$ precisely when $c=0$.

The arithmetic slice $a=2s$, $b=c=0$ now gives, with no omitted coefficient,

$$P_{2s,0,0}(r)=-2r^2-4s,\qquad \alpha=-2r.$$

Its root relation is exactly $s=-r^2/2$. For $r\neq 0$ the spatial inverse is the explicit map

$$\Gamma(r)=\left(-\frac{1}{2r},\ 3r,\ 26r^2\right),\qquad F(\Gamma(r))=(-r^2,0,0). \tag{49}$$

The inverse root is $y+1/x=3r-2r=r$. For $s\neq 0$, its two roots give two distinct finite points $\Gamma(r)$ and $\Gamma(-r)$, and the complete original fibre also has the point $(0,0,2s)$. For $s=0$ the polynomial has its double root at zero, both inverse chart points are excluded by $\alpha=0$, and the sole finite fibre point is $(0,0,0)$. The completed root point $r=0$ is not silently assigned the undefined spatial coordinate $x=-1/(2r)$.

For real t, every zero s_* of the specified $Z(t,\cdot)$ is nonzero by (45). It therefore has exactly two zero lifts r_* and $-r_*$ on the retained cover, and these have the finite spatial representatives in (49). The actual scalar $Z(t, F_1(x, y, w)/2)$ vanishes on all three points of the original fibre above $(2s_*, 0, 0)$. Thus the odd sheet observable is retained by the cover, and the extra $x = 0$ fibre point is separately recorded.

Multiplicity is also transported exactly. At any fixed complex time, write $Z(t, s) = (s - s_*)^m A(s)$ at a zero, where $A(s_*) \neq 0$. For $r_* \neq 0$ with $r_*^2 = -2s_*$,

$$f(t, r) = \left[-\tfrac{1}{2}(r - r_*)(r + r_*)\right]^m A(-r^2/2).$$

The factor $[-(r + r_*)/2]^m A(-r^2/2)$ is a unit near r_*, so the multiplicity is m there and also at $-r_*$. The local coordinate $s - s_* = -(r - r_*)(r + r_*)/2$ has derivative $-r_* \neq 0$; its holomorphic inverse is the corresponding branch of $r = \sqrt{-2s}$. It gives isomorphisms of the local function rings and of their quotients by the respective zero ideals. At $s_* = 0$ the exact ideal is instead $(s^m)\mathcal{O}_{\mathbb{C}_r,0} = (r^{2m})$; the factor $(-1/2)^m A(-r^2/2)$ is a unit. Thus the multiplicity doubles there. The quotient at that point is the rank-two module

$$\mathbb{C}\{r\}/(r^{2m}) = \mathbb{C}\{s\}/(s^m) \ \oplus\ r\mathbb{C}\{s\}/(s^m), \qquad s = -r^2/2.$$

Separating even and odd coefficients proves existence and uniqueness of this decomposition; the trace has precisely its odd summand as kernel. These local divisor rings do not assert that differentiation acts on a frozen global arithmetic spectral quotient.

For $r = u + iv$ with $u, v \in \mathbb{R}$, the original arithmetic point is

$$s = \frac{v^2 - u^2}{2} - iuv, \qquad \rho = \frac{1}{2} + uv + i\frac{v^2 - u^2}{2}. \tag{50}$$

It lies on $\operatorname{Re}\rho = 1/2$ exactly when $uv = 0$, which is exactly the union of the real and imaginary r axes. For real t there are no cover zeros on $u = v$ or $u = -v$, since these lines map into the imaginary s axis of (45). In particular an imaginary root coordinate alone does not give an off-critical arithmetic zero; the exact arithmetic test in this coordinate is $uv \neq 0$ at a zero of $f(0, r)$.

8.4 Simple zero motion with every analytic-unit term

Consider a point of the actual zero locus with $Z_s(t_0, s_0) \neq 0$. The holomorphic implicit function theorem applies because Z is jointly entire and this derivative is nonzero. It gives a unique holomorphic zero $s = \sigma(t)$ in a product of small disks. Differentiating $Z(t, \sigma(t)) = 0$ and using (43) gives the exact velocity

$$\sigma'(t) = \frac{Z_{ss}(t, \sigma(t))}{4Z_s(t, \sigma(t))}. \tag{51}$$

More generally write a local factorization $Z = AQ$, where A is a nonvanishing holomorphic function of (t, s) and Q is the full local zero factor under consideration. The product rule gives

$$Q_t = -\tfrac{1}{4}Q_{ss} - \frac{A_s}{2A}Q_s - \left(\frac{A_t}{A} + \frac{A_{ss}}{4A}\right)Q. \tag{52}$$

All functions in this equality are evaluated on that product of disks; dividing by A is valid there. At a simple root it gives

$$\sigma' = \frac{1}{4}\frac{Q_{ss}}{Q_s} + \frac{1}{2}\frac{A_s}{A}. \tag{53}$$

If $Q = \prod_{j=1}^m (s - \sigma_j(t))$ has distinct roots at the time in question, factor off $s - \sigma_j$ and differentiate the remaining product. This proves

$$\frac{Q_{ss}(\sigma_j)}{Q_s(\sigma_j)} = 2\sum_{k\neq j}\frac{1}{\sigma_j - \sigma_k}, \qquad \sigma_j' = \frac{1}{2}\sum_{k\neq j}\frac{1}{\sigma_j - \sigma_k} + \frac{1}{2}\frac{A_s(t, \sigma_j)}{A(t, \sigma_j)}.$$

For a single local root $Q = s - \sigma$, its entire velocity is $A_s/(2A)$. Removing the unit would remove this actual velocity.

Away from $r = 0$, a lifted simple root obeys $r^2 = -2\sigma$, so differentiation and the computed spatial derivatives give

$$r' = -\frac{\sigma'}{r} = -\frac{1}{4r}\frac{Z_{ss}}{Z_s} = \frac{1}{4r^2}\frac{f_{rr}}{f_r} - \frac{1}{4r^3}. \tag{54}$$

All ratios are evaluated at the moving zero. For a factorization $f = BV$ on a cover disk, the complete product equation is

$$\begin{aligned} V_t = & -\frac{1}{4r^2}V_{rr} + \left(\frac{1}{4r^3} - \frac{B_r}{2r^2 B}\right) V_r \\ & + \left(-\frac{B_{rr}}{4r^2 B} + \frac{B_r}{4r^3 B} - \frac{B_t}{B}\right) V, \end{aligned} \tag{55}$$

and at a simple root

$$r' = \frac{1}{4r^2}\frac{V_{rr}}{V_r} + \frac{B_r}{2r^2 B} - \frac{1}{4r^3}.$$

For the two complete sheet lifts of a single base zero, take $V = r^2 + 2\sigma(t)$ and $B(t, r) = -A(t, -r^2/2)/2$, where $Z = A(s - \sigma)$. At a root $V_{rr}/V_r = 1/r$. Its term $1/(4r^3)$ cancels the displayed $-1/(4r^3)$ exactly, while $B_r/B = -rA_s/A$. The result is $r' = -A_s/(2rA) = -\sigma'/r$. This computation retains both sheet roots and the analytic unit; it identifies the cancellation rather than deleting either term. The original spatial velocity is also explicit:

$$\frac{d}{dt}\Gamma(r(t)) = \left(\frac{r'}{2r^2},\ 3r',\ 52rr'\right), \qquad \frac{d}{dt}F(\Gamma(r(t))) = (2\sigma', 0, 0).$$

8.5 A double zero: exact pair equations and the unit drift

At an actual double zero (t_0, s_0), put $h = t - t_0$ and write

$$Z(t_0, s_0 + \delta) = a_2\delta^2 + a_3\delta^3 + O(\delta^4), \qquad a_2 \neq 0.$$

The local zero factor can be expressed as $Q = (s - M)^2 - D^2$, where M and D^2 are holomorphic in t, $M(t_0) = s_0$, and $D^2(t_0) = 0$. A construction not depending on the naming of the two roots is to take a small circle containing only the double zero and no boundary zero. The integrals

$$p_k(t) = \frac{1}{2\pi i}\int_{\text{circle}} s^k \frac{Z_s(t, s)}{Z(t, s)}\, ds \quad (k = 1, 2)$$

are holomorphic and equal the first two root power sums with multiplicity, by the residue computation for a local zero factor. Thus $M = p_1/2$ and $D^2 = p_2/2 - M^2$ are holomorphic. The two zeros, counted with multiplicity, are $M + D$ and $M - D$. The holomorphic unit can itself be constructed on a smaller concentric disk by the contour formula

$$A(t, s) = \frac{1}{2\pi i}\int_{\text{circle}} \frac{Z(t, \zeta)}{Q(t, \zeta)(\zeta - s)}\, d\zeta.$$

The denominator is nonzero on the contour for all the retained times, so the integral is jointly holomorphic. At each fixed time Z/Q extends holomorphically across its two zeros, counted with their multiplicities, by their one-variable Taylor factorizations. Cauchy's formula therefore identifies the displayed integral with Z/Q, proving $Z = AQ$. At $t = t_0$ it is $Z(t_0, s)/(s - s_0)^2$, nonzero at s_0, so A is nonvanishing after shrinking the disks.

Put $b(t,s) = A_s(t,s)/A(t,s)$. At every nearby time with $D \neq 0$ the proved simple-zero equation gives the exact pair dynamics

$$\begin{aligned}
(M+D)' &= \frac{1}{4D} + \tfrac{1}{2}b(t, M+D),\\
(M-D)' &= -\frac{1}{4D} + \tfrac{1}{2}b(t, M-D),\\
M' &= \tfrac{1}{4}\big[b(t,M+D) + b(t,M-D)\big],\\
(D^2)' &= \tfrac{1}{2} + \tfrac{D}{2}\big[b(t,M+D) - b(t,M-D)\big].
\end{aligned} \tag{56}$$

The sum and D times the difference are even holomorphic functions of D, hence holomorphic functions of D^2. These equations extend to $h = 0$, and there give

$$(D^2)'(t_0) = \tfrac{1}{2}, \qquad M'(t_0) = \frac{a_3}{2a_2} = \frac{Z_{sss}(t_0,s_0)}{6Z_{ss}(t_0,s_0)}.$$

Indeed $A(t_0,s_0) = a_2$ and $A_s(t_0,s_0) = a_3$. The full equation (52) still carries A_t and A_{ss}; their factors multiply Q and vanish only when evaluated at a zero. They have not been assumed to vanish on the neighborhood.

The following explicit expansion proves the pair's first unit term and its error order. The heat equation gives the terms of weighted degree at most three, with δ of weight one and h of weight two:

$$Z(t_0+h, s_0+\delta) = a_2(\delta^2 - h/2) + a_3(\delta^3 - 3h\delta/2) + O((|\delta| + |h|^{1/2})^4).$$

Set $h = \varepsilon^2$. Dividing by ε^2 after $\delta = \varepsilon y$ gives a holomorphic function of (ε, y) whose value at $\varepsilon = 0$ is $a_2(y^2 - 1/2)$. Its two simple roots are $y = \pm 1/\sqrt{2}$, so the holomorphic implicit function theorem gives two holomorphic y branches. Their next coefficient is found by substituting $\delta = \lambda\varepsilon + \mu\varepsilon^2$: $\lambda^2 = 1/2$ and $2a_2\lambda\mu + a_3(\lambda^3 - 3\lambda/2) = 0$. Hence $\mu = a_3/(2a_2)$ and

$$\sigma_\pm(t) = s_0 \pm \sqrt{h/2} + \frac{a_3}{2a_2}h + O(h^{3/2}). \tag{57}$$

This means convergent holomorphic expansions in ε, with a chosen $\varepsilon^2 = h$; reversing ε exchanges the root labels. It follows that, on either punctured time branch,

$$\sigma'_\pm = \pm\frac{1}{2\sqrt{2h}} + \frac{a_3}{2a_2} + O(h^{1/2}).$$

For real t_0, s_0 the coefficients are real. For small positive h both branches are real, by uniqueness and conjugation of their real ε power series. For small negative real h their imaginary parts are $\pm\sqrt{|h|/2} + O(|h|^{3/2})$, so they are a nonreal conjugate pair. This describes an actual double-zero event wherever it occurs; it does not insert such an event at an unspecified real arithmetic time.

At real time $s_0 \neq 0$ by (45). Choose either $r_0^2 = -2s_0$. The inverse branch has $dr/ds = -1/r_0$ and $d^2r/ds^2 = -1/r_0^3$. Taylor substitution in (57) yields the two roots near that particular sheet point,

$$r_\pm(t) = r_0 \mp \frac{1}{r_0}\sqrt{h/2} - \left(\frac{a_3}{2a_2 r_0} + \frac{1}{4r_0^3}\right)h + O(h^{3/2}). \tag{58}$$

The other sheet consists exactly of their negatives. Both the arithmetic unit drift and the curvature of the original root map occur in its displayed coefficient of h.

8.6 Every multiplicity and its interaction with ramification

At an actual zero of finite multiplicity $m \geq 1$ write $Z(t_0, s_0 + \delta) = \sum_{k \geq m} a_k \delta^k$, $a_m \neq 0$. There is finite multiplicity because $Z(t_0, \cdot)$ is not the zero function. For real t_0 positivity at zero already proves this. For complex t_0 put $K(u) = e^{t_0 u^2}\Phi(|u|)$ on $\mathbb{R}$. The proved decay shows that K is continuous, bounded, and integrable, and $K(0) = \Phi(0) > 0$. Its Fourier transform satisfies $\widehat{K}(\xi) = 2H_{t_0}(\xi)$ for real ξ, by evenness. If $Z(t_0, \cdot)$ were identically zero, this transform would vanish. For $\epsilon > 0$, the Gaussian integral gives

$$G_\epsilon(x) = \frac{e^{-x^2/(4\epsilon)}}{\sqrt{4\pi\epsilon}} = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\epsilon\xi^2} e^{ix\xi}\, d\xi.$$

Fubini is justified by $K \in L^1$ and the integrable Gaussian in ξ; it gives $(K * G_\epsilon)(0) = (2\pi)^{-1}\int \widehat{K}(\xi)e^{-\epsilon\xi^2}\, d\xi = 0$. On the other hand $(K * G_\epsilon)(0) \to K(0) > 0$ as $\epsilon \downarrow 0$: on $|u| < \delta$ continuity bounds $|K(u) - K(0)|$ arbitrarily small, while on its complement boundedness of K and the vanishing Gaussian tail bound the remaining integral. This contradiction proves the required nonidentity with the zero function for every complex time.

Repeated differentiation of the heat equation gives $\partial_t^\ell Z = (-1/4)^\ell \partial_s^{2\ell} Z$. The bivariate Taylor coefficient of $h^\ell \delta^j$ is therefore

$$\frac{(-1/4)^\ell}{\ell!}\frac{(j+2\ell)!}{j!} a_{j+2\ell}.$$

Set $h = \varepsilon^2$, $\delta = \varepsilon y/\sqrt{2}$, with the inverse substitutions retained whenever $\varepsilon \neq 0$. The holomorphic Taylor series is divisible by ε^m; grouping the terms of degree m and $m+1$ gives

$$\frac{Z(t_0 + \varepsilon^2, s_0 + \varepsilon y/\sqrt{2})}{a_m(\varepsilon/\sqrt{2})^m} = \mathrm{He}_m(y) + \frac{a_{m+1}}{a_m}\frac{\varepsilon}{\sqrt{2}}\mathrm{He}_{m+1}(y) + O(\varepsilon^2),$$

$$\mathrm{He}_m(y) = \sum_{\ell=0}^{\lfloor m/2 \rfloor} \frac{m!(-1)^\ell}{2^\ell \ell!(m-2\ell)!} y^{m-2\ell}. \tag{59}$$

The quotient extends holomorphically at $\varepsilon = 0$; its remainder is uniform on each bounded y set. This follows directly from the convergent bivariate Taylor series after substitution, so the error does not assume a finite polynomial initial datum.

For completeness the polynomial has m distinct real roots. Differentiating its displayed finite sum proves $\mathrm{He}_m' = m\,\mathrm{He}_{m-1}$ and $\mathrm{He}_{m+1} = y\,\mathrm{He}_m - m\,\mathrm{He}_{m-1}$. Starting with $\mathrm{He}_0 = 1$ and $\mathrm{He}_1 = y$, induction shows that the roots of two successive polynomials strictly interlace: at consecutive roots of He_m, the values $-m\,\mathrm{He}_{m-1}$ alternate sign, providing one root of He_{m+1} between each consecutive pair. At the largest root, $\mathrm{He}_{m-1} > 0$, so $\mathrm{He}_{m+1} < 0$ there and is positive sufficiently far to the right. At the smallest root, the sign is $(-1)^m$ and the far-left sign of its monic degree $m+1$ polynomial is $(-1)^{m+1}$, providing a further root to the left. There are thus $m+1$ distinct roots. The induction begins with the single root of He_1 and gives the claimed reality, simplicity, and interlacing. Let these roots be λ_j. At a root of He_m the recurrences also give

$$\frac{\mathrm{He}_{m+1}(\lambda_j)}{\mathrm{He}_m'(\lambda_j)} = -1.$$

Apply the holomorphic implicit function theorem to (59) at each $(0, \lambda_j)$. The resulting roots are

$$\sigma_j(t) = s_0 + \lambda_j\sqrt{h/2} + \frac{a_{m+1}}{2a_m}h + O(h^{3/2}). \tag{60}$$

They are all the nearby roots: a small fixed circle about s_0 has no boundary zero at t_0, and continuity followed by Rouché's theorem keeps its zero count at m; the m disjoint roots just constructed account for that count. Writing $Z(t_0, s) = (s - s_0)^m A_0(s)$ gives $a_{m+1}/(2a_m) = A_0'(s_0)/(2A_0(s_0))$. Thus the first common drift at every multiplicity is exactly the initial analytic-unit derivative. Higher coefficients remain determined by the full Taylor series and the exact unit equation (52).

For $s_0 \neq 0$, on either chosen sheet $r_0^2 = -2s_0$ the same inverse coordinate Taylor expansion proves

$$r_j(t) = r_0 - \frac{\lambda_j}{r_0}\sqrt{h/2} - \left(\frac{a_{m+1}}{2a_m r_0} + \frac{\lambda_j^2}{4r_0^3}\right) h + O(h^{3/2}).$$

At $s_0 = 0$, an event of the actual family is possible only at nonreal complex time, by (45). On this locus the exact evenness $Z(t, -s) = Z(t, s)$ forces $m = 2k$ and $a_{m+1} = 0$. Its Hermite roots are nonzero, since $\mathrm{He}_{2k}(0) = (-1)^k (2k)!/(2^k k!)$. Set $h = \eta^4$ and $\varepsilon = \eta^2$. Equation (60) now gives

$$\sigma_j(t_0 + \eta^4) = \frac{\lambda_j}{\sqrt{2}}\eta^2 + O(\eta^6), \qquad r_{j,\pm}(\eta) = \pm\eta\sqrt{-\sqrt{2}\lambda_j} + O(\eta^5).$$

The square root of each nonzero displayed constant has two choices. To justify the expansion, $-2\sigma_j(t_0 + \eta^4)/\eta^2$ extends holomorphically and has nonzero value $-\sqrt{2}\lambda_j$ at zero; either holomorphic square root of that unit gives exactly $r_{j,\pm} = \pm\eta\sqrt{-2\sigma_j/\eta^2}$. The $2m$ lifted roots account for the cover multiplicity $2m$ proved above. This is the precise fourth-root time ramification over an arithmetic multiple zero at the cover branch locus. It does not occur there at real Newman time, and it has no finite spatial inverse chart at its central point because $P' = 0$.

Verification status. The accompanying exact rational-polynomial checker verifies the chain rule coefficients, monomial pole coefficients, original spatial lift, analytic-unit product identities, and finite coefficients of the collision expansions. The convergence, positivity, local holomorphic factorizations, root counts, and complete multiplicity statements are proved in the text; the checker is not a substitute for those proofs. No off-critical zero of the distinguished arithmetic initial function is asserted by these calculations.

9 The actual arithmetic observable under spatial diffusion

Retain the original real spatial coordinates $q = (x, y, w)$ and the polynomial F in (1). The complex formulas below are their holomorphic extensions; a sum of derivative squares is a Euclidean squared norm only on the stated real locus. Define the exact observable

$$\sigma(q) = F_1(q)/2, \qquad \Psi(t, q) = Z_t(\sigma(q)).$$

The exact map of time and spatial coordinates is $\rho : \mathbb{C}_t \times \mathbb{C}_q^3 \to \mathbb{C}_t \times \mathbb{C}_s$, $\rho(t, q) = (t, \sigma(q))$, with pullback $\rho^* Z = \Psi$. For each differentiable time map $\vartheta : I_\tau \to \mathbb{C}_t$ its restriction is $\rho_\vartheta(\tau, q) = (\vartheta(\tau), \sigma(q))$. The chain rule gives $\partial_\tau \rho_\vartheta^* Z = \vartheta' \rho_\vartheta^* Z_t$ and $\partial_{q_i} \rho_\vartheta^* Z = (\partial_{q_i}\sigma)\rho_\vartheta^* Z_s$. These formulas specify the typed relation between Newman time and spatial time with its derivative factor. The already proved entire Z makes Ψ entire in $(t, q) \in \mathbb{C} \times \mathbb{C}^3$.

9.1 Original vector fields and the rank-one heat operator

Here is a direct determinant proof needed to define the vector fields. Use the already proved inverse-chart map

$$H(\alpha, r, c) = (\alpha^{-1}, r - \alpha, 5\alpha^2 - 3r\alpha - c\alpha^3), \quad \alpha \neq 0.$$

Its inverse on $x \neq 0$ is $(1/x, y + 1/x, F_3(q))$. The two derivative matrices, in the displayed coordinate order, are

$$DH = \begin{pmatrix} -\alpha^{-2} & 0 & 0 \\ -1 & 1 & 0 \\ -3c\alpha^2 + 10\alpha - 3r & -3\alpha & -\alpha^3 \end{pmatrix}, \quad D(F \circ H) = \begin{pmatrix} r & \alpha - 3cr^2 + 2r & -r^3 \\ 2 & 4 - 6cr & -3r^2 \\ 0 & 0 & 1 \end{pmatrix}.$$

The first determinant is α. Expansion along the last row of the second gives $r(4 - 6cr) - 2(\alpha - 3cr^2 + 2r) = -2\alpha$. The chain rule therefore proves $\det DF = -2$ on $x \neq 0$. Both sides are polynomials in (x, y, w), so their equality extends to $x = 0$ by continuity. Put $J = DF$, $W_i = J^{-1}e_i$, and

$$U = 2W_1 + 6F_3W_2.$$

These are globally defined polynomial vector fields, since $J^{-1} = -(1/2) \operatorname{adj} J$. With $\epsilon_{123} = 1$ and cyclic indices, the adjugate formula is $W_i = -\frac{1}{2}\nabla F_{i+1} \times \nabla F_{i+2}$. The divergence of each cross product is zero: its differentiated components contract a symmetric Hessian with an antisymmetric ϵ tensor. Also $W_iF_j = \delta_{ij}$, by $JW_i = e_i$. The product rule now gives

$$\operatorname{div} U = 2 \operatorname{div} W_1 + 6F_3 \operatorname{div} W_2 + 6W_2F_3 = 0, \qquad U\sigma = 1.$$

For every twice differentiable scalar $h(s)$, direct differentiation proves $U(h \circ \sigma) = h'(\sigma)$ and $U^2(h \circ \sigma) = h''(\sigma)$. Consequently the actual observable satisfies the exact operator identity

$$\partial_t\Psi + \tfrac{1}{4}U^2\Psi = 0. \tag{61}$$

Here the full spatial operator is

$$U^2h = U^iU^j\partial_i\partial_jh + U^i(\partial_iU^j)\partial_jh = \partial_i(U^iU^j\partial_jh),$$

where repeated indices run over the three original coordinates. The last equality uses the proved divergence identity. Its real principal tensor is $UU^\mathsf{T}/4$, of rank one because $U\sigma = 1$ implies $U \neq 0$. On $C_c^\infty(\mathbb{R}^3)$, one integration by parts gives $-\int \bar{h}U^2h = \int |Uh|^2$; polynomial coefficients are bounded on the support and there are no boundary terms. The explicit reversal $\Phi(\tau, q) = \Psi(-\tau, q)$ obeys $\partial_\tau\Phi = U^2\Phi/4$. This proves an exact forward spatial heat realization on the specified observable, with every first-order coefficient retained.

9.2 Every Euclidean diffusion coefficient

Let $Q = 1 + xy$, an abbreviation for the original polynomial factor. Differentiation before restriction to a fibre gives

$$\begin{aligned} g_1 &:= \partial_x\sigma = \tfrac{3}{2}yQ^2w + \tfrac{1}{2}y^3(7 + 6xy), \\ g_2 &:= \partial_y\sigma = \tfrac{3}{2}xQ^2w + \tfrac{1}{2}(8y + 21xy^2 + 12x^2y^3), \\ g_3 &:= \partial_w\sigma = \tfrac{1}{2}Q^3. \end{aligned}$$

Define $N = g_1^2 + g_2^2 + g_3^2$ and

$$L = \Delta_q\sigma = 3wQ(x^2 + y^2) + 3y^4 + 18x^2y^2 + 21xy + 4.$$

For example the second derivatives in the Laplacian are $\sigma_{xx} = 3wy^2Q + 3y^4$, $\sigma_{yy} = 3wx^2Q + 18x^2y^2 + 21xy + 4$, and $\sigma_{ww} = 0$, proving every term in L. The chain rule in each coordinate, followed by summation, proves

$$\Delta_q\Psi = N(q)Z_{ss}(t, \sigma(q)) + L(q)Z_s(t, \sigma(q)). \tag{62}$$

No term is suppressed by the one-variable presentation of Z. For real q, $N = |\nabla\sigma|^2 > 0$: an invertible J cannot have its first row zero.

On the original finite inverse lift $\Gamma(r) = (-1/(2r), 3r, 26r^2)$, $r \neq 0$, direct substitution gives

$$\nabla\sigma(\Gamma(r)) = (9r^3/4, 3r/8, -1/16)^\mathsf{T}, \qquad D^2\sigma(\Gamma(r)) = \begin{pmatrix} -108r^4 & 3r^2/4 & 9r/8 \\ 3r^2/4 & 13/4 & -3/(16r) \\ 9r/8 & -3/(16r) & 0 \end{pmatrix}.$$

Thus, retaining $\sigma(\Gamma(r)) = -r^2/2$,

$$\begin{aligned} N(\Gamma(r)) &= \frac{1296r^6 + 36r^2 + 1}{256}, \\ L(\Gamma(r)) &= \frac{13}{4} - 108r^4, \\ (\Delta_q\Psi)(t, \Gamma(r)) &= \frac{1296r^6 + 36r^2 + 1}{256} Z_{ss}(t, -r^2/2) + \left(\frac{13}{4} - 108r^4\right) Z_s(t, -r^2/2). \end{aligned} \tag{63}$$

These are identities of rational holomorphic functions for complex $r \neq 0$. Their Euclidean-norm interpretation uses real $r \neq 0$.

For a real differentiable time map $t = \theta(\tau)$ and the original viscosity $\nu > 0$, the complete advection–diffusion residual is

$$\begin{aligned} &(\partial_\tau + U \cdot \nabla_q - \nu\Delta_q)\Psi(\theta(\tau), q) \\ &\quad = \left(-\frac{\theta'(\tau)}{4} - \nu N(q)\right) Z_{ss}(\theta(\tau), \sigma(q)) + (1 - \nu L(q)) Z_s(\theta(\tau), \sigma(q)). \end{aligned} \tag{64}$$

The three preceding exact identities prove this formula term by term. It is the direct map between the actual arithmetic heat solution and the scalar spatial advection–diffusion operator of this polynomial flow. It does not assert that its displayed residual vanishes.

9.3 The particular arithmetic covector along the escaping orbit

Local existence and uniqueness of the smooth polynomial ODE $\partial_\tau X_\tau = U(X_\tau)$ follow on a closed finite ball from the contraction map $Y \mapsto q + \int_0^\tau U(Y(v))\,dv$ for sufficiently short time; bounded DU supplies its Lipschitz constant. Continuing these solutions defines the maximal local flow on its actual open domains. Smooth dependence follows by differentiating the integral equation; uniqueness supplies the inverse local flow. The identities $UF_1 = 2$, $UF_2 = 6F_3$, and $UF_3 = 0$ integrate to

$$F(X_\tau(q)) = (F_1(q) + 2\tau, F_2(q) + 6\tau F_3(q), F_3(q)).$$

In particular $\sigma(X_\tau(q)) = \sigma(q) + \tau$. Write $A_\tau = D_q X_\tau$ and $K_\tau = A_\tau^{-1} A_\tau^{-\mathsf{T}}$. Differentiating this last scalar identity gives the full covector map

$$A_\tau^\mathsf{T} \nabla\sigma(X_\tau(q)) = \nabla\sigma(q), \qquad \nu\nabla\sigma(q)^\mathsf{T} K_\tau \nabla\sigma(q) = \nu N(X_\tau(q)).$$

Every matrix is evaluated at the same initial point q and time τ; no arbitrary covector has replaced this actual one.

At $q_0 = (1, -3/2, 13/2) = \Gamma(-1/2)$ put $z = \sqrt{1 - 8\tau} > 0$, with inverse $\tau = (1 - z^2)/8$. For $-\infty < \tau < 1/8$ the explicit trajectory is $X_\tau(q_0) = \Gamma(-z/2) = (z^{-1}, -3z/2, 13z^2/2)$. To verify the ODE, its F image is $(-1/4 + 2\tau, 0, 0)$; differentiation gives $J\gamma' = (2, 0, 0) = JU(\gamma)$, so invertibility of J gives $\gamma' = U(\gamma)$. Its value at $\tau = 0$ is q_0, and its escape as $\tau \uparrow 1/8$ precludes a finite continuous extension of the ODE solution. For the specific initial covector

$$\ell_\sigma = \nabla\sigma(q_0) = (-9/32, -3/16, -1/16)^\mathsf{T}$$

we have the entire transported vector and cost

$$A_\tau^{-\mathsf{T}}\ell_\sigma = (-9z^3/32, -3z/16, -1/16)^\mathsf{T},$$
$$\nu\ell_\sigma^\mathsf{T} K_\tau \ell_\sigma = \nu\frac{81z^6+36z^2+4}{1024} \longrightarrow \frac{\nu}{256} \quad (z\downarrow 0).$$

This computes the precise diffusive cost of the covector selected by the arithmetic observable, even at the spatial escape endpoint. The zero condition itself remains $Z_t(\sigma(q)) = 0$ with its full analytic divisor as proved above; the transported cost is not an assertion of such a zero or of a Euclidean Navier–Stokes counterexample.

10 The source fourth coefficient and its exact Newman heat packet

The result supplied in the linked task uses the coupling symbol ξ. Keep its original finite open cubic box, $L \geq 2$, with $N = 3(2L)(2L+1)^2$ links and $M = 3(2L)^2(2L+1)$ elementary faces. On product Haar probability measure on $SU(2)^N$, the actual Hamiltonian and scales are

$$H = \kappa\sum_e E_e + b\sum_p (2 - W_p), \qquad \kappa = \frac{2g^2}{a}, \quad b = \frac{1}{2g^2a}, \quad \xi = \frac{b}{\kappa} = \frac{1}{4g^4}, \qquad a, g > 0.$$

Here a is the source lattice spacing, and the polynomial target also uses the letter $a = F_1$. Their exact typed relationship is as follows. The shared letters are retained through the following explicit product space and maps. Subscripts in this paragraph identify the original coordinate being evaluated: $a_{\mathrm{lat}} = a$ in the Hamiltonian, $b_{\mathrm{mag}} = b$ in its potential, and $(a_{\mathrm{pol}}, b_{\mathrm{pol}}, c_{\mathrm{pol}}) = (a, b, c)$ in the polynomial target. Put

$$\mathcal{P}_L = (0,\infty)_{a_{\mathrm{lat}}} \times (0,\infty)_g, \quad \mathcal{X}_L = \mathcal{P}_L \times \mathbb{C}^3_{x,y,w}, \quad \mathcal{Y}_L = \mathcal{P}_L \times \mathbb{C}^3_{a_{\mathrm{pol}},b_{\mathrm{pol}},c_{\mathrm{pol}}}.$$

The exact morphism is

$$\widehat{F}_L : \mathcal{X}_L \longrightarrow \mathcal{Y}_L, \qquad (a_{\mathrm{lat}}, g, q) \longmapsto (a_{\mathrm{lat}}, g, F(q)).$$

Its parameter projection is unchanged, its polynomial projection is the full F in (1), and its derivative in this coordinate order is $\mathrm{diag}(I_2, DF(q))$. These assertions follow by differentiating each displayed component. The arithmetic projection $\widehat{\sigma}(a_{\mathrm{lat}}, g, q) = F_1(q)/2$ therefore satisfies $\widehat{\sigma} = \frac{1}{2}\,\mathrm{pr}_{a_{\mathrm{pol}}}\widehat{F}_L$. In particular the lattice spacing is a retained parameter of the map, while the polynomial coordinate is evaluated by its explicit component; neither coordinate is set equal to a constant or erased.

The physical parameter chart and its globally defined positive-real inverse are

$$\chi : \mathcal{P}_L \longrightarrow (0,\infty)_\kappa \times (0,\infty)_\xi, \qquad \chi(a_{\mathrm{lat}}, g) = (2g^2/a_{\mathrm{lat}}, 1/(4g^4)),$$
$$\chi^{-1}(\kappa,\xi) = (1/(\kappa\sqrt{\xi}), (4\xi)^{-1/4}), \qquad b_{\mathrm{mag}} = \kappa\xi.$$

All roots here are the positive real roots. Indeed $g^2 = 1/(2\sqrt{\xi})$, so $2g^2/a_{\mathrm{lat}} = \kappa$ and $1/(2g^2a_{\mathrm{lat}}) = \kappa\xi$; substitution in the other direction returns the original a_{lat}, g. Thus this is an invertible parameter chart retaining both physical parameters. In particular a fixed-κ coupling path has $a_{\mathrm{lat}}(\xi) = 1/(\kappa\sqrt{\xi})$ and $g(\xi) = (4\xi)^{-1/4}$; it does not silently hold the lattice spacing fixed. The source's finite-box coefficient has the displayed dependence on these parameters and makes no continuum-limit assertion.

On the original generators the Pauli trace identity gives $-\operatorname{tr}(T_iT_j)/2 = \delta_{ij}/4$. Consequently the orthonormal frame for that metric is exactly $(2T_1, 2T_2, 2T_3)$ and its left derivatives are

$(2X_1, 2X_2, 2X_3)$. Their sum of squares is $4\sum_i X_i^2$, proving $-\Delta^c = 4E$ and the original electric coefficient $\kappa/4$. This records the metric map with its factor, as well as the parameter map. The electric operator is $E_e = -\sum_{a'=1}^{3} X_{e,a'}^2$ for $T_{a'} = -i\sigma_{a'}/2$; the prime distinguishes the generator index from both meanings of a. The potential trace W_p and the original operator and form domains are specified and proved in Appendix A. Let ψ_ξ be the positive unit vacuum, $\mathcal{E}$ its energy,

$$A_\xi = (H - \mathcal{E})/\kappa, \qquad v_e = (E_e - \langle \psi_\xi, E_e \psi_\xi \rangle)\psi_\xi.$$

For two distinct links e, f sharing no elementary face, retain the original ordered incidence count

$$a_{ef} = \#\{(p,q) : e \in \partial p,\ f \in \partial q, |\partial p \cap \partial q| = 1\}.$$

This is a count, not a continuum limit or a change of physical scale. Appendix A proves, in the full original box,

$$\begin{aligned} \langle v_e, v_f \rangle &= \frac{127 a_{ef}}{219024}\xi^4 + O_L(\xi^6), \\ \langle v_e, (H-\mathcal{E}) v_f \rangle &= 0. \end{aligned} \tag{65}$$

It retains both shared-link spin channels, their exact norms, the disconnected subtraction, and the full vacuum and spectral-band Gram factors. The complete source proof is included, not replaced by these two consequences.

For the energy variable λ define the distribution

$$\mathfrak{d} = c_0\delta_3 + \beta\delta_3' + c_1\delta_{9/2} + c_2\delta_{13/2}, \qquad \begin{aligned} c_0 &= -\frac{59}{275184}, \quad \beta = \frac{1}{336}, \\ c_1 &= \frac{1}{1296}, \quad c_2 = \frac{3}{132496}. \end{aligned} \tag{66}$$

The convention is $\delta_3'(h) = -h'(3)$, so every sign is fixed. The proved source functional is, for bounded continuous h differentiable at 3,

$$\lim_{\xi \to 0} \xi^{-4} \langle v_e, h(A_\xi) v_f \rangle = a_{ef}\mathfrak{d}(h). \tag{67}$$

The variable λ is the dimensionless energy ω/κ; the physical measure is its pushforward under $\lambda \mapsto \kappa\lambda$. Here is its complete typed map, including the derivative distribution. For each original $\kappa > 0$, let

$$p_\kappa : \mathbb{R}_\lambda \longrightarrow \mathbb{R}_\omega, \quad p_\kappa(\lambda) = \kappa\lambda, \qquad p_\kappa^{-1}(\omega) = \omega/\kappa.$$

For any compactly supported distribution d define $(p_{\kappa *} d)(h) = d(h \circ p_\kappa)$ for smooth h. This gives an isomorphism between the four-dimensional spaces

$$\begin{aligned} \mathcal{V} &= \operatorname{span}_{\mathbb{C}}\{\delta_3, \delta_3', \delta_{9/2}, \delta_{13/2}\}, \\ \mathcal{V}_\kappa &= \operatorname{span}_{\mathbb{C}}\{\delta_{3\kappa}, \delta_{3\kappa}', \delta_{9\kappa/2}, \delta_{13\kappa/2}\}. \end{aligned}$$

In these ordered bases the matrix and its inverse are

$$B_\kappa = \operatorname{diag}(1, \kappa, 1, 1), \qquad B_\kappa^{-1} = \operatorname{diag}(1, \kappa^{-1}, 1, 1).$$

To prove every column, evaluate a delta on $h(\kappa\lambda)$, and use $\delta_3'(h(\kappa\lambda)) = -\kappa h'(3\kappa)$ for the second column. Thus the physical fourth coefficient is exactly

$$p_{\kappa *}\mathfrak{d} = c_0\delta_{3\kappa} + \frac{\kappa}{336}\delta_{3\kappa}' + c_1\delta_{9\kappa/2} + c_2\delta_{13\kappa/2}. \tag{68}$$

The derivative coefficient retains its physical factor κ.

For the actual self-adjoint excitation operator $A_\xi = (H - \mathcal{E})/\kappa$, its projection-valued measure E_{A_ξ} obeys, for every Borel set $B \subset \mathbb{R}$,

$$E_{H-\mathcal{E}}(B) = E_{A_\xi}(p_\kappa^{-1}B), \qquad \Sigma^{\omega}_{ef}(B) = \langle v_e, E_{A_\xi}(p_\kappa^{-1}B)v_f\rangle = (p_{\kappa *}\Sigma^{\lambda}_{ef})(B).$$

Indeed the spectral integral for κA_ξ inserts $\kappa\lambda$, so its indicator function is $\mathbf{1}_B(\kappa\lambda)$, proving these identities. Integration of bounded Borel tests first for simple functions and then by dominated convergence proves the entire pushforward identity on the actual measures. For absolute Hamiltonian energy the additional exact affine coordinate is $\varepsilon = \mathcal{E} + \kappa\lambda$ and the same proof uses the preimage of B under this affine map. On the unchanged operator domain the inverse reconstruction is $H = \kappa A_\xi + \mathcal{E}I$, where $\mathcal{E} = 2bM + \kappa e(\xi)$; the scalar $2bM$ is recovered as well.

Multiplication by ω on $\mathcal{V}_\kappa$ has matrix

$$Q_\omega = \begin{pmatrix} 3\kappa & -1 & 0 & 0 \\ 0 & 3\kappa & 0 & 0 \\ 0 & 0 & 9\kappa/2 & 0 \\ 0 & 0 & 0 & 13\kappa/2 \end{pmatrix}.$$

Testing $\omega\delta'_{3\kappa}$ gives $-h(3\kappa) - 3\kappa h'(3\kappa)$ and proves its off-diagonal entry. With Q in (69), direct multiplication gives $Q_\omega B_\kappa = \kappa B_\kappa Q$. The zeroth moment is therefore unchanged, and the first moment is multiplied by its original κ, including the derivative term. The source proves that the original matrix-valued spectral measure is positive semidefinite on every Borel set. An off-diagonal entry retains its signs, and its fourth Taylor coefficient retains δ'_3.

10.1 A four-dimensional spectral distribution and its heat map

Let $\mathcal{V}$ be the ordered span $(\delta_3, \delta'_3, \delta_{9/2}, \delta_{13/2})$. Multiplication of a distribution by the unchanged energy coordinate acts in this basis by

$$Q = \begin{pmatrix} 3 & -1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 9/2 & 0 \\ 0 & 0 & 0 & 13/2 \end{pmatrix}. \tag{69}$$

Indeed $(\lambda\delta'_3)(h) = -3h'(3) - h(3)$, while the other three basis vectors are multiplied by their displayed energies. This proves the entire matrix, including its off-diagonal entry. The exact exponential of $tQ^2/4$ has its first block

$$e^{9t/4}\begin{pmatrix} 1 & -3t/2 \\ 0 & 1 \end{pmatrix}$$

and remaining entries $e^{81t/16}, e^{169t/16}$. To verify it, Q^2 has first block $9I + \left(\begin{smallmatrix} 0 & -6 \\ 0 & 0 \end{smallmatrix}\right)$; the second matrix has square zero, so its exponential series stops after its linear term. The two other entries are scalar exponentials.

Define the exact Fourier morphism, with its displayed domain and codomain,

$$\mathcal{F}_{\mathcal{V}} : \mathcal{V} \longrightarrow \operatorname{span}_{\mathbb{C}}\{e^{3is}, -ise^{3is}, e^{9is/2}, e^{13is/2}\}, \qquad \mathcal{F}_{\mathcal{V}}(d)(s) = d_\lambda(e^{is\lambda}).$$

Its basis images are $e^{3is}, -ise^{3is}, e^{9is/2}, e^{13is/2}$. It is injective. If a linear combination vanishes, divide it by e^{3is} and put $s = iy$, $y \to +\infty$. The remaining expression is a polynomial of degree at most one in y, plus constant multiples of $e^{-3y/2}$ and $e^{-7y/2}$. Its polynomial coefficients must vanish; multiplying the remaining identity by $e^{3y/2}$ then taking the same limit makes the third coefficient zero, and then the fourth is zero. Differentiation of the four displayed functions gives

$$\partial_s \mathcal{F}_{\mathcal{V}}(d) = i\mathcal{F}_{\mathcal{V}}(Qd), \qquad -\tfrac{1}{4}\partial_s^2 \mathcal{F}_{\mathcal{V}}(d) = \mathcal{F}_{\mathcal{V}}(Q^2 d/4).$$

Thus the source spectral coefficient has an exact heat intertwiner, not merely a shared dimension or notation.

Applying this map to $a_{ef} e^{tQ^2/4}\mathfrak{d}$ yields

$$J_t(s) = a_{ef}\Big[\Big(-\frac{59}{275184} - \frac{t}{224} - \frac{is}{336}\Big) e^{9t/4+3is} + \frac{1}{1296} e^{81t/16+9is/2} + \frac{3}{132496} e^{169t/16+13is/2}\Big]. \tag{70}$$

It is entire in both variables and satisfies $\partial_t J = -\partial_s^2 J/4$. Equivalently this follows directly from $J_t = a_{ef}\mathfrak{d}(e^{t\lambda^2/4+is\lambda})$. For the actual original spectral measure, this test is bounded on the real energy axis whenever $\operatorname{Re} t < 0$, for every complex s, and also when $t = 0, s \in \mathbb{R}$. Completing the real square in $\operatorname{Re}(t)\lambda^2/4 - \operatorname{Im}(s)\lambda$ proves that bound. Consequently (67) gives

$$\lim_{\xi \to 0} \xi^{-4} \langle v_e, e^{tA_\xi^2/4+isA_\xi} v_f\rangle = J_t(s)$$

on precisely these guaranteed domains. The finite coefficient (70) has the stated entire continuation; this does not assume Gaussian exponential moments of the full unbounded interacting spectral measure at positive real t.

Physical correlation time $\theta \geq 0$ and Newman time t have the following exact typed dynamical morphisms. The following coordinate maps prove the relationship between the two time actions. Put

$$\mathcal{W} = \operatorname{span}_{\mathbb{C}}\{e^{3is}, -ise^{3is}, e^{9is/2}, e^{13is/2}\} \subset \mathcal{O}(\mathbb{C}_s).$$

The previously proved Fourier morphism is the isomorphism $\mathcal{F}_{\mathcal{V}} : \mathcal{V} \to \mathcal{W}$. For physical time $\theta \geq 0$, define

$$P_{\kappa,\theta} = e^{-\kappa\theta Q} : \mathcal{V} \to \mathcal{V}, \quad S_{\kappa,\theta} : \mathcal{W} \to \mathcal{W}, \quad (S_{\kappa,\theta} f)(s) = f(s + i\kappa\theta).$$

The first block of $P_{\kappa,\theta}$ is

$$e^{-3\kappa\theta}\begin{pmatrix} 1 & \kappa\theta \\ 0 & 1 \end{pmatrix},$$

and the other diagonal entries are $e^{-9\kappa\theta/2}, e^{-13\kappa\theta/2}$. The nilpotent off-diagonal matrix squares to zero, so expanding its exponential proves the entire block, including $\kappa\theta$. For every $d \in \mathcal{V}$,

$$\mathcal{F}_{\mathcal{V}} P_{\kappa,\theta} d(s) = d(e^{-\kappa\theta\lambda} e^{is\lambda}) = \mathcal{F}_{\mathcal{V}} d(s + i\kappa\theta) = S_{\kappa,\theta}\mathcal{F}_{\mathcal{V}} d(s).$$

This proves the intertwiner on every vector, not only on the selected coefficient. Its infinitesimal identity is $\mathcal{F}_{\mathcal{V}}(-\kappa Q) = i\kappa\partial_s \mathcal{F}_{\mathcal{V}}$.

For Newman time $t \in \mathbb{C}$, the other action is $N_t = e^{tQ^2/4} : \mathcal{V} \to \mathcal{V}$, which intertwines with $e^{-t\partial_s^2/4} : \mathcal{W} \to \mathcal{W}$ by the proved matrix derivative identity. Both actions commute, since Q commutes with Q^2. This proves the full two-parameter correspondence, with the original physical and Newman generators retained. Finally evaluation at $s = 0$ after the coordinate map

$$\iota_\kappa : [0,\infty)_\theta \longrightarrow \mathbb{C}_t \times \mathbb{C}_s, \qquad \iota_\kappa(\theta) = (0, i\kappa\theta)$$

gives the displayed correlation coefficient $\iota_\kappa^* J(\theta) = J_0(i\kappa\theta)$. Its derivative is $i\kappa J_s(0, i\kappa\theta)$ by the chain rule.

The unchanged dimensionful correlation coefficient is

$$a_{ef}\mathfrak{d}(e^{-\kappa\theta\lambda}) = a_{ef}\left[\left(c_0 + \frac{\kappa\theta}{336}\right)e^{-3\kappa\theta} + c_1 e^{-9\kappa\theta/2} + c_2 e^{-13\kappa\theta/2}\right].$$

It equals $J_0(i\kappa\theta)$ by substitution, giving the exact time dictionary rather than identifying the two evolutions. Its value at zero is $127a_{ef}/219024$, and its first physical-time derivative is zero because $3c_0 - \beta + (9/2)c_1 + (13/2)c_2 = 0$. These are exactly (65), with the physical factor κ retained in the derivative.

11 The original fourth coupling coefficient as an arithmetic heat automorphism

This section uses the complete finite-box Wilson calculation retained in the complete source proof in Appendix A. Its parameter is $\xi = b/\kappa = 1/(4g^4)$, whereas the completed Riemann function is $\xi(\rho)$ and the unchanged arithmetic heat function is $Z_t(s)$. These notations retain their original meanings. We construct the actual operator-valued arithmetic observable, prove its fourth coupling coefficient, and give the inverse of the resulting arithmetic map.

11.1 Original source operators and the complete spectral functional

Fix the original open box with vertices $\{-L, \ldots, L\}^3$, integer $L \geq 2$, its $N = 3(2L)(2L+1)^2$ positively oriented physical edges, and its $M = 3(2L)^2(2L+1)$ contained elementary faces. On the gauge-invariant subspace of product Haar probability $L^2(SU(2)^N)$, retain

$$T_a = -i\sigma_a/2, \qquad E_e = -\sum_{a=1}^{3} X_{e,a}^2, \qquad H_0 = \sum_e E_e, \qquad S = \sum_p W_p,$$
$$H = \kappa H_0 + b\sum_p (2 - W_p) = \kappa(H_0 - \xi S) + 2bMI,$$
$$\kappa = \frac{2g^2}{a}, \quad b = \frac{1}{2g^2 a}, \quad \xi = \frac{1}{4g^4}, \quad a, g > 0. \tag{71}$$

Here $X_{e,a}$ is the original left derivative generated by T_a, and W_p is the original oriented fundamental Wilson trace. The metric identity is $-\Delta_e^c = 4E_e$, so the electric coefficient in that metric is $\kappa/4$. Neither this factor nor the scalar $2bM$ is changed. Let ψ_ξ be the positive unit ground vector and write

$$\mathcal{E}(\xi) = 2bM + \kappa e(\xi), \qquad A_\xi = \frac{H - \mathcal{E}(\xi)}{\kappa} = H_0 - \xi S - e(\xi)I,$$
$$\gamma_e(\xi) = \langle \psi_\xi, E_e\psi_\xi\rangle, \qquad v_e(\xi) = (E_e - \gamma_e(\xi))\psi_\xi. \tag{72}$$

The inner product is conjugate-linear in its first entry. For real ξ near zero, A_ξ is self-adjoint on the physical H^2 domain, nonnegative, and has the original smooth ground vector in its kernel. The formula for A_ξ is the exact physical excitation energy divided by its original κ; the scalar in H cancels its identical scalar in $\mathcal{E}$ in this displayed difference.

For distinct edges e, f sharing no face, let a_{ef} count the ordered-by-containment pairs of adjacent faces containing e and f, respectively. The complete spectral coefficient proved in the source calculation is, for bounded continuous complex $h : \mathbb{R} \to \mathbb{C}$ differentiable at 3,

$$\lim_{\xi\to 0} \xi^{-4}\langle v_e(\xi), h(A_\xi)v_f(\xi)\rangle = a_{ef}\mathcal{D}(h), \qquad \mathcal{D}(h) = c_0 h(3) - \frac{h'(3)}{336} + c_1 h(9/2) + c_2 h(13/2), \tag{73}$$

where every coefficient is retained:

$$c_0 = -\frac{59}{275184}, \qquad c_1 = \frac{1}{1296}, \qquad c_2 = \frac{3}{132496}. \tag{74}$$

Equivalently, $\mathcal{D} = c_0\delta_3 + \delta_3'/336 + c_1\delta_{9/2} + c_2\delta_{13/2}$, with $\delta_3'(h) = -h'(3)$. Thus the derivative term is part of the original spectral coefficient. The source proof obtains it from the actual first interacting spectral band and its full Gram isometry, together with both exterior spin channels; it is not replaced here by an atomic measure.

11.2 Bounded arithmetic functional calculus at every complex heat time

Lemma 11.1 (The actual arithmetic test and its operator integral)**.** *For every complex t, s, the function $h_{t,s}(q) = Z_t(s+q)$ is bounded and continuously differentiable on the whole real q-axis. On real ξ near zero, its bounded spectral calculus is exactly*

$$Z_t(s + A_\xi) = 4\int_{\mathbb{R}} k(v)e^{tv^2/4+isv}e^{ivA_\xi}\,dv. \tag{75}$$

The integral is the strong integral of bounded operators, defined on each Hilbert-space vector; its operator norm is bounded by

$$4\int_{\mathbb{R}} |k(v)|e^{(\operatorname{Re} t)v^2/4+|\operatorname{Im} s||v|}\,dv. \tag{76}$$

It is jointly entire in t, s, in the operator norm, and obeys $\partial_t Z_t(s + A_\xi) = -\frac{1}{4}\partial_s^2 Z_t(s + A_\xi)$.

Proof. Use the unchanged arithmetic identity (11):

$$h_{t,s}(q) = 4\int_{\mathbb{R}} k(v)e^{tv^2/4+isv}e^{iqv}\,dv.$$

For real q, the last factor has modulus one. The previously proved two-ended superexponential bound on k dominates the absolute integral in (76). Its product with $|v|^j$ is also integrable for every nonnegative integer j. This proves boundedness of every real q-derivative, uniformly in q, by differentiating the integral. On compact complex (t, s) sets the same dominator works after every specified t, s, q derivative.

For any Hilbert-space vector u, the function $v \mapsto e^{ivA_\xi}u$ is strongly continuous and has constant norm. Thus the right side of (75) is a Bochner integral on that vector and satisfies the stated norm bound. For any two vectors, the corresponding spectral measure has finite total variation, bounded by the product of their norms. Fubini therefore interchanges this measure integral with the absolute v integral and gives its matrix element as that of $h_{t,s}(A_\xi)$. This proves the operator identity, rather than postulating an unbounded entire functional calculus.

Expand the scalar exponential about any fixed (t, s). On every smaller compact polydisc, its absolutely summed Taylor series is dominated by the same integral with larger fixed quadratic and linear weights. The integral norm inequality proves convergence of the resulting operator Taylor series in norm. This establishes joint entire dependence. Differentiation in t inserts $v^2/4$, whereas two s derivatives insert $-v^2$, which proves the heat identity with the original sign and factor. ☐

Theorem 11.2 (The actual arithmetic fourth coefficient)**.** *Define the entire arithmetic function*

$$\mathcal{K}_t(s) = c_0 Z_t(s+3) - \frac{Z_t'(s+3)}{336} + c_1 Z_t(s+9/2) + c_2 Z_t(s+13/2). \tag{77}$$

For every complex t, s, the actual original observable

$$G_\xi(t,s) = \langle v_e(\xi), Z_t(s + A_\xi)v_f(\xi)\rangle \tag{78}$$

satisfies

$$\lim_{\xi\to 0} \xi^{-4}G_\xi(t,s) = a_{ef}\mathcal{K}_t(s). \tag{79}$$

The limit retains the actual Riemann initial datum:

$$\mathcal{K}_0(s) = c_0\Xi(s+3) - \frac{\Xi'(s+3)}{336} + c_1\Xi(s+9/2) + c_2\Xi(s+13/2), \tag{80}$$

$$\partial_t\mathcal{K}_t = -\tfrac{1}{4}\partial_s^2\mathcal{K}_t, \qquad \partial_t G_\xi = -\tfrac{1}{4}\partial_s^2 G_\xi. \tag{81}$$

Proof. Lemma 11.1 verifies every test-function hypothesis of (73) for the actual $h(q) = Z_t(s+q)$. Its values and derivative at the four specified source locations give (77) and (79) literally. Every translated term and its retained derivative is jointly entire. Translation by each of the original constants and ∂_s commute with ∂_s^2, so (12) proves the first identity in (81); the operator identity from Lemma 11.1 proves the second one. Finally $Z_0 = \Xi$ gives (80) with no change of arithmetic initial function. □

11.3 Actual coupling analyticity and the sixth-order remainder

Proposition 11.3 (Uniform remainder on compact arithmetic parameter sets)**.** *For each original finite box and each compact $\mathcal{Q} \subset \mathbb{C}^2_{t,s}$, there are numbers $r > 0$ and $C_{\mathcal{Q},L,e,f} < \infty$ such that, for real $|\xi| \le r/2$,*

$$\sup_{(t,s)\in\mathcal{Q}} |G_\xi(t,s) - a_{ef}\xi^4\mathcal{K}_t(s)| \le C_{\mathcal{Q},L,e,f}|\xi|^6. \tag{82}$$

The construction gives a jointly holomorphic extension in a complex coupling disc and all complex t, s. The radius and constant retain their dependence on the original finite box.

Proof. We first specify the analytic continuation of the vectors and bras. The full free ground eigenvalue is simple and the next full-space eigenvalue is $3/4$; the original perturbation has $\|S\| = 2M$. The resolvent on $|z| = 3/8$ has norm at most $8/3$, so the source Neumann construction supplies the analytic rank-one vacuum projection in a coupling disc contained in $|\xi| < 3/(16M)$. Write $\chi(\xi) = P(\xi)1$. Complex conjugation C on functions satisfies $CP(\xi)C = P(\bar\xi)$, since both H_0 and S have real coefficients. Thus $C\chi(\xi) = \chi(\bar\xi)$.

The scalar $\mathfrak{n}(\xi) = \langle\chi(\bar\xi), \chi(\xi)\rangle$ is holomorphic and equals one at zero. On a smaller symmetric disc, it is nonzero and has the holomorphic square root with value one at zero. Set $\psi(\xi) = \chi(\xi)/\sqrt{\mathfrak{n}(\xi)}$. For small real ξ this is the original positive unit vacuum: it is the real normalized eigenvector with positive overlap with the constant vector. For complex ξ, this formula fixes its analytic extension; no assertion that its Hilbert norm remains one off the real axis is needed. Its eigenvalue $e(\xi)$ is holomorphic and real on the real axis.

Analyticity of ψ holds in every fixed Sobolev norm. Indeed the exact equation

$$\psi(\xi) = (I + H_0)^{-1}[(1 + e(\xi) + \xi S)\psi(\xi)]$$

increases the Sobolev index by two, since multiplication by the original smooth S preserves that index and the elliptic inverse increases it by two. Starting with the Hilbert-space analytic projection proves the assertion inductively. Consequently

$$\gamma_e(\xi) = \langle\psi(\bar\xi), E_e\psi(\xi)\rangle, \qquad v_e(\xi) = (E_e - \gamma_e(\xi))\psi(\xi)$$

are holomorphic in every required Sobolev space. Their restrictions to real ξ are exactly (72). The conjugate-linear-first functional $u \mapsto \langle v_e(\bar{\xi}), u\rangle$ is holomorphic in ξ; it is the precise continuation of the original bra.

On the same physical domain put $A_\xi = H_0 - \xi S - e(\xi)I$, now for complex ξ. The bounded perturbation of the free unitary group gives e^{ivA_ξ} for every real v. Its Dyson series is obtained by inserting $-\xi S$ between the unchanged free groups. The term with n insertions has norm at most $(2M|\xi||v|)^n/n!$; the ordered integrals are strong integrals on each vector. The resulting bounded-operator series converges in norm, uniformly on compact coupling sets. The scalar shift commutes with every term and contributes $e^{-ive(\xi)}$. Hence

$$\|e^{ivA_\xi}\| \le \exp\{|v|(2M|\xi| + |\operatorname{Im} e(\xi)|)\}. \tag{83}$$

This proves norm-holomorphic dependence on ξ for fixed real v, with the entire original bounded potential retained.

Define the continuation of (78) by

$$\widetilde{G}(\xi, t, s) = 4\int_{\mathbb{R}} k(v) e^{tv^2/4+isv} \langle v_e(\bar{\xi}), e^{ivA_\xi} v_f(\xi)\rangle \, dv. \tag{84}$$

Choose a closed coupling disc $|\xi| \le r$ strictly inside the disc just constructed. Put

$$B_r = \sup_{|\xi|\le r} \|v_e(\bar{\xi})\| \, \|v_f(\xi)\|, \qquad E_r = \sup_{|\xi|\le r} |\operatorname{Im} e(\xi)|.$$

Both numbers are finite by continuity on that closed disc. On a compact arithmetic parameter set $\mathcal{Q}$, set $T_{\mathcal{Q}} = \max_{\mathcal{Q}} \operatorname{Re} t$ and $S_{\mathcal{Q}} = \max_{\mathcal{Q}} |\operatorname{Im} s|$. The absolute integral in (84) is uniformly bounded by the explicit finite number

$$M_{r,\mathcal{Q}} = 4B_r \int_{\mathbb{R}} |k(v)| e^{T_{\mathcal{Q}} v^2/4 + (S_{\mathcal{Q}} + 2Mr + E_r)|v|} \, dv. \tag{85}$$

The superexponential arithmetic kernel absorbs all these retained weights. On larger compact arithmetic sets it also absorbs the weights introduced by each parameter derivative. The locally uniformly convergent Dyson and scalar exponential series, or their Cauchy integral formulas dominated by this same bound, therefore prove joint holomorphy of $\widetilde{G}$. For real coupling it equals the original observable by Lemma 11.1.

The source central-link unitary is

$$(\mathcal{Z}F)(U) = F\big(((-1)^{\sum_{j<i} n_j} U_{(n,i)})_{(n,i)}\big).$$

It preserves Haar measure, commutes with every original E_e, and changes the sign of every Wilson trace. Consequently $\mathcal{Z}(H_0 - \xi S)\mathcal{Z} = H_0 + \xi S$. Uniqueness and positivity of the real vacuum imply $\psi_{-\xi} = \mathcal{Z}\psi_\xi$, $e(-\xi) = e(\xi)$, $v_e(-\xi) = \mathcal{Z} v_e(\xi)$, and $A_{-\xi} = \mathcal{Z} A_\xi \mathcal{Z}$. Thus $G_{-\xi}(t,s) = G_\xi(t,s)$ for real coupling. The identity theorem extends this equality to the complex disc. No time or arithmetic coordinate changes in this symmetry.

Write the coupling Taylor series as $\widetilde{G} = \sum_{n\ge 0} g_n(t,s)\xi^n$. The actual limit (79) on the real axis forces $g_0 = g_1 = g_2 = g_3 = 0$ and $g_4 = a_{ef}\mathcal{K}_t(s)$: otherwise division by ξ^4 could not have the proved finite limit. Evenness makes every odd coefficient vanish. The Cauchy bound on $|\xi| = r$ gives $\sup_{\mathcal{Q}} |g_n| \le M_{r,\mathcal{Q}} r^{-n}$. For $|\xi| \le r/2$, summing the remaining series yields

$$\sup_{\mathcal{Q}} |\widetilde{G} - a_{ef}\xi^4\mathcal{K}| \le 2M_{r,\mathcal{Q}} r^{-6} |\xi|^6.$$

This proves (82), for example with the specified constant $2M_{r,\mathcal{Q}} r^{-6}$. All box dependence in M, N, r, B_r, E_r remains present. The estimate does not exchange a volume limit with a coupling derivative. □

11.4 An invertible arithmetic map on the full retained Fourier domain

Retain the exact spaces $\mathscr{E}, \mathscr{A}$ and transform T already defined above, including every seminorm

$$p_{A,B,m,j}(\phi) = \int_{\mathbb{R}} |\phi^{(j)}(v)|(1+|v|)^m e^{Av^2+B|v|}\, dv, \qquad T\phi(s) = 4\int_{\mathbb{R}} \phi(v)e^{isv}\, dv.$$

Define the source multiplier and arithmetic operator by

$$m(v) = c_0 e^{3iv} - \frac{iv}{336}e^{3iv} + c_1 e^{9iv/2} + c_2 e^{13iv/2}, \tag{86}$$

$$(\mathcal{T}_{\mathcal{D}} f)(s) = c_0 f(s+3) - \frac{f'(s+3)}{336} + c_1 f(s+9/2) + c_2 f(s+13/2). \tag{87}$$

Lemma 11.4 (A uniform exact lower bound for the original multiplier)**.** *For every real v,*

$$|m(v)| \geq \delta, \qquad \delta = \frac{575}{1752192} > 0. \tag{88}$$

In particular its real-axis reciprocal exists everywhere, with no branch choice or removed spectral point.

Proof. Factoring the nonzero phase gives the exact identity

$$e^{-3iv}m(v) = c_0 - \frac{iv}{336} + c_1 e^{3iv/2} + c_2 e^{7iv/2}.$$

For $|v| \leq 1/2$, the elementary inequality $\cos x \geq 1 - x^2/2$ yields

$$\begin{aligned}\operatorname{Re}(e^{-3iv}m(v)) &= c_0 + c_1\cos(3v/2) + c_2\cos(7v/2)\\ &\geq c_0 + \frac{23}{32}c_1 - \frac{17}{32}c_2 = \frac{575}{1752192}.\end{aligned}$$

For completeness, this cosine bound follows directly by integration: for $x \geq 0$, $1 - \cos x = \int_0^x \sin u\, du \leq \int_0^x u\, du$, using $\sin u \leq u$; the latter follows from $u - \sin u = \int_0^u (1-\cos w)\, dw \geq 0$. Evenness gives the negative case. Since the real part is positive, its displayed lower bound also bounds the modulus.

For $|v| \geq 1/2$, the triangle inequality with the original linear term gives

$$\begin{aligned}|m(v)| &\geq \frac{|v|}{336} - (|c_0| + c_1 + c_2)\\ &\geq \frac{1}{672} - \frac{10825}{10732176} = \frac{10291}{21464352} > \frac{575}{1752192}.\end{aligned}$$

All coefficients in these estimates are the original rational coefficients. The two intervals cover the whole real axis and prove (88). □

Theorem 11.5 (Exact arithmetic automorphism and heat conjugacy)**.** *Multiplication by m is a continuous automorphism of $\mathscr{E}$. Its inverse is literal pointwise multiplication by $1/m$ on the unchanged real Fourier axis. The operator $\mathcal{T}_{\mathcal{D}}$ is a continuous automorphism of $\mathscr{A}$, with exact inverse*

$$\mathcal{T}_{\mathcal{D}}^{-1} f = T\left(\frac{T^{-1}f}{m}\right). \tag{89}$$

For every complex t,

$$\mathcal{T}_{\mathcal{D}} = TM_mT^{-1}, \qquad \mathcal{T}_{\mathcal{D}}U_t = U_t\mathcal{T}_{\mathcal{D}}, \qquad \mathcal{T}_{\mathcal{D}}^{-1}U_t = U_t\mathcal{T}_{\mathcal{D}}^{-1}, \tag{90}$$

$$\mathcal{K}_t = \mathcal{T}_{\mathcal{D}}Z_t, \qquad Z_t = \mathcal{T}_{\mathcal{D}}^{-1}\mathcal{K}_t. \tag{91}$$

Thus the actual arithmetic function is exactly recoverable from the propagated source coefficient, on this fully specified domain.

Proof. Every derivative of m has at most linear growth on the real axis. More explicitly, for $j \geq 0$ it satisfies $|m^{(j)}(v)| \leq C_j(1+|v|)$, where one may take

$$C_j = |c_0|3^j + c_1(9/2)^j + c_2(13/2)^j + \frac{3^j + \mathbf{1}_{j\geq 1} j3^{j-1}}{336}.$$

The product rule gives the seminorm estimate

$$p_{A,B,m,j}(m\phi) \leq \sum_{\ell=0}^{j} \binom{j}{\ell} C_\ell p_{A,B,m+1,j-\ell}(\phi).$$

This proves continuity of M_m on the original $\mathscr{E}$.

Let $r(v) = 1/m(v)$, which is smooth by Lemma 11.4. The exact product identity $mr = 1$ and its derivatives give

$$r^{(j)} = -\frac{1}{m}\sum_{\ell=1}^{j}\binom{j}{\ell} m^{(\ell)} r^{(j-\ell)} \quad (j \geq 1), \qquad |r| \leq \delta^{-1}.$$

Set $Q_0 = \delta^{-1}$ and recursively define the finite positive constants

$$Q_j = \delta^{-1}\sum_{\ell=1}^{j}\binom{j}{\ell} C_\ell Q_{j-\ell}.$$

Induction proves $|r^{(j)}(v)| \leq Q_j(1+|v|)^j$: the term with index $\ell \geq 1$ has exponent at most $1 + j - \ell \leq j$, and the denominator is bounded by δ^{-1}. Therefore

$$p_{A,B,m,j}(r\phi) \leq \sum_{\ell=0}^{j}\binom{j}{\ell} Q_\ell p_{A,B,m+\ell,j-\ell}(\phi).$$

This proves that reciprocal multiplication is continuous on exactly the same $\mathscr{E}$. The two pointwise compositions are the identity, proving the Fourier automorphism and its inverse.

Applying the retained transform to each term of $m\phi$ gives the three translations and the derivative in (87); the factor iv is exactly the Fourier multiplier of ∂_s. The injectivity of T already proved above makes its inverse on $\mathscr{A}$ unambiguous. This proves the first identity in (90) and (89). Multiplication by m and by $1/m$ each commutes pointwise with multiplication by $e^{tv^2/4}$. All three maps preserve $\mathscr{E}$, so applying T proves the two full-domain heat commutation identities for every complex t. Finally (77) is exactly $\mathcal{T}_{\mathcal{D}} Z_t$, and applying the proved inverse gives (91). $\square$

The source factor acts by the typed scalar map $M_{a_{ef}} : \mathscr{A} \to \mathscr{A},\ f \mapsto a_{ef} f$. For $a_{ef} > 0$ its inverse is $g \mapsto g/a_{ef}$, by direct composition; for $a_{ef} = 0$ its kernel is all of $\mathscr{A}$ and its image is $\{0\}$. This retains the complete pair observation map $M_{a_{ef}}\mathcal{T}_{\mathcal{D}}$, including its zero case. When $a_{ef} > 0$, its division also recovers the unscaled $\mathcal{K}$ from the coefficient in (79). The original boxes contain such pairs. In particular, for $e = ((0,0,0),1)$ and $f = ((0,2,0),1)$, there is exactly one common edge-adjacency neighbor, $((0,1,0),1)$. To verify uniqueness directly, a direction-1 neighbor of e is its parallel translate by one step in direction 2 or 3; the corresponding lists for e and f meet only at the displayed midpoint edge. A direction-2 neighbor has its direction-2 base coordinate in $\{-1,0\}$ for e, and in $\{1,2\}$ for f, so these lists are disjoint. A direction-3 neighbor has direction-2 coordinate zero for e and two for f, also disjoint. All faces in the unique midpoint channel are contained in every original box with $L \geq 2$. The exact source channel bijection gives $a_{ef} = 1$, including these open boundaries.

Equations (84) and (91) provide the complete source-to-arithmetic heat correspondence. They assert an invertible map of the specified arithmetic function space. The coordinate-level map on every pointwise zero jet and analytic unit is proved next. The exact morphism on zero data uses the transported jet maps below. Fix $z \in \mathbb{C}$ and an integer $n \geq 1$. Define continuous linear maps

$$\begin{aligned} \mathcal{J}_{z,n} &: \mathscr{A} \longrightarrow \mathbb{C}^n, & \mathcal{J}_{z,n} f &= (f(z), f'(z), \dots, f^{(n-1)}(z)), \\ \mathcal{L}_{z,n} &: \mathscr{A} \longrightarrow \mathbb{C}^n, & \mathcal{L}_{z,n} g &= \mathcal{J}_{z,n} \mathcal{T}_{\mathcal{D}}^{-1} g. \end{aligned}$$

For $g = T\phi$ their individual coordinate functionals are exactly

$$(\mathcal{L}_{z,n} g)_j = 4 \int_{\mathbb{R}} (iv)^j \frac{\phi(v)}{m(v)} e^{izv}\, dv, \qquad j = 0, \dots, n-1. \tag{92}$$

This follows by differentiating the proved inverse integral; it retains the factor 4, the original coordinate z and every derivative factor $(iv)^j$. Absolute values are bounded by

$$4\delta^{-1} \int_{\mathbb{R}} |\phi(v)||v|^j e^{|\operatorname{Im} z||v|}\, dv,$$

where $\delta = 575/1752192$. One defining seminorm with an integer linear weight at least $|\operatorname{Im} z|$ dominates this integral. The same estimate on compact sets justifies every displayed derivative and proves continuity. It also proves continuity of $\mathcal{J}_{z,n}$ by omitting the factor $1/m$.

Both maps are onto. To prove this without assuming a jet-extension theorem on $\mathscr{A}$, set $u_z(s) = \Xi(s-z)$. This lies in $\mathscr{A}$, since its kernel is $k(v)e^{-izv}$ and all differentiated weighted seminorms are bounded by finitely many seminorms of k with a larger linear weight. Each $(s-z)^j u_z(s)$ also lies in $\mathscr{A}$ by the already proved invariance under multiplication by s. Its first n derivatives at z form a triangular matrix with diagonal entries $j!\Xi(0) \neq 0$, $j = 0, \dots, n-1$. The nonzero value was proved by the positive kernel integral. Back substitution therefore constructs every specified jet exactly. This proves surjectivity of $\mathcal{J}_{z,n}$, and that of $\mathcal{L}_{z,n}$ follows by composition with the automorphism. The finite back-substitution formula is also a continuous linear right inverse from $\mathbb{C}^n$ into the displayed finite-dimensional span.

Put $\mathcal{M}_{z,n} = \ker \mathcal{J}_{z,n}$. The exact zero-jet correspondence is

$$\mathcal{T}_{\mathcal{D}} \mathcal{M}_{z,n} = \ker \mathcal{L}_{z,n}, \qquad \mathscr{A}/\mathcal{M}_{z,n} \longrightarrow \mathscr{A}/\ker \mathcal{L}_{z,n}, \quad [f] \longmapsto [\mathcal{T}_{\mathcal{D}} f].$$

The first equality follows in both directions by applying the inverse automorphism. It makes the quotient map well-defined and gives the inverse $[g] \mapsto [\mathcal{T}_{\mathcal{D}}^{-1} g]$; both are continuous by their quotient topologies. Both quotients have the displayed coordinates in $\mathbb{C}^n$, and the map is the identity in those coordinates because $\mathcal{L}_{z,n}\mathcal{T}_{\mathcal{D}} = \mathcal{J}_{z,n}$.

For the actual $\mathcal{K}_t = \mathcal{T}_{\mathcal{D}} Z_t$ this proves every jet identity $\mathcal{L}_{z,n}\mathcal{K}_t = \mathcal{J}_{z,n} Z_t$. Thus a zero of order exactly n of Z_t is expressed on the transformed function by vanishing of coordinates $0, \dots, n-1$ of $\mathcal{L}_{z,n+1}\mathcal{K}_t$ and nonvanishing of coordinate n. At every such zero the complete analytic unit is recovered as the convergent series

$$\frac{Z_t(s)}{(s-z)^n} = \sum_{j=0}^{\infty} \frac{(\mathcal{L}_{z,n+j+1}\mathcal{K}_t)_{n+j}}{(n+j)!} (s-z)^j.$$

It is the Taylor expansion of the recovered entire function, so the identity and convergence follow from its holomorphy. No derivative, unit, coordinate or multiplicity is discarded by this morphism.

12 Propagation through the original inverse fibres and spatial flow

Use the exact per-channel arithmetic heat function $\mathcal{K}_t(s)$ defined by (77), with all source constants in (66). Retain the incidence count by setting $G_t(s) = a_{ef}\mathcal{K}_t(s)$. Define

$$g_t(r) = G_t(-r^2/2), \qquad \Psi^{(4)}(t, q) = G_t(\sigma(q)), \quad \sigma(q) = F_1(q)/2.$$

The underlying typed maps and their exact relation are

$$\pi : \mathbb{C}_r \to \mathbb{C}_s, \quad \pi(r) = -r^2/2, \qquad \sigma : \mathbb{C}^3_q \to \mathbb{C}_s, \quad \sigma(q) = F_1(q)/2,$$

$$\Gamma : \mathbb{C}^*_r \to \mathbb{C}^3_q, \quad \Gamma(r) = (-1/(2r), 3r, 26r^2), \qquad \sigma \circ \Gamma = \pi|_{\mathbb{C}^*}.$$

The last equality follows by the full polynomial substitution $F(\Gamma(r)) = (-r^2, 0, 0)$ proved in (2). Their pullbacks are respectively $\pi^* : \mathcal{O}(\mathbb{C}_s) \to \mathcal{O}_{\text{even}}(\mathbb{C}_r)$ and $\sigma^* : \mathcal{O}(\mathbb{C}_s) \to \mathcal{O}(\mathbb{C}^3_q)$. The first is the isomorphism with inverse proved in Section 8.2. The second is injective with explicit left inverse η^*, where $\eta : \mathbb{C}_s \to \mathbb{C}^3_q$, $\eta(s) = (0, 0, 2s)$: indeed $F_1(0, 0, 2s) = 2s$, so $\sigma\eta = \text{id}$ and $\eta^*\sigma^* = \text{id}$. Composition of these polynomial or entire maps proves continuity for uniform convergence on compact sets. In particular $\Gamma^*\sigma^* = \pi^*$ on $\mathbb{C}^*$, with the entire extension on the right retained. These are maps on the full functions and all their derivatives, not only on zero sets. These are entire compositions of the actual transformed arithmetic function; they retain the incidence factor a_{ef}. Since the new transform commutes with the actual Newman heat evolution, the already proved coordinate chain rules apply to it verbatim and give

$$\begin{aligned}
\partial_t g_t &= -\frac{1}{4r^2}\partial_r^2 g_t + \frac{1}{4r^3}\partial_r g_t \qquad (r \neq 0),\\
\partial_t \Psi^{(4)} + \tfrac{1}{4}U^2\Psi^{(4)} &= 0,\\
\Delta_q \Psi^{(4)} &= N(q)G_{ss}(t, \sigma(q)) + L(q)G_s(t, \sigma(q)).
\end{aligned} \tag{93}$$

Every coefficient N, L, U is the original one proved in Section 9; no Euclidean diffusion coefficient has been replaced by the root-coordinate generator. For any retained time map $t = \theta(\tau)$, the full residual is explicitly

$$(\partial_\tau + U \cdot \nabla_q - \nu\Delta_q)G_{\theta(\tau)}(\sigma(q)) = \left(-\frac{\theta'}{4} - \nu N\right) G_{ss} + (1 - \nu L)G_s,$$

with the derivatives on the right evaluated at $(\theta(\tau), \sigma(q))$. The proof is the time chain rule and the three identities in (93), so the force-sized remainder is calculated rather than assumed absent.

The three spectral translations also have exact maps in the retained original root coordinates. For $\lambda \in \{3, 9/2, 13/2\}$ define the entire algebraic correspondence

$$\mathcal{C}_\lambda = \{(r, u) \in \mathbb{C}^2 : u^2 = r^2 - 2\lambda\}.$$

Its morphisms are the coordinate projections $p_\lambda : \mathcal{C}_\lambda \to \mathbb{C}_r$, $(r, u) \mapsto r$, and $q_\lambda : \mathcal{C}_\lambda \to \mathbb{C}_u$, $(r, u) \mapsto u$. Both are defined on the entire stated algebraic curve. With $\tau_\lambda : \mathbb{C}_s \to \mathbb{C}_s$, $s \mapsto s+\lambda$, the exact commuting identity is $\pi q_\lambda = \tau_\lambda \pi p_\lambda$, proved by its defining equation. Pullback of any entire function therefore gives $q^*_\lambda \pi^* = p^*_\lambda \pi^* \tau^*_\lambda$; the derivative channel is computed below in the same u coordinate. It satisfies $-u^2/2 = -r^2/2 + \lambda$. Consequently

$$Z_t(-r^2/2 + \lambda) = f_t(u), \qquad Z_s(t, -r^2/2 + 3) = -\frac{\partial_u f_t(u)}{u} \quad ((r, u) \in \mathcal{C}_3,\ u \neq 0),$$

where $f_t(u) = Z_t(-u^2/2)$ is the original function, already proved even and entire. The second formula follows from $\partial_u f_t = -uZ_s(t, -u^2/2)$; thus its quotient has a unique entire continuation at $u = 0$ equal to $-Z_s(t, 0)$. Neither formula depends on the choice of sign of u.

In these actual correspondences the full propagated function is

$$g_t(r) = a_{ef}\left[c_0 f_t(u_3) + \beta\frac{\partial_u f_t(u_3)}{u_3} + c_1 f_t(u_{9/2}) + c_2 f_t(u_{13/2})\right], \qquad u^2_\lambda = r^2 - 2\lambda.$$

Each potentially branched expression descends to the displayed entire function of r, by the proven evenness and removable quotient. The finite inverse point for a nonzero u_λ is exactly $\Gamma(u_\lambda) = (-1/(2u_\lambda), 3u_\lambda, 26u_\lambda^2)$; its target is

$$F(\Gamma(u_\lambda)) = (-r^2 + 2\lambda, 0, 0).$$

At $u_\lambda = 0$ the completed root point still has no finite inverse in the $x \neq 0$ chart, as its unchanged first coordinate shows. The additional finite point with $x = 0$ is retained by the earlier complete fibre calculation. The three new branch values are precisely $r^2 = 6, 9, 13$; they arise from the source's three specified energies.

All of this also retains the full original cubic coefficient family. For fixed b, c and $s(r) = (cr^3 - 2r^2 + br)/4$, set $g_{t;b,c}(r) = G_t(s(r))$. The exact same chain rule gives

$$\partial_t g_{t;b,c} = -\frac{4}{P_r^2}\partial_r^2 g_{t;b,c} + \frac{4P_{rr}}{P_r^3}\partial_r g_{t;b,c} \quad (P_r = 3cr^2 - 4r + b \neq 0).$$

Thus the source fourth coefficient is propagated into the coefficient family as well as its $b = c = 0$ slice. This is a completed map of the source observable and its actual heat dynamics. For the per-channel function, recovery of the original Ξ uses the full-domain inverse automorphism proved in Section 11. For any actual pair with $a_{ef} > 0$, the recovered input is $Z_t = \mathcal{T}_{\mathcal{D}}^{-1}(G_t/a_{ef})$; the explicit separated pair in the source proof has $a_{ef} \geq 1$. At $a_{ef} = 0$ this fourth coefficient is exactly zero and carries no pair observation. A zero of a finite linear combination is not silently substituted for a zero of its recovered arithmetic input.

13 The second Euclidean diffusion of the actual arithmetic observable

The first diffusion coefficient in the preceding calculation has a finite limit along the escaping inverse-fibre trajectory. We compute the next Euclidean diffusion without dropping any derivative of that coefficient. All spatial derivatives in this section refer to the original Cartesian coordinates (x, y, w), with their Euclidean metric. Newman time t remains distinct from the fluid trajectory time τ.

Keep

$$Q = 1 + xy, \qquad \sigma(x, y, w) = \tfrac{1}{2}\big[Q^3 w + y^2 Q(4 + 3xy)\big], \qquad \Psi_t(x, y, w) = Z(t, \sigma(x, y, w)).$$

Here Z is the actual entire function defined by (42); its initial datum is the specified completed zeta function. Its analyticity and all differentiated integral bounds were proved above. In particular, for each real t,

$$Z_s(t, 0) = Z_{sss}(t, 0) = 0, \qquad Z_{ss}(t, 0) = -32M_1(t) < 0.$$

The strict inequality retains the positive original moment $M_1(t) = \int_0^\infty u^2 e^{tu^2}\Phi(u)\,du$.

13.1 The full fourth-order chain rule

Write $g = \nabla\sigma$, $H = D^2\sigma$, $N = g \cdot g$, and $L = \Delta\sigma$, where $\Delta = \partial_x^2 + \partial_y^2 + \partial_w^2$. For any entire function h of one variable, ordinary differentiation gives

$$\Delta(h \circ \sigma) = Nh''(\sigma) + Lh'(\sigma), \tag{94}$$

$$\begin{aligned}\Delta^2(h \circ \sigma) = {} & N^2 h^{(4)}(\sigma) + (4g^\mathsf{T} Hg + 2NL)h^{(3)}(\sigma) \\ & + (2\|H\|_\mathrm{F}^2 + 4g \cdot \nabla L + L^2)h''(\sigma) + (\Delta L)h'(\sigma).\end{aligned} \tag{95}$$

Here $\|H\|_{\mathrm{F}}^2 = \sum_{i,j=1}^3 H_{ij}^2$ on the real locus. To prove every coefficient, differentiate the first formula by the product rule:

$$\begin{aligned}\Delta(Nh'') &= (\Delta N)h'' + 2(g\cdot\nabla N)h^{(3)} + NLh^{(3)} + N^2h^{(4)},\\ \Delta(Lh') &= (\Delta L)h' + 2(g\cdot\nabla L)h'' + L^2h'' + LNh^{(3)}.\end{aligned}$$

In these lines every derivative of h is evaluated at σ. For the remaining two identities, write $N = \sum_i(\partial_i\sigma)^2$ and differentiate each summand:

$$\partial_j N = 2\sum_i(\partial_i\sigma)(\partial_j\partial_i\sigma), \qquad \Delta N = 2\sum_{i,j}(\partial_j\partial_i\sigma)^2 + 2\sum_i(\partial_i\sigma)\partial_i\Delta\sigma.$$

Thus $g\cdot\nabla N = 2g^{\mathsf{T}}Hg$ and $\Delta N = 2\|H\|_{\mathrm{F}}^2 + 2g\cdot\nabla L$. Substitution proves (95), including its second- and first-derivative terms.

13.2 All coefficients on the unchanged inverse trajectory

For $z > 0$ set

$$\gamma(z) = (z^{-1}, -3z/2, 13z^2/2), \qquad \tau = (1-z^2)/8.$$

The inverse is $z = \sqrt{1-8\tau}$ on $\tau < 1/8$. This is precisely the original trajectory, parametrized by a positive coordinate; it has not been rescaled. Direct substitution gives $Q(\gamma) = -1/2$ and $\sigma(\gamma) = -z^2/8$. For clarity the original first derivatives and Laplacian before substitution are

$$\begin{aligned}g_1 &= \tfrac{3}{2}yQ^2w + \tfrac{1}{2}y^3(7+6xy),\\ g_2 &= \tfrac{3}{2}xQ^2w + \tfrac{1}{2}(8y + 21xy^2 + 12x^2y^3),\\ g_3 &= \tfrac{1}{2}Q^3,\\ L &= 3wQ(x^2+y^2) + 3y^4 + 18x^2y^2 + 21xy + 4.\end{aligned}$$

Differentiating these polynomials, followed by the stated substitution, gives

$$g(\gamma) = \begin{pmatrix}-9z^3/32\\ -3z/16\\ -1/16\end{pmatrix}, \qquad H(\gamma) = \begin{pmatrix}-27z^4/4 & 3z^2/16 & -9z/16\\ 3z^2/16 & 13/4 & 3/(8z)\\ -9z/16 & 3/(8z) & 0\end{pmatrix}.$$

The full contractions needed in (95) are

$$\begin{aligned}N(\gamma) &= \frac{81z^6+36z^2+4}{1024}, \qquad\qquad L(\gamma) = \frac{13-27z^4}{4},\\ \|H(\gamma)\|_{\mathrm{F}}^2 &= \frac{729z^8}{16} + \frac{9z^4}{128} + \frac{81z^2}{128} + \frac{169}{16} + \frac{9}{32z^2},\\ (g\cdot\nabla L)(\gamma) &= \frac{9477z^8}{512} - \frac{405z^4}{64} + \frac{27z^2}{128} + \frac{81}{32} + \frac{3}{32z^2},\\ (g^{\mathsf{T}}Hg)(\gamma) &= -\frac{2187z^{10}}{4096} + \frac{81z^6}{4096} - \frac{81z^4}{4096} + \frac{117z^2}{1024} + \frac{9}{1024},\\ (\Delta L)(\gamma) &= -111z^2 + \frac{36}{z^2}.\end{aligned}$$

These are polynomial differentiation and finite matrix products, with both occurrences of each off-diagonal Hessian entry retained in its squared Frobenius norm. They also give a direct way to check each term in the following fully expanded result:

$$(\Delta^2\Psi_t)(\gamma(z)) = \sum_{j=1}^4 C_j(z)\,\partial_s^j Z(t, -z^2/8), \tag{96}$$

where

$$
\begin{aligned}
C_4(z) &= \left(\frac{81z^6 + 36z^2 + 4}{1024}\right)^2, \\
C_3(z) &= -\frac{6561z^{10}}{2048} + \frac{243z^6}{2048} - \frac{135z^4}{1024} + \frac{351z^2}{512} + \frac{31}{512}, \\
C_2(z) &= \frac{26973z^8}{128} - \frac{4419z^4}{64} + \frac{135z^2}{64} + \frac{669}{16} + \frac{15}{16z^2}, \\
C_1(z) &= -111z^2 + \frac{36}{z^2}.
\end{aligned}
$$

For example, the coefficient of z^{-2} in C_2 is $2(9/32) + 4(3/32) = 15/16$; the L^2 term has no negative power. Thus this coefficient includes the differentiated first-order coefficient as well as the Hessian square.

Proposition 13.1 (Two successive Euclidean diffusion limits)**.** *For every fixed real Newman time t,*

$$
\lim_{z\downarrow 0}(\Delta\Psi_t)(\gamma(z)) = \frac{Z_{ss}(t,0)}{256} = -\frac{M_1(t)}{8}, \tag{97}
$$

$$
\lim_{z\downarrow 0} z^2(\Delta^2\Psi_t)(\gamma(z)) = \frac{15}{16}Z_{ss}(t,0) = -30M_1(t) < 0. \tag{98}
$$

In particular $\Delta^2\Psi_t$ is unbounded on $\mathbb{R}^3$, although Ψ_t is real analytic and smooth at every finite spatial point.

Proof. At this fixed t, the entire even Taylor expansion of $Z(t,s)$ and its termwise derivatives give, with $d_j = \partial_s^j Z(t,0)$,

$$
\begin{aligned}
Z_s(t, -z^2/8) &= -d_2 z^2/8 + O(z^6), \\
Z_{ss}(t, -z^2/8) &= d_2 + O(z^4), \\
Z_{sss}(t, -z^2/8) &= -d_4 z^2/8 + O(z^6), \\
Z_{ssss}(t, -z^2/8) &= d_4 + O(z^4).
\end{aligned}
$$

The remainders follow from convergence on a fixed closed disk around zero; they are bounded there by the convergent Taylor tails. Equation (94), $N(\gamma) \to 1/256$, and $L(\gamma) \to 13/4$ prove (97). In (96), C_4 and C_3 are bounded as $z \downarrow 0$, $z^2C_2 \to 15/16$, and $z^2C_1 \to 36$. Multiplying that equation by z^2 and using the four Taylor estimates proves (98). The last term tends to zero because $Z_s(t, -z^2/8) \to 0$; its unmultiplied finite contribution is $-36d_2/8 + O(z^4)$ and has not been discarded from the exact formula. The moment identity and positivity give the displayed strict sign. Consequently, for sufficiently small $z > 0$ the bilaplacian is at most $-15M_1(t)z^{-2}$, proving unboundedness. Entire composition proves smoothness at each finite point. Finally $|\gamma(z)| \geq z^{-1} \to \infty$, so this sequence is explicitly at spatial infinity. □

13.3 The exact relation to the material diffusion operator

Retain the original divergence-free polynomial flow X_τ and its Jacobian A_τ from the material-generator construction. On each of its open flow domains, X_τ is a smooth diffeomorphism, $\det A_\tau = 1$, and $K_\tau = A_\tau^{-1}A_\tau^{-\mathsf{T}}$. The scalar operator proved there is $\mathcal{L}_\tau = \operatorname{div}(K_\tau\nabla)$. For every smooth physical scalar v on the image domain the full chain and cofactor identity give

$$
\mathcal{L}_\tau(v \circ X_\tau) = (\Delta v) \circ X_\tau.
$$

Apply this same identity a second time, now to the smooth scalar Δv. At this fixed τ it proves the exact intertwining

$$\mathcal{L}_\tau^2(v \circ X_\tau) = (\Delta^2 v) \circ X_\tau.$$

The square is operator composition; it differentiates every spatial coefficient of K_τ. It is not the square of a coefficient evaluated at one point.

The already established original flow identity $F_1(X_\tau(q)) = F_1(q) + 2\tau$ gives

$$\Psi_t(X_\tau(q)) = Z(t, \sigma(q) + \tau).$$

At $q_0 = (1, -3/2, 13/2)$ and $\tau < 1/8$ one has $X_\tau(q_0) = \gamma(\sqrt{1 - 8\tau})$. Therefore (97) and (98) give, without interchanging the two time variables,

$$\lim_{\tau \uparrow 1/8} \mathcal{L}_\tau[Z(t, \sigma + \tau)](q_0) = -M_1(t)/8,$$
$$\lim_{\tau \uparrow 1/8} (1 - 8\tau)\mathcal{L}_\tau^2[Z(t, \sigma + \tau)](q_0) = -30M_1(t).$$

For the physical operator $\nu\Delta$ with the unchanged $\nu > 0$, the respective limits acquire the factors ν and ν^2. The scalar in brackets itself has an entire extension in τ. Its material coefficients nevertheless have the displayed second diffusion behavior at the end of this incomplete chart. These identities are an exact relation between the arithmetic observable and Euclidean viscosity. They do not give a singular velocity at a finite physical point, admissible initial velocity or forcing, or a counterexample to the stated Navier–Stokes alternatives.

13.4 The full material-coordinate intertwiner

Here is a complete coordinate proof of the material identity used above. At a fixed permitted real flow time τ, write $X = X_\tau$ on an open flow domain Ω_τ, and let $\Omega'_\tau = X(\Omega_\tau)$. The map

$$X^* : C^\infty(\Omega'_\tau) \longrightarrow C^\infty(\Omega_\tau), \qquad v \longmapsto v \circ X$$

is a linear homeomorphism for the usual compact derivative seminorms; its inverse is composition with the smooth inverse flow. The chain rule on compact sets proves both continuity assertions. Put $A_{ij} = \partial_{q_j} X_i$. Differentiating the flow equation gives $\partial_\tau A = (DU)(X)A$ and $A_0 = I$. On every interval of local invertibility, the determinant derivative is $\partial_\tau \det A = (\operatorname{div} U)(X) \det A = 0$. The proved zero divergence and the initial value consequently give $\det A = 1$ throughout the flow domain.

For completeness the needed cofactor identity, in the unchanged index order, is

$$(A^{-1})_{ji} = \frac{1}{2}\epsilon_{iab}\epsilon_{jcd}(\partial_{q_c} X_a)(\partial_{q_d} X_b), \qquad \sum_j \partial_{q_j}(A^{-1})_{ji} = 0.$$

The first formula is the cofactor formula with $\det A = 1$. In the derivative of its first factor, the symmetric pair (j, c) contracts against ϵ_{jcd}; in the derivative of its second factor the symmetric pair (j, d) does so. Each term is zero, proving the second formula. Therefore every smooth vector field w on Ω'_τ obeys

$$\operatorname{div}_q(A^{-1}(w \circ X)) = \sum_{i,j,k} (A^{-1})_{ji}(\partial_{x_k} w_i)(X)A_{kj} = (\operatorname{div}_x w) \circ X.$$

The gradient chain rule gives $\nabla_q(v \circ X) = A^\mathsf{T}(\nabla_x v) \circ X$. With the full matrix $K_\tau = A^{-1}A^{-\mathsf{T}}$ it follows that

$$\mathcal{L}_\tau X^* = X^*\Delta_x, \qquad \mathcal{L}_\tau^2 X^* = X^*\Delta_x^2, \qquad \mathcal{L}_\tau = \operatorname{div}_q(K_\tau \nabla_q). \tag{99}$$

The second equality is the first one applied to $\Delta_x v$; all derivatives of K_τ are part of the composed operator. This proves the typed isomorphism of the differential operators, including every spatial coefficient, on the stated smooth domains.

13.5 The general entire-function residue and the actual fourth coefficient

Let $\mathcal{O}(\mathbb{C}_s)$ carry its compact-convergence topology. The original polynomial pullback σ^* has domain $\mathcal{O}(\mathbb{C}_s)$ and codomain $\mathcal{O}(\mathbb{C}_q^3)$, and is injective with the left inverse η^* already proved above. The four coefficients C_j in (96) are independent of the particular entire input. The complete chain rule therefore gives the following equality for every $h \in \mathcal{O}(\mathbb{C}_s)$:

$$(\Delta_q^2 \sigma^* h)(\gamma(z)) = \sum_{j=1}^{4} C_j(z) h^{(j)}(-z^2/8), \tag{100}$$

$$\mathfrak{r}(h) := \lim_{z\downarrow 0} z^2 (\Delta_q^2 \sigma^* h)(\gamma(z)) = 36h'(0) + \frac{15}{16} h''(0). \tag{101}$$

To prove the limit without discarding a derivative, all four $h^{(j)}(-z^2/8)$ converge to $h^{(j)}(0)$; meanwhile $z^2C_4, z^2C_3 \to 0$, $z^2C_2 \to 15/16$ and $z^2C_1 \to 36$. These four separate limits prove the displayed equality. This is a continuous linear map $\mathfrak{r} : \mathcal{O}(\mathbb{C}_s) \to \mathbb{C}$: the Cauchy formula on any fixed circle about zero bounds both derivative evaluations by its compact supremum.

Its domain can also be restricted to the original $\mathscr{A}$ or the common space $C = \mathscr{A} \cap \mathcal{H}$. It is continuous there because weighted Fourier seminorms bound both derivatives. An explicit right inverse in those very spaces is

$$E : \mathbb{C} \longrightarrow C, \qquad E(a)(s) = \frac{8a}{15\Xi(0)} s^2 \Xi(s), \qquad \mathfrak{r}E(a) = a. \tag{102}$$

Indeed $s^2\Xi \in C$, $E(a)'(0) = 0$ and $E(a)''(0) = 16a/15$. Its scalar-to-function continuity holds for the specified topological vector spaces. Thus the precise kernel is $\{h : 36h'(0) + (15/16)h''(0) = 0\}$, the image is all of $\mathbb{C}$, and $h \mapsto (h - E\mathfrak{r}(h), \mathfrak{r}(h))$ is a continuous linear isomorphism onto $\ker \mathfrak{r} \times \mathbb{C}$, with inverse $(b, a) \mapsto b + E(a)$, on each of these domains.

In particular retain $\mathcal{K}_t = \mathcal{T}_{\mathcal{D}} Z_t$ and the physical incidence factor a_{ef} from (77). Define $\mathcal{R}(t) = \mathfrak{r}(\mathcal{K}_t)$. The exact value is

$$\begin{aligned}\mathcal{R}(t) = c_0 \left(36Z_t'(3) + \frac{15}{16} Z_t''(3)\right) - \frac{1}{336}\left(36Z_t''(3) + \frac{15}{16} Z_t'''(3)\right) \\ + c_1 \left(36Z_t'(9/2) + \frac{15}{16} Z_t''(9/2)\right) + c_2 \left(36Z_t'(13/2) + \frac{15}{16} Z_t''(13/2)\right).\end{aligned} \tag{103}$$

Every derivative and shift remains present. The original evenness of Z_t supplies no deletion of the derivative at any of these nonzero arguments. Equations (100)–(103) prove

$$\lim_{z\downarrow 0} z^2 (\Delta_q^2 \Psi^{(4)})(t, \gamma(z)) = a_{ef} \mathcal{R}(t). \tag{104}$$

In the material coordinates the exact same number is

$$\lim_{\tau\uparrow 1/8} (1 - 8\tau) \mathcal{L}_\tau^2 [a_{ef} \mathcal{K}_t(\sigma + \tau)](q_0) = a_{ef} \mathcal{R}(t),$$

by (99) and the original inverse $z = \sqrt{1 - 8\tau}$, $\tau = (1 - z^2)/8$. For $(\nu\Delta_q)^2$ and $(\nu\mathcal{L}_\tau)^2$ the right sides acquire exactly ν^2, with the same original $\nu > 0$.

This calculation is also a full coordinate morphism through the invertible fourth-coefficient operator, on all of $\mathscr{A}$:

$$\mathfrak{r}\mathcal{T}_{\mathcal{D}} : \mathscr{A} \to \mathbb{C}, \qquad \ker(\mathfrak{r}\mathcal{T}_{\mathcal{D}}) = \mathcal{T}_{\mathcal{D}}^{-1}(\ker \mathfrak{r}), \qquad (\mathfrak{r}\mathcal{T}_{\mathcal{D}})(\mathcal{T}_{\mathcal{D}}^{-1} E) = \mathrm{id}_{\mathbb{C}}.$$

Its coordinate formula is (103) with Z_t replaced by its actual input h. The first equality of kernels is the definition tested in both directions, and the last equality uses the two proved inverses, so surjectivity and a continuous right inverse follow without an assertion about pointwise zero preservation.

13.6 A strictly nonzero transferred residue at negative Newman times

Set

$$d_0 = c_0 + c_1 + c_2 = \frac{127}{219024} > 0, \qquad d_1 = 3c_0 - \frac{1}{336} + \frac{9}{2}c_1 + \frac{13}{2}c_2 = 0.$$

These are the original zeroth and first energy moments of $\mathcal{D}$, including its derivative delta. For the unchanged Fourier multiplier $m(v)$ from Section 11, $m(0) = d_0$ and $m'(0) = id_1 = 0$. Keep the original kernel k, with

$$k(0) = \sum_{n\geq 1} \frac{\pi}{2} n^2 (2\pi n^2 - 3)e^{-\pi n^2} > 0.$$

The sum converges by Gaussian decay and is strictly positive term by term. No constant kernel has replaced $k(v)$.

Proposition 13.2 (The full transferred asymptotic with a remainder bound)**.** *There is a finite explicitly defined constant C_* below such that, for every real $T \geq 1$,*

$$\mathcal{R}(-T) = -15 d_0 k(0)\sqrt{\pi}\, T^{-3/2} + \mathcal{E}(T), \qquad |\mathcal{E}(T)| \leq 96 C_* \sqrt{\pi}\, T^{-5/2}. \tag{105}$$

Consequently $\mathcal{R}(-T) < 0$ for all sufficiently large T. For every original pair with $a_{ef} > 0$, the actual transferred bilaplacian is unbounded along the original escaping trajectory at each such fixed Newman time.

Proof. Differentiation of the original Fourier integral, justified by its full weighted bounds, gives

$$\mathcal{R}(-T) = 4 \int_{\mathbb{R}} e^{-Tv^2/4} k(v) \left(-\frac{15}{16}v^2 + 36iv\right) m(v)\, dv.$$

The original coefficients are real, so $m(-v) = \overline{m(v)}$. The kernel k is real and even. Write

$$F(v) = k(v)\, \mathrm{Re}\left[\left(-\frac{15}{16}v^2 + 36iv\right) m(v)\right], \qquad c = \frac{15}{16} d_0 k(0) > 0.$$

Then F is smooth, real and even. Its Taylor coefficients through degree two are $F(0) = F'(0) = 0$ and $F''(0)/2 = -c$. Indeed the real quadratic coefficient from $36ivm(v)$ is $-36d_1v^2 = 0$; the other quadratic coefficient is $-(15/16)d_0v^2$. This retains and computes the derivative-channel contribution. Evenness also gives $F'''(0) = 0$.

All constants in the following definition are finite:

$$K_* := \sup_{v\in\mathbb{R}} |k(v)|, \quad A_* := |c_0| + c_1 + c_2, \quad B_* := 1/336,$$

$$C_* := \max\left\{\frac{1}{24}\max_{|v|\leq 1} |F^{(4)}(v)|,\ K_*\left(\frac{15}{16} + 36\right)(A_* + B_*) + c\right\}.$$

The proved two-ended decay of k makes K_* finite, and smoothness makes the displayed compact derivative maximum finite. For $|v| \leq 1$, Taylor's theorem with its integral remainder gives $|F(v) + cv^2| \leq C_*v^4$. For $|v| \geq 1$, use the complete bound $|m(v)| \leq A_* + B_*|v|$ and

$$|F(v)| \leq K_*\left(\frac{15}{16}v^2 + 36|v|\right)(A_* + B_*|v|).$$

Every power on the right is at most $|v|^4$ on that set. The second entry in the maximum consequently gives the same bound $|F(v) + cv^2| \leq C_*v^4$ there. It is therefore a global remainder bound for the original integrand, not only a formal Taylor expansion. The imaginary integrand is odd and absolutely integrable, so its integral vanishes by the substitution $v \mapsto -v$.

The Gaussian integral and two derivatives with respect to its positive parameter give, in the original v coordinate,

$$\int_{\mathbb{R}} v^2 e^{-Tv^2/4}\,dv = 4\sqrt{\pi}T^{-3/2}, \qquad \int_{\mathbb{R}} v^4 e^{-Tv^2/4}\,dv = 24\sqrt{\pi}T^{-5/2}.$$

Differentiation is justified on each compact positive parameter interval by a polynomial times a fixed Gaussian. Integration of the global remainder bound gives $\mathcal{R}(-T) = -16c\sqrt{\pi}T^{-3/2} + \mathcal{E}(T)$ with the stated error. Substitution of c proves (105). For example every $T > \max\{1, 6C_*/c\}$ makes this value negative. Equation (104) then gives unboundedness, since $z^2 > 0$ and $a_{ef}\mathcal{R}(-T) < 0$. $\square$

Every such observable is still entire at each finite spatial point. The exact coordinate map giving the unbounded sequence is $z \mapsto \gamma(z) = \Gamma(-z/2)$, with inverse $z = 1/x$ on its image and $|\gamma(z)| \geq z^{-1} \to \infty$. In material coordinates the same sequence is represented by (τ, q_0) with $\tau = (1 - z^2)/8$, and (99) proves the full operator correspondence there. The newly proved divergence is thus a property of these actual scalar observables and their complete diffusion operators along that image. It supplies no off-critical zero of Z_0 and no finite-point singular classical velocity.

14 The common heat space and the two moving quotient coordinates

The full continuation in Appendices B–I uses $\mathcal{H}_{\leq 1} = \mathcal{H}_{\leq 1}$ for the entire functions of order at most one. This is the same underlying function set denoted by $\mathcal{H}$ in Section 5.1. Its source topology is τ, its original relation is $N_{\mathrm{ar}} = N_{\mathrm{C}}$, and its original quotient is $Q_{\mathrm{ar}} = Q_{\mathrm{C}}$. These equalities identify the unchanged source objects. The symbol $\mathscr{E}$ elsewhere in this manuscript still denotes the weighted smooth Fourier-kernel space, and $\mathscr{A} = T\mathscr{E}$ retains its transported topology. In this section write $N_t^{\mathscr{A}}, Q_t^{\mathscr{A}}$ for the earlier Fourier relation and quotient, and $N_t^{\mathcal{H}}, Q_t^{\mathcal{H}}$ for those in (164). Thus every coordinate in the comparison has a specified source and target. The topology called κ in the continuation is its compact-convergence topology; the positive physical energy scale $\kappa = 2g^2/a$ remains the separate real parameter of Section 10.

Theorem 14.1 (Equality of the heat maps on the common representatives)**.** *On $C = \mathscr{A} \cap \mathcal{H}$, the two globally defined maps*

$$U_t^{\mathscr{A}} f = T(e^{tv^2/4}T^{-1}f), \qquad U_t^{\mathcal{H}} f = \sum_{k=0}^{\infty} \frac{(-t/4)^k}{k!}\partial_s^{2k} f$$

agree as entire functions, for every $t \in \mathbb{C}$. In particular C is invariant under each map and its inverse. Give $C_t = C$ the topology induced by

$$C_t \longrightarrow \mathscr{A} \times (\mathcal{H}, \tau_t), \qquad f \longmapsto (f, f).$$

Then $U_t : C_0 \longrightarrow C_t$ is a linear homeomorphism with inverse U_{-t}; the parameter in the inverse indicates the function map from C_t back to C_0.

Proof. Take $f = T\phi \in C$. For $|t| \leq T_0$ and $|s| \leq R_0$, the absolute value of the Fourier integrand is bounded by $4|\phi(v)|\exp(T_0v^2/4 + R_0|v|)$. Its integral is finite by the defining seminorms of $\mathscr{E}$. The sum of the absolute values of the time-exponential terms has this same integrable bound. Absolute integration of the series therefore gives, at every (t, s),

$$\begin{aligned} U_t^{\mathscr{A}} f(s) &= 4 \sum_{k=0}^{\infty} \frac{(t/4)^k}{k!} \int_{\mathbb{R}} v^{2k} \phi(v) e^{isv}\, dv \\ &= \sum_{k=0}^{\infty} \frac{(-t/4)^k}{k!} \partial_s^{2k} f(s). \end{aligned}$$

The second identity uses $(iv)^{2k} = (-1)^k v^{2k}$, and the same weighted bounds justify every spatial derivative. The series on the right is precisely the convergent γ-series proved in Lemma C.1; convergence in γ implies convergence at every s. Hence the two maps agree with all constants and signs as displayed. Their common output is in both underlying spaces. Applying this conclusion at $-t$ proves invariance under the inverse.

The product map $U_t^{\mathscr{A}} \times U_t^{\mathcal{H}}$ is a homeomorphism from $\mathscr{A} \times (\mathcal{H}, \tau)$ to $\mathscr{A} \times (\mathcal{H}, \tau_t)$: the first assertion is the proved Fourier heat group, and the second is the exact definition of τ_t. The pointwise equality just proved maps the first diagonal onto the second diagonal. Restricting the product map and its continuous inverse proves the claimed homeomorphism. $\square$

Theorem 14.2 (Simultaneous moving quotient correspondence)**.** *Put*

$$\mathscr{K}_t = N_t^{\mathscr{A}} \cap N_t^{\mathcal{H}}, \qquad \mathscr{R}_t = C_t / \mathscr{K}_t.$$

Then $\mathscr{K}_t = U_t \mathscr{K}_0$, and the following maps have the stated types:

$$\overline{U}_t : \mathscr{R}_0 \longrightarrow \mathscr{R}_t, \qquad [f]_{\mathscr{K}_0} \longmapsto [U_t f]_{\mathscr{K}_t}, \tag{106}$$

$$J_t : \mathscr{R}_t \longrightarrow Q_t^{\mathscr{A}} \times Q_t^{\mathcal{H}}, \qquad [g]_{\mathscr{K}_t} \longmapsto ([g]_{N_t^{\mathscr{A}}}, [g]_{N_t^{\mathcal{H}}}). \tag{107}$$

The first map is a homeomorphism and the second a continuous injection. Its two coordinate images and kernels are exactly

$$\operatorname{im} \pi_{\mathscr{A},t} = \{[g]_{N_t^{\mathscr{A}}} : g \in C\}, \qquad \ker \pi_{\mathscr{A},t} = (N_t^{\mathscr{A}} \cap \mathcal{H}) / \mathscr{K}_t,$$

$$\operatorname{im} \pi_{\mathcal{H},t} = \{[g]_{N_t^{\mathcal{H}}} : g \in C\}, \qquad \ker \pi_{\mathcal{H},t} = (\mathscr{A} \cap N_t^{\mathcal{H}}) / \mathscr{K}_t.$$

The first image is dense in $Q_t^{\mathscr{A}}$. For $B_t = M_s - (t/2)\partial_s$, the induced continuous operator on $\mathscr{R}_t$ satisfies the complete coordinate identity

$$J_t([B_t g]_{\mathscr{K}_t}) = ([B_t g]_{N_t^{\mathscr{A}}}, [B_t g]_{N_t^{\mathcal{H}}}). \tag{108}$$

Proof. For $f \in \mathscr{K}_0$, both definitions of the moving relation put $U_t f$ in $\mathscr{K}_t$. Conversely, $g \in \mathscr{K}_t$ lies in both function spaces; Theorem 14.1 identifies both inverse images as the same $U_{-t} g$. Each moving-relation definition puts this inverse in its original relation, so it belongs to $\mathscr{K}_0$. This proves the equality, including both inclusions. The homeomorphism of C_0 and C_t consequently descends to the homeomorphism (106), with inverse $[g] \mapsto [U_{-t} g]$.

The two coordinate quotient maps are continuous on the product and therefore on its diagonal. Their joint kernel on C_t is exactly $\mathscr{K}_t$. The quotient universal property gives continuity and injectivity of J_t. Testing its first coordinate for zero means precisely $g \in N_t^{\mathscr{A}}$ in addition to $g \in C$; testing the second means $g \in N_t^{\mathcal{H}}$. This proves both kernel identities and the displayed image formulas, without an assumption about a topology on the image of the injection. Density follows from the cutoff proof in Section 5.1: C is dense in $\mathscr{A}$, and its image under the continuous surjective quotient map is dense in $Q_t^{\mathscr{A}}$.

At time zero, multiplication M_s is continuous on both spaces, preserves C, and preserves both closed original relations. It therefore preserves $\mathscr{K}_0$. Conjugation by the proved homeomorphism gives a continuous operator on C_t preserving $\mathscr{K}_t$. In both coordinates its entire-function formula is $U_t M_s U_{-t} = M_s - (t/2)\partial_s$. Passing to representatives proves (108) in both quotient coordinates. $\square$

This also gives the exact map from the earlier fixed-second-coordinate presentation (28) to the present one. If $\mathcal{U}_t^{\mathcal{H}} : Q_{\mathrm{C}} \to Q_t^{\mathcal{H}}$ is the source heat homeomorphism, then

$$(\mathrm{id}_{Q_t^{\mathscr{A}}} \times \mathcal{U}_t^{\mathcal{H}}) j_t = J_t \overline{U}_t. \tag{109}$$

Both sides send $[f]_{\mathscr{K}_0}$ to $([U_t f]_{N_t^{\mathscr{A}}}, [U_t f]_{N_t^{\mathcal{H}}})$; this proves the identity with its domain, codomain and inverse coordinate change.

The actual fourth coefficient in the same function space. The transported coefficient retains its complete formula

$$\begin{aligned} K_t(s) = T_D Z_t(s) = & -\frac{59}{275184} Z_t(s+3) - \frac{1}{336} Z_t'(s+3) \\ & + \frac{1}{1296} Z_t(s+9/2) + \frac{3}{132496} Z_t(s+13/2). \end{aligned}$$

The resulting map is explicitly

$$C \longrightarrow \mathscr{R}_t \xrightarrow{J_t} Q_t^{\mathscr{A}} \times Q_t^{\mathcal{H}}, \qquad f \longmapsto [f]_{\mathscr{K}_t} \longmapsto ([f]_{N_t^{\mathscr{A}}}, [f]_{N_t^{\mathcal{H}}}),$$

applied to the actual $f = K_t$. Its joint kernel is $\mathscr{K}_t$, as proved above. For this application K_t lies in C: $Z_t = U_t \Xi \in C$, and the four displayed operations preserve both underlying function spaces. On $\mathscr{A}$, a translation by h multiplies the kernel by e^{ihv} and differentiation multiplies it by iv; every seminorm remains finite by Leibniz's rule and the weights already defining $\mathscr{E}$. On $\mathcal{H}$, the translation bound is, for $b = a/2^{p-1}$,

$$\|f(\,\cdot + h)\|_{p,a} \le e^{a|h|^p} \|f\|_{p,b},$$

because $|s+h|^p \le 2^{p-1}(|s|^p + |h|^p)$; differentiation has the bound (161). Thus $T_D : C \to C$ is a well-defined linear map on the common underlying representatives. The formula does not require that T_D preserve either quotient relation. Its exact inverse on $\mathscr{A}$, the entire jet quotients and the retained analytic-unit recovery were proved earlier; the present calculation places the actual coefficient in both moving quotient coordinates. The incidence factor remains $a_{ef} K_t$ in the original fourth-order covariance, so these same linear maps retain a_{ef}.

15 The original three special points and an exact real torus quotient

The input for this section is the three-point period and gluing data in the supplied unpublished manuscript *A compact complex threefold fibred by tori over the projective line, and the six-sphere*, its independent TeX reconstruction and its accompanying workbench proofs. The precise source loci are the reconstruction's Section 3 and `candidate_geometry.tex`, `exact_algebra.tex`, and `integral_leray.tex` in that workbench. We use their explicitly marked local period family and overlap formulas as source input. No identification of the global source manifold with a sphere is used in the following quotient calculation.

15.1 The retained base points, periods and real marking

The original base and its distinguished points are

$$B = \mathbb{P}_t^1, \qquad p_1 = 0, \quad p_2 = 1, \quad p_0 = \infty, \qquad t_c = 1/t, \qquad B^\circ = B \setminus \{p_0, p_1, p_2\}.$$

The finite orders are $m_1 = 3$ and $m_2 = 4$. On the marked orbifold cover $\pi : \mathfrak{h}_z \to B \setminus \{p_0\}$, put $g_0 = (g_1 g_2)^{-1}$. The chosen lifts z_j of p_j have coordinates

$$s_j(z) = \frac{z - z_j}{z - \overline{z_j}}, \qquad s_j(g_j z) = e^{-2\pi i/m_j} s_j(z), \qquad t_j \circ \pi = s_j^{m_j}.$$

The inverse of the displayed Cayley coordinate is $z = (z_j - s_j \overline{z_j})/(1 - s_j)$. The modular period is the separate map $\tau : \mathfrak{h}_z \to \mathfrak{h}_\tau$, with

$$j(\tau) = 1728(t \circ \pi), \qquad \tau(z_1) = \rho = e^{\pi i/3}, \qquad \tau(z_2) = i,$$
$$\mu(z_1) = \frac{2 - \rho}{3}, \qquad \mu(z_2) = \frac{1 - i}{2}.$$

Its exact local forms are $\tau - \rho = s_1 U_1(s_1)$ and $\tau - i = s_2^2 U_2(s_2)$ with $U_1(0)U_2(0) \neq 0$. These units and the local base-coordinate changes $t = t_1 V_1(t_1)$ and $t - 1 = t_2 V_2(t_2)$ remain part of the original data; $V_j(0) \neq 0$ does not assign them the value one.

The ordered source lattices and periods are

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \qquad \Lambda = V^* = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle,$$
$$\Pi(z) = [Z(z) \mid I_2], \qquad Z(z) = \begin{pmatrix} 6\mu(z) & \tau(z) \\ \beta(z) & \mu(z) \end{pmatrix}.$$

Vectors are columns in this ordered basis. The real map underlying the period map is exactly

$$\Pi(z)(a, b, c, d) = (6\mu a + \tau b + c,\ \beta a + \mu b + d). \tag{110}$$

The source admissible period parameter ensures

$$\operatorname{Im} \tau > 0, \qquad D = \operatorname{Im} \beta - \frac{6(\operatorname{Im} \mu)^2}{\operatorname{Im} \tau} < 0.$$

Thus the real determinant in the output order $(\operatorname{Re} \zeta_1, \operatorname{Im} \zeta_1, \operatorname{Re} \zeta_2, \operatorname{Im} \zeta_2)$ is $(\operatorname{Im} \tau) D \neq 0$. The admissible constant in β is retained; no special numerical value is selected here. For an ordinary lifted base open set U, the marking is the real analytic isomorphism

$$\mathcal{M}_U : U \times (\mathbb{R}^4/\mathbb{Z}^4) \longrightarrow \mathcal{T}|_U, \qquad (z, x + \mathbb{Z}^4) \longmapsto [z, \Pi(z)x]. \tag{111}$$

The inverse uses the real inverse of (110). This keeps the full source complex torus in the domain of the next map.

The original inverse-transpose monodromies on Λ are

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix},$$
$$M_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}. \tag{112}$$

The source relations are $A_1^3 = A_2^4 = I_4$ and $A_1 A_2 M_0 = I_4$. The two original fixed twist vectors are

$$v_1 = (1, 2, -4, 0)^{\mathsf{T}}, \qquad v_2 = (-1, -3, 3, 0)^{\mathsf{T}}, \qquad A_j v_j = v_j. \tag{113}$$

In particular the fourth marked coordinate is present in every original matrix and twist before taking a quotient.

15.2 The exact circle quotient and its section defects

Let e_1, e_2, e_3, e_4 denote the four vectors of the original ordered real marking, and use $\bar{e}_1, \bar{e}_2, \bar{e}_3$ in the quotient. Define

$$P : \mathbb{R}^4 \longrightarrow \mathbb{R}^3, \quad P(a,b,c,d) = (a,b,c), \qquad \iota : \mathbb{R}^3 \longrightarrow \mathbb{R}^4, \quad \iota(a,b,c) = (a,b,c,0).$$

Their matrices are $P = [I_3 \mid 0]$ and $\iota = [I_3 \mid 0]^\mathsf{T}$. Since $P\mathbb{Z}^4 = \mathbb{Z}^3$, they induce smooth homomorphisms

$$\overline{P} : \mathbb{T}^4 \longrightarrow \mathbb{T}^3, \quad [x] \longmapsto [Px], \qquad \bar{\iota} : \mathbb{T}^3 \longrightarrow \mathbb{T}^4, \quad [y] \longmapsto [\iota y], \qquad \mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n. \tag{114}$$

Proposition 15.1 (Typed quotient, kernel and full monodromy intertwining)**.** *The maps in* (114) *give the split exact sequence of marked compact real Lie groups*

$$0 \longrightarrow \mathbb{R}e_4/\mathbb{Z}e_4 \longrightarrow \mathbb{T}^4 \xrightarrow{\overline{P}} \mathbb{T}^3 \longrightarrow 0, \qquad \overline{P}\,\bar{\iota} = \mathrm{id}_{\mathbb{T}^3}. \tag{115}$$

The quotient monodromies in the original first three coordinates are

$$\bar{A}_1 = \begin{pmatrix} 1 & 0 & 0 \\ 6 & 0 & 1 \\ -6 & -1 & -1 \end{pmatrix}, \qquad \bar{A}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ -6 & 1 & 0 \end{pmatrix}, \qquad \bar{M}_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}. \tag{116}$$

They satisfy

$$PA_j = \bar{A}_j P, \quad PM_0 = \bar{M}_0 P, \quad \bar{A}_1^3 = \bar{A}_2^4 = I_3, \quad \bar{A}_1\bar{A}_2\bar{M}_0 = I_3.$$

The section has the exact defects

$$\begin{aligned} (A_1\iota - \iota\bar{A}_1)(a,b,c) &= (-2a+b)e_4, \\ (A_2\iota - \iota\bar{A}_2)(a,b,c) &= (3a+c)e_4, \\ (M_0\iota - \iota\bar{M}_0)(a,b,c) &= -ae_4. \end{aligned} \tag{117}$$

Proof. For $[x] \in \ker \overline{P}$, the first three entries of x are integers. Subtracting that vector of $\mathbb{Z}^4$ leaves precisely de_4. Two such representatives give the same class exactly when their fourth coordinates differ by an integer. This proves the stated kernel. Surjectivity and the section identity follow by applying P to $\iota(a,b,c)$. The same equations show that $\mathbb{T}^4/(\mathbb{R}e_4/\mathbb{Z}e_4) \to \mathbb{T}^3$, $[[x]] \mapsto [Px]$, is bijective with inverse $[y] \mapsto [[\iota y]]$. Both maps are smooth: in the ordinary local covering coordinates they are the displayed linear projection and inclusion followed by the quotient.

Each matrix in (112) fixes e_4, and its first three rows have zero fourth entry. Taking those rows proves all three intertwiners and the matrices in (116) by literal multiplication. Applying P to the source product and power identities, and then composing with the surjection P, proves the corresponding identities in dimension three. The lower row of each full matrix applied to $(a,b,c,0)$ gives exactly (117). These defects are generally nonzero even as circle classes; thus the section is a right inverse of the marked quotient without being a monodromy-equivariant section. Indeed there is no real linear section equivariant even for M_0 alone. Any linear section has the form $\iota_L(y) = (y, Ly)$ for a row L. Equivariance would require $L(\bar{M}_0 - I_3) = (-1,0,0)$. Its left side is $(0, L_3, 0)$, contradicting the first coordinate of the required equality. $\square$

On the source fibre, this gives the concrete map

$$p_z : \mathbb{C}^2/\Pi(z)\Lambda \longrightarrow \mathbb{T}^3, \qquad [\zeta] \longmapsto [P\Pi(z)^{-1}\zeta], \tag{118}$$

where the inverse is real linear. Its kernel is the original circle $\mathbb{R}\Pi(z)e_4/\mathbb{Z}\Pi(z)e_4$; here $\Pi(z)e_4 = (0,1)^\mathsf{T}$ exactly. The map $[y] \mapsto [\Pi(z)\iota y]$ is its marked right inverse. Equation (111) proves the smooth dependence of the quotient and section on the original base variable. This construction concerns a three-real-dimensional fibre quotient. Keeping the complex base produces a five-real-dimensional family; no identification of that total space with a three-dimensional physical domain is implicit.

15.3 The finite affine actions and the complete logarithmic overlaps

Put

$$c_1 = P(v_1/3) = \tfrac{1}{3}(1,2,-4)^\mathsf{T}, \qquad c_2 = P(v_2/4) = \tfrac{1}{4}(-1,-3,3)^\mathsf{T}. \tag{119}$$

For $j = 1, 2$ the full marked map $L_j(x) = A_jx + v_j/m_j$ on $\mathbb{T}^4$ descends to

$$\bar{L}_j([y]) = [\bar{A}_jy + c_j], \qquad \overline{P}L_j = \bar{L}_j\overline{P}.$$

All matrices and translations are those obtained above. Since $\bar{A}_jc_j = c_j$, induction gives

$$\bar{L}_j^k([y]) = [\bar{A}_j^ky + kc_j]. \tag{120}$$

At $k = m_j$ the matrix is the identity and the translation is the integer vector Pv_j, so the map is the identity on $\mathbb{T}^3$. The first coordinate of every $\bar{A}_j^ky$ is y_1. For $0 < k < 3$, the first-coordinate translation of $\bar{L}_1^k$ is $k/3 \notin \mathbb{Z}$; for $0 < k < 4$, that of $\bar{L}_2^k$ is $-k/4 \notin \mathbb{Z}$. Thus every nonidentity power is fixed-point free, and the two affine torus actions have exactly orders three and four. On $\Delta_j \times \mathbb{T}^3$ the corresponding maps are $(z,[y]) \mapsto (g_jz, \bar{L}_j[y])$, and the same argument proves freeness also over the fixed base point. The quotient fibre over p_j is therefore the smooth compact real quotient $\mathbb{T}^3/\langle\bar{L}_j\rangle$.

The overlap is a varying translation whose complete real expression must also be retained. Define

$$\begin{aligned} J_z &= \Pi(z)^{-1}i\Pi(z) : \mathbb{R}^4 \to \mathbb{R}^4, \qquad \alpha(z) = \frac{\log s_j(z)}{2\pi i}, \\ a_j(z) &= \operatorname{Re}\alpha(z)\, v_j + \operatorname{Im}\alpha(z)\, J_zv_j, \qquad \bar{a}_j(z) = Pa_j(z). \end{aligned} \tag{121}$$

Here is an explicit coordinate formula for the full J_z, retaining all entries of the period matrix. Write $\mu_R = \operatorname{Re}\mu$, $\mu_I = \operatorname{Im}\mu$, and likewise for τ, β. For $x = (a,b,c,d)$ set

$$p = 6\mu a + \tau b + c, \qquad q = \beta a + \mu b + d, \qquad E = 6\mu_I^2 - \tau_I\beta_I = -\tau_ID \neq 0.$$

Then $J_zx = (a', b', c', d')$ is exactly

$$\begin{aligned} a' &= \frac{\mu_I \operatorname{Re}p - \tau_I \operatorname{Re}q}{E}, \qquad & b' &= \frac{-\beta_I \operatorname{Re}p + 6\mu_I \operatorname{Re}q}{E}, \\ c' &= -\operatorname{Im}p - 6\mu_Ra' - \tau_Rb', & d' &= -\operatorname{Im}q - \beta_Ra' - \mu_Rb'. \end{aligned} \tag{122}$$

Indeed the imaginary parts of $\Pi(z)J_zx = i(p,q)$ are the two linear equations for a', b' with determinant E; its real parts give c', d'. For the two actual twist inputs one uses, without changing their signs,

$$(p,q)_{v_1} = (6\mu + 2\tau - 4, \beta + 2\mu), \qquad (p,q)_{v_2} = (-6\mu - 3\tau + 3, -\beta - 3\mu).$$

Equations (121) and (122) consequently specify every coordinate of the descended translation.

The source period covariance gives $J_{g_jz}A_j = A_jJ_z$. Clockwise continuation of the chosen logarithm changes α by $-1/m_j$, so

$$a_j(g_jz) = A_ja_j(z) - v_j/m_j, \qquad \bar{a}_j(g_jz) = \bar{A}_j\bar{a}_j(z) - c_j. \tag{123}$$

To verify the first identity directly, the real part of α acquires the displayed subtraction; its imaginary part stays fixed, and $A_j v_j = v_j$ gives $J_{g_j z} v_j = A_j J_z v_j$. Applying the proved intertwiner P gives the second identity.

On the punctured lifted family put

$$\bar{h}_j(z,[y]) = (z,[y+\bar{a}_j(z)]).$$

Changing $\log s_j$ by $2\pi i n$ changes $\bar{a}_j$ by the integer vector nPv_j, proving branch independence. Its inverse is translation by $-\bar{a}_j(z)$. Applying (123) to the three successive maps gives the exact conjugacy

$$\bar{h}_j\,(g_j,\bar{L}_j)\,\bar{h}_j^{-1} : (z,[y]) \longmapsto (g_j z,[\bar{A}_j y]). \tag{124}$$

The cancelled translation is $-\bar{A}_j\bar{a}_j(z) + c_j + \bar{a}_j(g_j z) = 0$. This proves descent of the complete logarithmic gluing to the real circle quotient, including its inverse and every period-dependent coefficient. Circle cosets are preserved because the full linear maps fix e_4 and fibre translations preserve cosets. No preservation of a chosen Euclidean or Hermitian metric is required or asserted.

There is a concrete reason to retain these overlap translations when combining the three source points. In a single uncorrected marked fibre, compose the two finite affine formulas and the linear cusp formula. Their linear product is $\bar{A}_1\bar{A}_2\bar{M}_0 = I_3$, but their translation is

$$c_1 + \bar{A}_1 c_2 = (1/12,-1/12,1/6)^{\mathsf{T}} \notin \mathbb{Z}^3.$$

Thus those three uncorrected affine formulas are not the three meridians in the same middle-family marking. The exact marking change is (124); it identifies each finite formula with its linear middle-family representative, where the product relation holds. This computes the discrepancy and the maps correcting it.

15.4 The original cusp direction and three independent shear data

The cusp's complete period expression is

$$e^{2\pi i\tau} = t_c u_1(t_c), \quad u_1(0) = 1/1728, \quad h = (2\pi i)^{-1}\log u_1, \quad s = \tau - h, \quad b = \beta + \tau,$$

$$Z = sB_0 + C(t_c), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C(t_c) = \begin{pmatrix} 6\mu(t_c) & h(t_c) \\ b(t_c) - h(t_c) & \mu(t_c) \end{pmatrix}. \tag{125}$$

The full holomorphic unit u_1 and all entries of C remain present. Define the original cusp sublattice and quotient with their ordered bases:

$$\Lambda_{\mathrm{tor}} = \mathbb{Z}\langle e_3, e_4\rangle, \qquad \bar{\Lambda} = \Lambda/\Lambda_{\mathrm{tor}} = \mathbb{Z}\langle [e_1],[e_2]\rangle.$$

On the original marked lattice,

$$(M_0 - I)(a,b,c,d) = (0,0,b,-a), \qquad \ker(M_0 - I) = \operatorname{im}(M_0 - I) = \mathbb{Z}\langle e_3, e_4\rangle.$$

The quotient sends this cusp image lattice onto $\mathbb{Z}\bar{e}_3$, with kernel $\mathbb{Z}e_4$. In particular the unchanged source vector e_2 supplies

$$c_0 = P((M_0 - I)e_2) = \bar{e}_3 = (0,0,1)^{\mathsf{T}}. \tag{126}$$

The other primitive image $(M_0 - I)e_1 = -e_4$ is killed by this precise quotient. This records the lost cusp direction explicitly. Translation by the integer vector c_0 is the identity on $\mathbb{T}^3$; the same vector defines a nonzero tangent direction there. These two statements follow respectively from taking its class modulo $\mathbb{Z}^3$ and from the differential of the covering $\mathbb{R}^3 \to \mathbb{T}^3$.

The relation with the original two-dimensional cusp torus is therefore the following exact restriction. Denote its full marked torus by

$$\mathbb{T}^2_{\mathrm{tor}} = (\mathbb{R}e_3 \oplus \mathbb{R}e_4)/(\mathbb{Z}e_3 \oplus \mathbb{Z}e_4), \qquad K_4 = \mathbb{R}e_4/\mathbb{Z}e_4, \quad K_3 = \mathbb{R}\bar{e}_3/\mathbb{Z}\bar{e}_3.$$

In addition to (115), one has

$$\begin{aligned} &0 \longrightarrow K_4 \longrightarrow \mathbb{T}^2_{\mathrm{tor}} \xrightarrow{[re_3+ue_4]\mapsto[r\bar{e}_3]} K_3 \longrightarrow 0, \\ &0 \longrightarrow \mathbb{R}e_4 \longrightarrow \mathbb{R}e_3 \oplus \mathbb{R}e_4 \xrightarrow{(r,u)\mapsto r} \mathbb{R}\bar{e}_3 \longrightarrow 0. \end{aligned} \tag{127}$$

The vertical inclusions of the middle and final terms into $\mathbb{T}^4$ and $\mathbb{T}^3$ make a commuting diagram with (115): on every representative, $P(re_3 + ue_4) = r\bar{e}_3$. The same equality proves commutation of the differential sequences. Their kernels, surjections and sections $[r] \mapsto [re_3]$ follow by testing the displayed coordinates and their integer lattices. The original B_0 is also retained in this diagram: the map $\bar{\Lambda} \to \Lambda_{\mathrm{tor}}$ sends $(a, b) \mapsto be_3 - ae_4$, and composition with P is $(a, b) \mapsto b\bar{e}_3$. This proves the exact relationship between the two real cusp coordinates, their two-torus, and the one-dimensional image inside the three-dimensional marked quotient.

The cusp punctured-overlap map on the original complex covering is

$$(z, \zeta_1, \zeta_2) \longmapsto (e^{2\pi i\zeta_1}, e^{2\pi i\zeta_2}, e^{2\pi i s(z)}).$$

In these coordinates the e_4 circle acts by $(x_1, x_2, t_c) \mapsto (x_1, e^{2\pi i r}x_2, t_c)$, $r \in \mathbb{R}/\mathbb{Z}$. This follows by exponentiating $\Pi(z)e_4 = (0, 1)$. It commutes with the full toric multipliers

$$(x, t_c) \longmapsto (e^{2\pi i C(t_c)\bar{\lambda}} t_c^{B_0\bar{\lambda}} x, t_c), \qquad \bar{\lambda} \in \mathbb{Z}^2.$$

Thus the overlap identifies the same circle quotient on the punctured cusp model. The calculation makes no assertion that this circle action remains free on the singular central fibre; the defined three-dimensional marked fibre quotient is on the smooth torus locus.

Finally, in the original order (p_1, p_2, p_0) the three directions form

$$[c_1\ c_2\ c_0] = \begin{pmatrix} 1/3 & -1/4 & 0 \\ 2/3 & -3/4 & 0 \\ -4/3 & 3/4 & 1 \end{pmatrix}, \qquad \det[c_1\ c_2\ c_0] = -\frac{1}{12}. \tag{128}$$

The integer wave covectors in the same order are

$$k_1 = (4, 0, 1), \qquad k_2 = (3, 0, 1), \qquad k_0 = (0, 1, 0), \qquad \det\begin{pmatrix} k_1 \\ k_2 \\ k_0 \end{pmatrix} = -1. \tag{129}$$

Direct multiplication gives $k_j c_j = 0$ for all three labels: explicitly $(4-4)/3 = 0$, $(-3+3)/4 = 0$, and $0 = 0$. The first determinant is the upper-left minor $-1/4 + 1/6 = -1/12$; expansion along the last row of the second matrix gives $-(4 - 3) = -1$. Hence the source directions span the whole quotient tangent space and the wave covectors form an integral basis of its dual lattice. The direction map $(r_1, r_2, r_0) \mapsto r_1c_1 + r_2c_2 + r_0c_0$ has the exact inverse matrix

$$[c_1\ c_2\ c_0]^{-1} = \begin{pmatrix} 9 & -3 & 0 \\ 8 & -4 & 0 \\ 6 & -1 & 1 \end{pmatrix}.$$

Multiplying this displayed matrix on either side by (128) gives I_3, proving the inverse in the original coordinates. For each j, each scalar amplitude a_j and each smooth one-periodic function

F_j, the literal quotient field $y \mapsto a_j c_j F_j(k_j y)$ is periodic and has divergence $a_j(k_j c_j)F_j'(k_j y) = 0$. This final derivative identity uses the explicit quotient coordinates; constructing and checking its subsequent Navier–Stokes dynamics is a further calculation on these defined fields. These individual modes are fields on the marked $\mathbb{T}^3$. For clarity, their first two wave covectors obey $k_1\bar{A}_1 = (-2,-1,-1)$ and $k_2\bar{A}_2 = (-3,1,0)$, whereas $k_0\bar{M}_0 = k_0$. Therefore no automatic descent of the individual finite-point modes to $\mathbb{T}^3/\langle \bar{L}_j\rangle$ is asserted. There is an explicit field map to that quotient: for a smooth one-periodic real function F, the finite orbit sum

$$\mathcal{R}_j F(y) = c_j \sum_{\ell=0}^{m_j-1} F(k_j \bar{A}_j^{\ell} y)$$

is invariant as a vector field under $\bar{L}_j$. Indeed $\bar{A}_j c_j = c_j$ and $k_j \bar{A}_j^{\ell} c_j = k_j c_j = 0$; substitution of $\bar{A}_j y + c_j$ cycles the sum and gives $\mathcal{R}_j F(\bar{L}_j y) = \bar{A}_j \mathcal{R}_j F(y)$. Each summand has zero divergence by the same coordinate derivative calculation. The local quotient covering then gives a unique smooth field downstairs: at $[y]$ use its differential on $\mathcal{R}_j F(y)$; the displayed covariance proves independence of the lift. This supplies the exact field descent map while keeping its differential-operator and metric questions separate for the subsequent dynamics calculation.

15.5 Three source polarizations in the original Euclidean three-torus

Fix an ordinary lifted base point z and the real quotient map p_z in (118). The physical three-torus below is its marked target. Write t_B for the unchanged source base coordinate and t for fluid time; their product is $\mathbb{P}^1_{t_B} \times I_t$, with the original base recovered by projection to t_B. The fields below are defined on this regular marked target and use the data of all three special points as their labelled polarization inputs.

Use the three vectors from the preceding exact lattice quotient, in the retained order $(1,2,0)$:

$$c_1 = \tfrac{1}{3}(1,2,-4)^{\mathsf{T}}, \quad c_2 = \tfrac{1}{4}(-1,-3,3)^{\mathsf{T}}, \quad c_0 = (0,0,1)^{\mathsf{T}}.$$

The divisors $3, 4$ are the actual finite filling orders. Choose the following integral character rows on the quotient torus:

$$k_1 = (4,0,1), \qquad k_2 = (3,0,1), \qquad k_0 = (0,1,0).$$

They obey $k_j c_j = 0$. For $\nu > 0$ and any three real smooth amplitudes a_j on a time interval, define on the unit Euclidean torus $\mathbb{T}^3_x = \mathbb{R}^3/\mathbb{Z}^3$

$$u(t,x) = \sum_{j\in\{1,2,0\}} a_j(t) c_j \sin(2\pi k_j x). \tag{130}$$

This is an explicit linear map from the amplitude space to smooth vector fields. Its inverse on its image is

$$a_j(t) = \frac{2}{|c_j|^2} \int_{[0,1]^3} u(t,x)\cdot c_j \sin(2\pi k_j x)\, dx. \tag{131}$$

To prove it, the integral of a nonconstant integer exponential is zero by integrating a coordinate with nonzero frequency. Product-to-sum then gives zero for distinct displayed sine frequencies and $1/2$ for the square of each. This proves the inverse and also continuity in every compact-time smooth seminorm in both directions. Direct differentiation gives $\operatorname{div} u = 0$ and

$$\Delta_x u = -4\pi^2 \sum_j |k_j|^2 a_j c_j \sin(2\pi k_j x), \qquad (|k_1|^2, |k_2|^2, |k_0|^2) = (17, 10, 1).$$

The three phases are themselves a typed torus automorphism:

$$W : \mathbb{T}^3_x \longrightarrow \mathbb{T}^3_\theta, \qquad \theta = (4x_1 + x_3, 3x_1 + x_3, x_2),$$
$$W^{-1}(\theta) = (\theta_1 - \theta_2, \theta_0, -3\theta_1 + 4\theta_2).$$

Both matrices are integral and their composition is the identity; $\det W = -1$, so no finite covering kernel or factor in volume is introduced. The complete Euclidean metric transformation is

$$g_\theta = W^{-\mathsf{T}} W^{-1}, \qquad g_\theta^{-1} = WW^{\mathsf{T}} = G = \begin{pmatrix} 17 & 13 & 0 \\ 13 & 10 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \Delta_x = W^*(G^{ab}\partial_{\theta_a}\partial_{\theta_b})(W^*)^{-1}. \tag{132}$$

Indeed $\partial_{x_j} = \sum_a W_{aj}\partial_{\theta_a}$; applying this twice proves the last identity, including its mixed coefficient $26\partial_{\theta_1}\partial_{\theta_2}$. For $v = Wu \circ W^{-1}$, $p_\theta = p \circ W^{-1}$ and $f_\theta = Wf \circ W^{-1}$, the full momentum equation becomes

$$\partial_t v + (v \cdot \nabla_\theta)v - \nu G^{ab}\partial_{\theta_a}\partial_{\theta_b} v + G\nabla_\theta p_\theta = f_\theta, \qquad \operatorname{div}_\theta v = 0.$$

The derivative chain rule proves convection and divergence; multiplying $\nabla_x p = W^{\mathsf{T}}\nabla_\theta p_\theta$ by W proves the pressure term. Thus the phase coordinates retain the original Euclidean viscosity rather than changing its matrix.

At every time when all a_j are nonzero, the field depends on all three spatial directions: if a constant real vector b obeys $(b \cdot \nabla_x)u = 0$, Fourier independence gives $k_j b = 0$ for all three j. Invertibility of W gives $b = 0$. Its exact kinetic energy on the unit cell is

$$\frac{1}{2}\int_{[0,1]^3} |u|^2\, dx = \frac{1}{4}\left(\frac{7}{3}a_1^2 + \frac{19}{16}a_2^2 + a_0^2\right). \tag{133}$$

This follows from the same orthogonality and the unchanged lengths $|c_1|^2 = 7/3$, $|c_2|^2 = 19/16$, $|c_0|^2 = 1$.

15.6 Every mixed interaction, the pressure and the force

Use ordered pairs $(i, j) = (1, 2), (1, 0), (2, 0)$ and define

$$k^{\pm}_{ij} = k_i \pm k_j, \qquad B^{\pm}_{ij} = (c_i \cdot k_j)c_j \pm (c_j \cdot k_i)c_i, \qquad P_\ell b = b - \ell^{\mathsf{T}}\frac{\ell b}{|\ell|^2}.$$

Here each k or ℓ is a row and each B or b a column. Every displayed cross frequency is nonzero. The six frequencies, in the same pair order and then $+,-$, are

$$(7, 0, 2),\ (1, 0, 0),\ (4, 1, 1),\ (4, -1, 1),\ (3, 1, 1),\ (3, -1, 1).$$

They are distinct up to sign and none is a base frequency. Differentiating (130) before collecting terms gives

$$N := (u \cdot \nabla)u = 2\pi \sum_{i \neq j} a_i a_j (c_i \cdot k_j)c_j \sin(2\pi k_i x)\cos(2\pi k_j x).$$

The self terms are zero by $k_j c_j = 0$. Applying both exact product-to-sum identities to each ordered pair gives

$$N = \pi \sum_{i<j} a_i a_j \sum_{\epsilon \in \{+,-\}} B^{\epsilon}_{ij} \sin(2\pi k^{\epsilon}_{ij} x). \tag{134}$$

The notation $i < j$ here means the three ordered pairs just fixed, not numerical ordering of their names.

Choose the following zero-mean, unit-periodic pressure and force:

$$p(t,x) = \frac{1}{2}\sum_{i<j} a_i a_j \sum_{\epsilon\in\{+,-\}} \frac{k_{ij}^\epsilon B_{ij}^\epsilon}{|k_{ij}^\epsilon|^2}\cos(2\pi k_{ij}^\epsilon x), \tag{135}$$

$$\begin{aligned} f(t,x) &= \sum_j (a_j' + 4\pi^2\nu|k_j|^2 a_j)c_j \sin(2\pi k_j x) \\ &\quad + \pi\sum_{i<j} a_i a_j \sum_{\epsilon\in\{+,-\}} P_{k_{ij}^\epsilon} B_{ij}^\epsilon \sin(2\pi k_{ij}^\epsilon x). \end{aligned} \tag{136}$$

The gradient of each cosine pressure term is minus the longitudinal part of the corresponding term in (134). Hence

$$\partial_t u + (u\cdot\nabla)u - \nu\Delta u + \nabla p = f, \qquad \operatorname{div} u = \operatorname{div} f = 0.$$

This proves every component of an exact forced Navier–Stokes solution, with the original positive viscosity and periodic pressure. It also proves the exact residual map $(a_1, a_2, a_0) \mapsto (u, p, f)$ and its inverse on the velocity coordinate (131).

The six transverse interaction columns are explicitly

$$\begin{aligned} &P_{k_{12}^+} B_{12}^+ = (-7/318, 1/12, 49/636)^\mathsf{T}, && P_{k_{12}^-} B_{12}^- = (0, 5/12, -7/12)^\mathsf{T}, \\ &P_{k_{10}^+} B_{10}^+ = (1, 16, -20)^\mathsf{T}/27, && P_{k_{10}^-} B_{10}^- = (-17, -16, 52)^\mathsf{T}/27, \\ &P_{k_{20}^+} B_{20}^+ = (7, -27, 6)^\mathsf{T}/44, && P_{k_{20}^-} B_{20}^- = (29, 27, -60)^\mathsf{T}/44. \end{aligned}$$

They follow by inserting the three original c_j into the displayed definitions; all are nonzero. For an independent component-level check of the non-gradient interaction,

$$B_{12}^+ = (0, 1/12, 1/12)^\mathsf{T}, \qquad (k_{12}^+)^\mathsf{T} \times B_{12}^+ = (-1/6, -7/12, 7/12)^\mathsf{T} \neq 0.$$

The coefficient of $\cos(2\pi(7x_1 + 2x_3))$ in $\operatorname{curl} N$ is exactly $2\pi^2 a_1 a_2$ times this vector. Distinct frequencies preclude cancellation with another pair. Thus the three-point image has actual transverse nonlinear forcing, with all pair interactions explicitly calculated.

The complete mean quadratic stress and its exact image cone are

$$\int_{[0,1]^3} u\otimes u\,dx = \frac{1}{2}C\operatorname{diag}(a_1^2, a_2^2, a_0^2)C^\mathsf{T}, \qquad C = (c_1, c_2, c_0), \quad \det C = -1/12. \tag{137}$$

Indeed all mixed sine products integrate to zero. A real symmetric matrix belongs to this image exactly when $C^{-1}RC^{-\mathsf{T}}$ is diagonal with nonnegative entries. Necessity is the displayed identity; for sufficiency take the three amplitudes to be either sign of the square roots of twice those entries. This proves both directions and all amplitude fibres, including the unique zero choice for a zero entry. It quantifies precisely the stress supplied by these three specific source channels.

15.7 The exact forcing class and what fixed modes permit

Define the original scalar function

$$b(t) = \begin{cases} \exp[-1/(t(1-t))], & 0 < t < 1, \\ 0, & t \notin (0,1). \end{cases}$$

Every derivative on $(0,1)$ is the same exponential times a rational function with poles only at $0, 1$. For every positive integer n, $y^n e^{-y} \to 0$ as $y \to \infty$, for example by comparison with the $(n+1)$st exponential-series term. Consequently each derivative tends to zero at both endpoints. Extending by zero proves $b \in C_c^\infty(\mathbb{R})$, with support $[0,1]$. Take $a_1 = a_2 = a_0 = b$ in (130)–(136). This gives an actual three-dimensional field starting from rest, a smooth periodic pressure, and a force compactly supported in time. All spatial and temporal derivatives of the force are bounded on its compact time support, and therefore satisfy every bound $C_{\alpha m K}(1+t)^{-K}$ on $t \geq 0$. Formula (133) proves a uniform finite kinetic-energy bound. The velocity and pressure are globally smooth, so this explicit admissible forced example itself is not a breakdown example.

For the entire fixed three-mode image there is an exact stronger calculation. Suppose an input amplitude curve on $[0,T)$ has a prescribed momentum force F, with any periodic pressure. Projecting the momentum equation onto the divergence-free base sine mode and using the six distinct cross frequencies gives

$$a_j'(t) + \lambda_j a_j(t) = b_j(t), \quad \lambda_j = 4\pi^2\nu|k_j|^2 > 0, \qquad b_j(t) = \frac{2}{|c_j|^2}\int_{[0,1]^3} F(t,x)\cdot c_j \sin(2\pi k_j x)\,dx. \tag{138}$$

The pressure projection is zero by periodic integration by parts and $\operatorname{div}(c_j \sin(2\pi k_j x)) = 0$. Multiplication by $e^{\lambda_j t}$ and integration give the complete inverse formula

$$a_j(t) = e^{-\lambda_j t} a_j(0) + \int_0^t e^{-\lambda_j(t-s)} b_j(s)\,ds. \tag{139}$$

If F extends smoothly through a finite T, so do its integrals b_j and the explicit functions (139); differentiation under their finite integrals proves all time derivatives. The finite mode sum then extends smoothly as a velocity. In particular,

$$|a_j(t)| \leq |a_j(0)| + \lambda_j^{-1} \sup_{0\leq s\leq T} |b_j(s)| \quad (0 \leq t < T).$$

This proves directly that bounded terminal forcing cannot generate velocity blowup within this fixed finite image. No choice of its periodic pressure changes the conclusion.

For zero force the exact classification is even more explicit: (139) gives $a_j(t) = a_j(0)e^{-\lambda_j t}$. Every cross-mode projection in (136) must also vanish. The six nonzero transverse columns and distinct frequencies force $a_i a_j = 0$ for each of the three pairs. Hence at most one initial amplitude is nonzero. Conversely any one such decaying mode, with zero pressure, solves the unforced equations directly. This proves both directions of the unforced classification for the actual image. The full integer frequency group generated by the three source characters is nevertheless $\mathbb{Z}^3$, since W^{-1} is integral. The six calculated cross frequencies give the first required extra modes of a nonlinear extension; the group identity does not assert that every possible mode has already been dynamically constructed.

15.8 Compact spatial realization with every additional force term

Regard the periodic functions above as functions on $\mathbb{R}^3_x$ and fix any real $\chi \in C_c^\infty(\mathbb{R}^3)$ equal to one on a chosen open cube. The exact vector potential is

$$A(t,x) = \sum_j \frac{a_j(t)}{2\pi|k_j|^2}(k_j^\top \times c_j)\cos(2\pi k_j x), \qquad \operatorname{curl} A = u.$$

Indeed differentiating its cosine yields $-k_j^\top \times (k_j^\top \times c_j)/|k_j|^2 = c_j$ by $k_j c_j = 0$. Define

$$V = \operatorname{curl}(\chi A) = \chi u + d, \qquad d = \nabla\chi \times A, \qquad p_{\mathrm{loc}} = \chi p.$$

Then V is smooth, compactly supported and divergence-free. The complete force for this spatially localized field is

$$
\begin{aligned}
F_{\mathrm{loc}} = {} & \chi f + (\chi^2 - \chi)N + \partial_t d + (\chi(u\cdot\nabla\chi) + d\cdot\nabla\chi)u \\
& + \chi(u\cdot\nabla)d + \chi(d\cdot\nabla)u + (d\cdot\nabla)d \\
& - \nu\left(2\sum_l(\partial_l\chi)\partial_l u + (\Delta\chi)u + \Delta d\right) + p\nabla\chi.
\end{aligned} \tag{140}
$$

To verify every term, expand $\partial_t(\chi u + d)$, expand the two factors of $(\chi u + d)\cdot\nabla(\chi u + d)$, and use $\Delta(\chi u) = \chi\Delta u + 2\sum_l \chi_l u_l + (\Delta\chi)u$ and $\nabla(\chi p) = \chi\nabla p + p\nabla\chi$. The terms multiplied by χ form exactly χf; the remaining terms are precisely (140). Thus

$$\partial_t V + (V\cdot\nabla)V - \nu\Delta V + \nabla p_{\mathrm{loc}} = F_{\mathrm{loc}}.$$

On the cube where $\chi = 1$, the map recovers (u, p, f) exactly. It is a continuous map between the indicated smooth function spaces because each output is a finite sum of products and derivatives. For the specific compact-time amplitudes b, all three localized outputs have compact space-time support and $V(0) = 0$. Each force derivative therefore satisfies every whole-space bound $C_{\alpha m K}(1 + |x| + t)^{-K}$: multiply it by the displayed weight and take its finite maximum on the fixed compact support. The same argument proves the initial-data decay bounds and a uniform finite energy bound. This is an explicit whole-space counterpart with the complete forcing cost, again giving a smooth forced solution rather than asserting breakdown.

16 Exact arithmetic profiles driven by a supplied incompressible flow

16.1 Public source, retained input class and coordinates

The public manuscript *Finite Time Blowup for Navier–Stokes* [8, Theorem 1.1] claims a smooth forced velocity on $\mathbb{R}^3 \times [0, 1)$, with one fixed compact spatial support, initial velocity zero, bounded kinetic energy and unbounded terminal supremum norm, for every $\nu > 0$. Its complete proof and formal verification are not imported as independently certified results in this section. We construct and prove an operator correspondence for its stated smooth input class, which also contains the explicit localized fields in Section 15.8. The correspondence itself is established below without assuming any unproved endpoint. The first source statement is an attribution; the formulas and estimates that follow are our proofs.

For attribution, Buckmaster's primary statement [9, p. 1] identifies his collaboration with Levent Alpöge on smooth-forced Euler and credits Diego Córdoba and Luis Martínez-Zoroa for the underlying forced-blowup programme. The revised NS manuscript [8, pp. 2–3] now discusses that programme and the hypodissipative extension with Zheng explicitly. These statements identify the respective work; they do not establish a dependency on the three-point period construction used above.

Let $\mathcal{U}_{K,T,\nu}$ be the set of smooth real triples (u, p, f) on $[0, T) \times \mathbb{R}^3$ satisfying, component by component,

$$\partial_t u_i + \sum_{j=1}^{3} u_j \partial_j u_i = \nu \sum_{j=1}^{3} \partial_j^2 u_i - \partial_i p + f_i, \qquad \sum_i \partial_i u_i = 0, \qquad \operatorname{supp} u(t,\cdot) \subset K, \tag{141}$$

where $K \subset \mathbb{R}^3$ is compact, $0 < T < \infty$, and $\nu > 0$ is retained. We do not require compact support of p, f for the following map. The particular compactly supported inputs already constructed above satisfy these conditions on every finite interval. No assertion of existence of a singular member is part of the definition of this set.

In the original real polynomial coordinates $x = (x_1, x_2, x_3) = (x, y, w)$, retain every coefficient of F and put

$$S(x) = (F_1(x)/2, F_2(x), F_3(x))^{\mathsf{T}}, \quad J(x) = DS(x) = \operatorname{diag}(1/2, 1, 1)DF(x). \tag{142}$$

The earlier determinant proof gives $\det J = -1$, with the exact inverse

$$J^{-1} = (DF)^{-1} \operatorname{diag}(2, 1, 1) = [2W_1\ W_2\ W_3]. \tag{143}$$

Thus $S_1 = \sigma$ is precisely the earlier arithmetic coordinate. The other two coordinates retain the entire original target. No global injectivity of S or F is used; J^{-1} is a pointwise matrix inverse. Both J and J^{-1} have polynomial entries. The first-coordinate observation alone has the exact velocity kernel $\ker(d\sigma)_x = \operatorname{span}_{\mathbb{R}}\{W_2(x), W_3(x)\}$. Indeed $d\sigma(W_i) = \delta_{1i}/2$ and the W_i form a basis, by the full inverse matrix. The retained two additional target coordinates recover precisely those two directions in (143).

16.2 Flow, inverse and volume form on every preterminal interval

For an input of (141), define $X_t(q)$ by

$$\partial_t X_t(q) = u(t, X_t(q)), \qquad X_0(q) = q.$$

Here $q \in \mathbb{R}^3$ is the retained initial label and $x = X_t(q)$ the physical coordinate. This is a global smooth diffeomorphism for each $t < T$. To prove existence with its actual domain, fix $T_0 < T$. On $[0, T_0] \times K$, u and every spatial derivative are bounded by compactness; they vanish outside K. In particular u is globally Lipschitz in space, uniformly on this time interval. On a sufficiently short time interval, the map $Y \mapsto q + \int_0^t u(s, Y(s))\, ds$ is a contraction in the uniform norm on continuous curves, because its Lipschitz constant is at most the interval length times $\sup |Du|$. Successive contractions cover $[0, T_0]$; bounded u prevents escape in each such interval. The same argument backward in time gives the inverse flow. Differentiation of the integral equation gives successive linear integral equations for each label derivative, proving their existence and continuity by the same local iteration, and proves smoothness by induction. Since T_0 is arbitrary this proves all the assertions on $[0, T)$.

Set $A = D_q X_t$. Differentiation gives

$$A' = Du(t, X_t)A, \qquad A(0) = I_3, \qquad (\det A)' = (\operatorname{div} u)(t, X_t) \det A = 0.$$

The determinant identity follows first near $t = 0$ by the product rule for the determinant and A^{-1}; it gives $\det A = 1$, which prevents loss of invertibility and continues that identity to every $t < T$. Thus $\det A = 1$ with its original positive orientation. Every point outside K has the constant trajectory, by uniqueness. If a trajectory starting in K ended outside K, the backward constant trajectory from its endpoint would contradict the inverse-flow uniqueness. Consequently

$$X_t(K) = K, \qquad X_t(q) = q\ (q \notin K), \qquad X_t(L) \subset K \cup L \quad \text{for every compact } L \subset \mathbb{R}^3. \tag{144}$$

The inverse equality follows in the same way. These conclusions require no bound on $\sup_{t<T} |u|$.

16.3 An invertible profile correspondence with a complex spectral coordinate

Write $\mathcal{B} = C^\infty(\mathbb{R}^3; \mathcal{O}(\mathbb{C}_\zeta))$, with seminorms given by uniform suprema of finitely many real q derivatives and complex ζ derivatives on compact products. Let

$$s_{j,t}(q) = S_j(X_t(q)), \qquad \mathcal{P}_t : \mathcal{O}(\mathbb{C})^3 \longrightarrow \mathcal{B}^3, \quad (h_j)_j \longmapsto \big(h_j(s_{j,t}(q)+\zeta)\big)_j.$$

Every $s_{j,t}$ is real. The domain variables of h_j are complex; the spatial labels remain real, since a general smooth NS velocity need not have a holomorphic extension in space. The map on the full product is explicitly

$$H_{j,t} : \mathbb{R}^3 \times \mathbb{C}_\zeta \longrightarrow \mathbb{R}^3 \times \mathbb{C}_s, \quad (q,\zeta) \mapsto (q, s_{j,t}(q)+\zeta), \qquad H_{j,t}^{-1}(q,s) = (q, s - s_{j,t}(q)). \tag{145}$$

These are smooth in the real variables and biholomorphic on every complex fibre. For any fixed label q_*, an exact left inverse of $\mathcal{P}_t$, and inverse on its image, is

$$(\mathcal{R}_t G)_j(s) = G_j(q_*, s - s_{j,t}(q_*)), \qquad \mathcal{R}_t \mathcal{P}_t = I. \tag{146}$$

Substitution proves both directions on the image. On each compact product the image argument stays in a compact subset of $\mathbb{C}$, and all the finitely many derivatives of $s_{j,t}$ involved are bounded. The chain and product rules prove continuity of $\mathcal{P}_t$; evaluation and translation prove continuity of $\mathcal{R}_t$. Thus the image carries a topology for which the stated maps are an actual topological isomorphism, namely its subspace topology in $\mathcal{B}^3$. The map retains the full original complex profile.

For comparison with the earlier spatial observable, let $e_0(q) = (q, 0)$. The precise restriction is

$$e_0^*(\mathcal{P}_t h)_1(q) = h_1(\sigma(X_t(q))) = X_t^*(\sigma^* h_1)(q).$$

The inverse in (146) uses the complex fibre; no holomorphic spatial flow is silently supplied to this real restriction.

16.4 Full first and second time generators, and Euclidean diffusion

Let $\vartheta : [0,T) \to \mathbb{R}$ be smooth. It is the chosen Newman-time map, independent of physical t; no value for it is inferred from NS. Take either $h_\theta = Z_\theta$ or the full four-term coefficient $h_\theta = K_\theta$ of Section 11, so that $\partial_\theta h_\theta = -\partial_s^2 h_\theta/4$ with its original factor and sign. Set

$$\Psi_j(t,q,\zeta) = h_{\vartheta(t)}(s_{j,t}(q)+\zeta), \qquad v_j(t,q) = [\nabla S_j \cdot u](t, X_t(q)).$$

Then $\partial_t s_{j,t} = v_j$ and the chain rule proves the exact equation

$$\partial_t \Psi_j = -\frac{\vartheta'}{4} \partial_\zeta^2 \Psi_j + v_j \partial_\zeta \Psi_j. \tag{147}$$

The exact connection on the moving image of $\mathcal{P}_t$ is $\nabla_t = \partial_t - \operatorname{diag}(v_j)\partial_\zeta$: $\nabla_t(\mathcal{P}_t h_t) = \mathcal{P}_t(\partial_t h_t)$. This follows by the same chain rule for every smooth curve of entire profiles. It is a connection between the displayed moving images; multiplication by $v_j(q)$ is not being asserted to preserve a fixed image of the profile space. There is no ζ dependence in v_j. Differentiating this complete equation once more, including the time derivatives of its coefficients, gives

$$\partial_t^2 \Psi_j = \frac{(\vartheta')^2}{16} \partial_\zeta^4 \Psi_j - \frac{\vartheta' v_j}{2} \partial_\zeta^3 \Psi_j + \left(v_j^2 - \frac{\vartheta''}{4}\right) \partial_\zeta^2 \Psi_j + a_j \partial_\zeta \Psi_j, \tag{148}$$

where the acceleration coefficient is, with all NS terms retained,

$$a_j = \partial_t v_j = \left[\sum_i (\nu \Delta u_i - \partial_i p + f_i)\partial_i S_j + \sum_{i,l} u_i u_l \partial_i \partial_l S_j\right](t, X_t(q)). \tag{149}$$

Indeed apply $\partial_t + u \cdot \nabla_x$ to $\sum_i u_i \partial_i S_j$ and substitute (141); the derivative of $\partial_i S_j$ is exactly the displayed Hessian term. In (148), the cubic derivative occurs twice, each with coefficient $-\vartheta' v_j/4$, proving its coefficient $-\vartheta' v_j/2$.

The full physical Laplacian has an exact material counterpart. Define

$$G_t = A^{-1}A^{-\mathsf{T}}, \qquad \mathcal{L}_t = \sum_{a,b} \partial_{q_a}(G_t^{ab}\partial_{q_b}).$$

For every smooth scalar $b(x)$,

$$\mathcal{L}_t(X_t^* b) = X_t^*(\Delta_x b). \tag{150}$$

Here is a proof retaining the first-order terms in $\mathcal{L}_t$. The gradient chain rule gives $\nabla_q X_t^* b = A^{\mathsf{T}}(\nabla_x b) \circ X_t$. For a compactly supported smooth test ϕ, integration by parts and the change of variables $x = X_t(q)$, whose Jacobian is one, give

$$\int \phi\, \mathcal{L}_t X_t^* b\, dq = -\int \nabla_q \phi \cdot A^{-1}(\nabla_x b) \circ X_t\, dq = -\int \nabla_x(\phi \circ X_t^{-1}) \cdot \nabla_x b\, dx = \int \phi\, X_t^* \Delta_x b\, dq.$$

All functions are smooth, so the distributional equality is pointwise. Apply this at fixed ζ to the full profile to obtain

$$\mathcal{L}_t \Psi_j = (|\nabla S_j|^2 \circ X_t)\partial_\zeta^2 \Psi_j + (\Delta S_j \circ X_t)\partial_\zeta \Psi_j. \tag{151}$$

Consequently its complete advective viscous residual is

$$(\partial_t - \nu\mathcal{L}_t)\Psi_j = -\left(\frac{\vartheta'}{4} + \nu|\nabla S_j|^2 \circ X_t\right)\partial_\zeta^2 \Psi_j + (v_j - \nu\Delta S_j \circ X_t)\partial_\zeta \Psi_j.$$

These formulas identify the two differential operators through an explicit coordinate map and calculate their remaining coefficients; they do not delete the viscosity or equate two time variables.

16.5 Recovering the actual velocity from the arithmetic temporal defect

For real θ the functions Z_θ, K_θ and their spectral derivatives are Schwartz on the real s line, by the weighted Fourier space proved above. With the original real kernel k and multiplier m, their Fourier representations are

$$Z_\theta(s) = 4\int_{\mathbb{R}} k(\omega)e^{\theta\omega^2/4}e^{is\omega}\, d\omega, \qquad K_\theta(s) = 4\int_{\mathbb{R}} m(\omega)k(\omega)e^{\theta\omega^2/4}e^{is\omega}\, d\omega.$$

No source energy, derivative channel, or Fourier factor is changed. Define $d_h(\theta)^2 = \int_{\mathbb{R}} |h_\theta'(s)|^2\, ds$. Then

$$\begin{aligned} d_Z(\theta)^2 &= 32\pi \int_{\mathbb{R}} \omega^2 k(\omega)^2 e^{\theta\omega^2/2}\, d\omega, \\ d_K(\theta)^2 &= 32\pi \int_{\mathbb{R}} \omega^2 |m(\omega)|^2 k(\omega)^2 e^{\theta\omega^2/2}\, d\omega. \end{aligned} \tag{152}$$

For completeness, the factor follows without a Fourier convention change. For a Schwartz amplitude b, insert $e^{-\epsilon s^2}$ in the squared norm of $4\int b(\omega)e^{is\omega}d\omega$ and use Fubini. The result is $16\sqrt{\pi/\epsilon}\iint b(\omega)\overline{b(\eta)}e^{-(\omega-\eta)^2/(4\epsilon)}d\omega d\eta$. The Gaussian convolution, with mass 2π, tends to $2\pi b$; splitting at any fixed small distance and using continuity and the Gaussian tail proves this convergence under the integral. Thus the limit is $32\pi\int |b|^2$. On the original side dominated convergence applies because the transform is Schwartz. Use respectively $b = i\omega k e^{\theta\omega^2/4}$ and $b = i\omega m k e^{\theta\omega^2/4}$ to prove (152).

Both numbers are strictly positive. The previously proved $k(0) > 0$ and continuity give an interval of nonzero k containing nonzero ω. For the second integral the established full-real bound $|m| \geq 575/1752192$ applies. The weighted bounds also prove continuity of these integrals in θ by domination on each compact parameter interval. Hence for each compact interval $I \subset \mathbb{R}$ there are finite constants $0 < d_- \leq d_h(\theta) \leq d_+ < \infty$ on I.

Define the actual arithmetic temporal defect by

$$D_j(t, q, s) = \partial_t \Psi_j(t, q, s) + \frac{\vartheta'}{4} \partial_s^2 \Psi_j(t, q, s), \qquad s \in \mathbb{R}. \tag{153}$$

Equation (147) gives $D_j = v_j h'_\vartheta(s + s_{j,t})$. Real translation preserves Lebesgue measure, so

$$\sum_j \|D_j(t, q, \cdot)\|^2_{L^2(\mathbb{R})} = d_h(\vartheta(t))^2 |v(t, q)|^2. \tag{154}$$

Moreover the inverse is a literal integral followed by the original matrix inverse:

$$\begin{aligned} v_j &= \frac{\int_{\mathbb{R}} \overline{h'_\vartheta(s + s_{j,t})} D_j(t, q, s)\, ds}{d_h(\vartheta)^2}, \\ u(t, X_t(q)) &= 2W_1(X_t(q))v_1 + W_2(X_t(q))v_2 + W_3(X_t(q))v_3. \end{aligned} \tag{155}$$

Substitution proves both inverse identities. The correspondence retains $X_t(q)$ as part of both sides: $(X_t, u \circ X_t) \leftrightarrow (X_t, D)$. The data D alone are not claimed to select a globally unique inverse sheet of the polynomial map. At fixed (t, q), the formula identifies $\mathbb{R}^3$ with the three-dimensional real subspace of $L^2(\mathbb{R}; \mathbb{C})^3$ spanned componentwise by the three displayed shifted profiles. It is not merely an analogy between their norms.

For a nonempty K, set $M = \max_K \|J\|_{\mathrm{op}}$ and $N = \max_K \|J^{-1}\|_{\mathrm{op}}$. They are finite and positive. If $\vartheta([0, T)) \subset I$, the exact map satisfies, for all $t < T$,

$$\frac{d_-}{N} \|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq \sup_q \left(\sum_j \|D_j(t, q, \cdot)\|_2^2 \right)^{1/2} \leq d_+ M \|u(t)\|_{L^\infty(\mathbb{R}^3)}. \tag{156}$$

Indeed $|Ju| \leq M|u|$ and $|u| \leq N|Ju|$ on K, while both u and v vanish off K; (144) preserves K and surjectivity preserves the supremum. Combine this with (154). This proves equivalence of unbounded terminal velocity and unbounded supremum of this specific arithmetic temporal-defect norm for every input in the defined class. It supplies an exact detection map even though no singular member is established here. At $u = 0$ all these defects vanish, as the same formulas require. The full space–spectral energy has an exact identity too. Since $v = 0$ off K, (154) is integrable in q and the volume-preserving change of variables gives

$$\sum_j \int_{\mathbb{R}^3} \int_{\mathbb{R}} |D_j(t, q, s)|^2\, ds\, dq = d_h(\vartheta(t))^2 \int_{\mathbb{R}^3} |J(x)u(t, x)|^2\, dx.$$

The same constants therefore bound its square root between $(d_-/N)\|u(t)\|_2$ and $d_+ M \|u(t)\|_2$. In particular the map transfers a uniform kinetic-energy bound into a uniform joint $L^2(q, s)$ bound, while (156) detects concentration through the separate supremum in q. This conclusion follows from the displayed integrals without a spatial truncation of the defect. In particular $\vartheta(t) = 0$ for every physical time is an explicit allowed choice. Then the entire input stays equal to the actual Ξ (or K_0), and $D_j = \partial_t \Psi_j$; the constants in (156) use the single strictly positive value $d_h(0)$.

For an actual separated-link pair, the full coefficient is $a_{ef} K$. The defect is then $a_{ef} D^K$, and its squared norm acquires exactly a_{ef}^2; division in (155) is available for $a_{ef} \neq 0$. If $a_{ef} = 0$ this pair supplies the zero observation, not a spurious inverse. For a pair with $a_{ef} > 0$, recovery of Z from the profiles of K additionally uses the already proved full-domain inverse $\mathcal{T}_{\mathcal{D}}^{-1}$; its translation and derivative commutation hold because multiplication by m commutes with $e^{is_{j,t}\omega}$ and $i\omega$ in the unchanged Fourier domain.

16.6 Bounded profile values and the exact transported zero divisor

For compact label set L, compact spectral set $E \subset \mathbb{C}$, and bounded $\vartheta([0,T)) \subset I$, (144) puts every argument $s_{j,t}(q) + \zeta$ in the compact set $S_j(K \cup L) + E$. The jointly entire Z and K are bounded on that set times I. Therefore

$$\sup_{t<T,\ q\in L,\ \zeta\in E} |\Psi_j(t, q, \zeta)| < \infty.$$

The identical argument bounds every pure spectral derivative. It does not bound time or spatial-label derivatives; (156) specifies a quantity that can grow even while these values stay bounded.

Let $h_\theta(s)$ have a zero at z_* of multiplicity m and write its actual local factorization $h_\theta(s) = (s - z_*)^m A(s)$, $A(z_*) \neq 0$. The full inverse of (145) takes that zero to

$$\zeta_*(q) = z_* - s_{j,t}(q), \qquad \Psi_j = (\zeta - \zeta_*(q))^m A(s_{j,t}(q) + \zeta).$$

Thus it preserves multiplicity and the whole analytic unit on every complex fibre. Its complete jet formula is

$$\partial_\zeta^{m+n} \Psi_j(t, q, \zeta_*(q)) = \frac{(m+n)!}{n!} A^{(n)}(z_*), \qquad n \geq 0,$$

with all lower derivatives zero. This follows by multiplying the Taylor series of the unchanged A by $(\zeta - \zeta_*)^m$; no unit coefficient has been discarded. Since $s_{j,t}(q)$ is real, $\operatorname{Im}\zeta_*(q) = \operatorname{Im} z_*$. At Newman time $\vartheta = 0$ with $h = Z$, the full map to the completed-zeta argument is

$$\rho = \tfrac{1}{2} + i(s_{j,t}(q) + \zeta), \qquad \zeta = -i(\rho - \tfrac{1}{2}) - s_{j,t}(q), \qquad \operatorname{Re}\rho = \tfrac{1}{2} - \operatorname{Im}\zeta.$$

Consequently this NS-induced real translation preserves each zero's distance from the critical line exactly; the full arithmetic time evolution is still the specified Z_ϑ. For $h = K$ the same coordinate proof transports the zeros of the four-term profile; these are converted to a statement about Z only by actually applying the proved inverse operator to the whole profile. The inverse does not identify individual zeros of two different entire functions. These explicit maps give both the velocity detection and the precise zero transport of the construction, without deducing an off-critical zero from growth of the temporal defect.

17 Provenance and reproducibility

The Newman convention and its public source are Rodgers–Tao, [1, Section 1, equations (1)–(4)]. The polynomial is Tao's displayed map [3, equation (1)]; its coordinate identities used here are reproved above, so no hidden local reference is required to check them. The actual topological Connes quotient is described in [2, Section 3.6, Proposition 3.6]. Section 5.1 proves the coordinate-level Mellin/Fourier map, the common-representative quotient correspondence with its complete kernels, and its heat and multiplication intertwiners, retaining both original topologies.

The readable proofs establish the analytic assertions. The scripts `scripts/check_heat_connection.py` and `scripts/zero_transport_check.py`, together with `scripts/check_euclidean_observable.py`, replay exact algebraic identities using rational polynomial arithmetic. Their JSON receipts state precisely which identities were checked. They are not Lean certificates or certificates of an off-critical zero. No Lean, Lake, or Elan run was used. The work was prepared with Codex and independent agent checks; the cited human literature remains the scholarly source. The fourth-order transfer is also checked by `scripts/check_xi4_transport.py`. The explicit physical, temporal, spatial and zero-jet morphisms are also checked by `scripts/check_xi4_typed_maps.py`. Appendix A contains the complete supplied spectral-coefficient proof; the original Markdown and its independent verification are preserved under `sources/xi4_source/`. The coupling notation there is ξ^4. The full arithmetic quotient continuation is preserved byte-for-byte under `sources/connes_continuation/`; its full proof is included in Appendices B–I. Its `checks/verify_exact.py` supplies 206 supplementary exact checks; the replay receipt is `checks/connes_continuation_replay.json`. Section 14 proves the common-space identification and its simultaneous moving quotient intertwiners. The complete incoming Cartesian bilaplacian proof is preserved in `sources/ns_bilaplacian/` and included in Section 13. Its fourth-coefficient continuation in Sections 13.4–13.6 proves the full material intertwiner, residue map and negative-time asymptotic; `scripts/check_xi4_bilaplacian.py` verifies the exact algebra. Section 15 retains the three actual marked special points of the supplied period construction and proves its real circle quotient, including the full logarithmic overlaps. Sections 15.5–15.8 give the resulting three-dimensional force, pressure, all mixed interactions, the exact finite-mode classification and every spatial localization term. The source provenance is retained under `sources/s6_three_points/`; the independent algebra replays are `scripts/check_s6_three_point_forcing.py` and `scripts/check_s6_three_point_fields.py`. The exact supplied-flow arithmetic correspondence is proved in Section 16; its source versions are identified in `sources/ns_public/provenance.json`, and its independent algebra checks are in `scripts/check_ns_arithmetic_flow.py`.

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earlier version has 165 pages. Both versions are identified by hashes in the accompanying provenance file; their full equivalence is not asserted here.

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A Complete source proof of the separated-link fourth coefficient

8 September 2026. Two links which share no elementary face nevertheless develop a nonzero connected electric covariance through two adjacent faces. This note calculates that coefficient in the actual full-box vacuum. The calculation keeps both intermediate spin channels and the disconnected expectation subtraction. It does not assume a volume-uniform remainder.

A.1 Original Hilbert space and the coefficient to be calculated

Fix $L \geq 2$, vertices $\{-L, \ldots, L\}^3$, all positive contained links e, and all contained elementary faces p. Write their complete counts as $N = 3(2L)(2L+1)^2$ and $M = 3(2L)^2(2L+1)$. Link variables are $U_e \in SU(2)$, with the inverse on reverse traversal. The face trace is

$$W_{(n;i,j)} = \mathrm{tr}\{U_i(n)U_j(n+\mathbf{e}_i)U_i(n+\mathbf{e}_j)^{-1}U_j(n)^{-1}\}, \qquad i<j.$$

On product Haar probability space use $T_a = -i\sigma_a/2$, left derivatives $X_{e,a}$, $E_e = -\sum_a X_{e,a}^2$, and

$$H = \kappa\sum_e E_e + b\sum_p(2-W_p), \quad \kappa = 2g^2/a, \quad b = 1/(2g^2a), \quad \xi = b/\kappa = 1/(4g^4). \tag{S1}$$

The fundamental Casimir is 3/4. In the original metric $-\mathrm{tr}(AB)/2$, the Laplacian is four times the sum-of-squares Laplacian in these generators, so the electric coefficient in that metric is $\kappa/4$. Neither this factor nor the Wilson scalar $2bM$ is changed.

The compact connected configuration manifold and bounded smooth potential give a self-adjoint compact-resolvent operator of domain H^2, form domain H^1, and unique smooth positive unit ground vector ψ_ξ. For precision, the modulus inequality supplies a nonnegative form minimizer, elliptic regularity and the maximum principle make it positive, and the identity $q_{H-\mathcal{E}}(\psi f) = \kappa\sum_{e,a}\int\psi^2|X_{e,a}f|^2$ implies uniqueness by division by ψ. Gauge transformations preserve this vector and every link Casimir. All following vectors are thus physical.

Set $r_e = \#\{p : e \in \partial p\}$, $\gamma_e = \langle\psi, E_e\psi\rangle$, $v_e = (E_e - \gamma_e)\psi$, and

$$C_{ef} = \langle v_e, v_f\rangle = \langle\psi, E_eE_f\psi\rangle - \gamma_e\gamma_f. \tag{S2}$$

The operators commute for different links. Smoothness makes (S2) legitimate as an operator identity on the vacuum, not only a formal product of forms.

For distinct e,f that share no face, define the integer

$$a_{ef} = \#\{(p,q) : e \in \partial p,\ f \in \partial q, |\partial p \cap \partial q| = 1\}. \tag{S3}$$

The pair in (S3) is ordered by which face contains e and which contains f; there is no double counting because no face contains both links. All faces are those of the original open box. The result is

$$\boxed{C_{ef} = \frac{127}{219024}\,a_{ef}\xi^4 + O_L(\xi^6) \quad (e,f \text{ share no face}).} \tag{S4}$$

In particular the coefficient is positive whenever a two-face channel exists, even though the energy matrix between these links is exactly zero. The latter assertion is proved in Appendix A.4.

A.2 The full second vacuum coefficient, including both shared-link channels

Put $H_0 = \sum E_e$ and $S = \sum W_p$, so $H = \kappa(H_0 - \xi S) + 2bM$. On $|z| = 3/8$ in the full Hilbert space the resolvent of H_0 has norm at most $8/3$, since its first nonzero eigenvalue is $3/4$. Also $\|S\| = 2M$. Therefore the bounded perturbation Neumann series converges for $|\xi| < 3/(16M)$, gives the analytic rank-one bottom projection, and gives an analytic positive unit eigenvector for real ξ near zero. Elliptic regularity applied repeatedly to the eigen-equation upgrades this to analyticity in each fixed Sobolev space. Its radius and estimates may depend on L. This suffices for the finite derivative calculations below.

Product Haar integration gives $\int W_p = 0$, $\langle W_p, W_q\rangle = \delta_{pq}$, and $H_0 W_p = 3W_p$. Indeed a link in only one of two distinct face boundaries has odd central parity; for the same face integrate its Haar holonomy and use the fundamental character squared norm one. Comparing coefficients in the actual equation $(H_0 - \xi S)\psi = e(\xi)\psi$, including its unit-vector condition, gives

$$\psi = 1 + \xi A_1 + \xi^2 A_2 + O_{H^k,L}(\xi^3), \quad A_1 = S/3, \quad e(\xi) = -M\xi^2/3 + O_L(\xi^3),$$
$$H_0 A_2 = (S^2 - M)/3, \qquad \int A_2 = -M/18. \tag{S5}$$

The inverse in (S5) can be resolved entirely. For adjacent distinct faces p,q sharing g put

$$Z_{pq,0} = \int W_p W_q \, dU_g, \qquad Z_{pq,1} = W_p W_q - Z_{pq,0}.$$

After reversing one real trace if needed, their words are $\mathrm{tr}(U_g B)$, $\mathrm{tr}(U_g^{-1} D)$, with six distinct outer links. The fundamental Haar identity gives $Z_{pq,0} = \mathrm{tr}(BD)/2$. The shared-edge spin tensor $1/2 \otimes 1/2$ consists of spins zero and one, while all outer links have spin $1/2$. Thus

$$H_0 Z_{pq,0} = \tfrac{9}{2} Z_{pq,0}, \quad H_0 Z_{pq,1} = \tfrac{13}{2} Z_{pq,1}, \quad \|Z_{pq,0}\|^2 = \tfrac{1}{4}, \quad \|Z_{pq,1}\|^2 = \tfrac{3}{4}. \tag{S6}$$

The norms use orthogonal Haar projection on g and total product norm one. A repeated face has $W_p^2 - 1 = \chi_1(U_p)$ with free energy 8 and norm one. An edge-disjoint pair has energy 6 and norm one, even when the faces meet at a vertex. Consequently the full coefficient is

$$\boxed{\begin{aligned} A_2 = & -\frac{M}{18} + \sum_p \frac{W_p^2 - 1}{24} \\ & + \sum_{\{p,q\}\ \text{edge-disjoint}} \frac{W_p W_q}{9} \\ & + \sum_{\{p,q\}\ \text{adjacent}} \left(\frac{4}{27} Z_{pq,0} + \frac{4}{39} Z_{pq,1}\right). \end{aligned}} \tag{S7}$$

All distinct-pair sums in (S7) are unordered; the factor two in S^2 has already been included. No finite-face subsystem is substituted for H.

Different unordered pairs in (S7) have different odd-link parity supports. If their symmetric boundaries agreed, their symmetric face difference would be a nonempty closed mod-two collection of at most four squares. A face in that collection needs another face on each of its four boundary edges; two distinct elementary squares share at most one edge, so at least four other squares would be needed, a contradiction. Haar integration on a link where parity differs makes the cross term zero. Repeated-face terms are even everywhere, and different repeated-face terms have different joint link spins. Thus all the displayed components, with their two spin channels, are mutually orthogonal. This remains true after application of any link Casimir, which preserves those parity and spin subspaces.

A.3 Exact connected fourth coefficient

Suppose e,f share no face. Then $E_eE_f1 = E_eE_fA_1 = 0$: each term in A_1 is a single face and depends on at most one of these links. Self-adjointness moves E_eE_f to either of these vectors, so every term of degree less than four in $\langle\psi, E_eE_f\psi\rangle$ vanishes, and its degree-four term is exactly

$$\langle A_2, E_eE_fA_2\rangle. \tag{S8}$$

In particular no third or fourth unknown coefficient of the vacuum is being omitted from (S8): its possible pairing with 1 or A_1 is identically zero for the stated reason. Similarly

$$\gamma_e = \xi^2\langle A_1, E_eA_1\rangle + O_L(\xi^3) = \frac{r_e}{12}\xi^2 + O_L(\xi^3). \tag{S9}$$

Only a pair p,q with $e \in p$, $f \in q$ contributes to (S8). Repeated faces contribute zero. There are exactly r_er_f such distinct unordered pairs when they are labeled as in (S3). For an edge-disjoint pair, both e and f carry fundamental spin, so its contribution is $(3/4)^2/9^2 = 1/144$. For an adjacent pair, neither e nor f can be its shared edge: otherwise the face containing the other link would contain both e and f. They are therefore both outer fundamental edges in each of the two channels (S6), and the contribution is

$$\left(\frac{3}{4}\right)^2\left[\left(\frac{4}{27}\right)^2\frac{1}{4} + \left(\frac{4}{39}\right)^2\frac{3}{4}\right] = \frac{1}{324} + \frac{3}{676}. \tag{S10}$$

Equation (S9) subtracts $r_er_f/144$ at degree four in (S2). Hence every edge-disjoint pair cancels its disconnected contribution exactly; every adjacent pair leaves

$$\frac{1}{324} + \frac{3}{676} - \frac{1}{144} = \frac{127}{219024} > 0. \tag{S11}$$

This proves the coefficient in (S4).

For its stated even error, the center transformation $U_{(n,i)} \mapsto (-1)^{\sum_{j<i} n_j} U_{(n,i)}$ changes every face trace sign, commutes with every E_e and preserves Haar measure. It intertwines $H_0 - \xi S$ with $H_0 + \xi S$ and hence the positive unit vacua at ξ and$-\xi$. Both terms in (S2) are consequently even analytic functions of ξ near zero. Their next possible order after four is six. This proves the error in (S4); it does not supply a volume-uniform analytic radius.

A.4 Exact support map and relation to the energy matrix

Define the edge adjacency matrix B by $B_{eg} = 1$ when e,g are distinct physical links belonging to a common elementary face, and zero otherwise. Two distinct links belong to at most one elementary square: perpendicular incident links determine its two plane directions and base, while parallel links in a square must be opposite sides and also determine that square. Thus, for the e,f considered above,

$$\boxed{a_{ef} = (B^2)_{ef}.} \tag{S12}$$

Indeed a pair p,q counted in (S3) has a unique shared edge g, which is a common neighbor of e,f. Conversely a common neighbor g determines the unique p containing e,g and q containing g,f; these are distinct because e,f share no face, and their unique shared edge is g. These two maps are inverse. The covariance coefficient thus has support at graph distance two, including boundary incidences, and no support beyond distance two at this order. It makes no all-orders finite-range claim about C.

There are actual such channels in every $L \geq 2$. Take $e = ((0,0,0),1)$ and $f = ((0,2,0),1)$. They share no face, but the faces in directions $1,2$ based at $(0,0,0)$ and $(0,1,0)$ share the edge $((0,1,0),1)$, giving $a_{ef} \geq 1$. All of their vertices are in the original box, including at L=2.

Put $\mathcal{N}_{ef} = \langle v_e, (H - \mathcal{E})v_f\rangle$. Commuting the link operators on the smooth real vacuum gives

$$\mathcal{N}_{ef} = \tfrac{1}{2}\langle \psi, [E_e, [H, E_f]]\psi\rangle. \tag{S13}$$

To check (S13), expand the double commutator, use $H\psi = \mathcal{E}\psi$, commute E_e,E_f, and use the real symmetry of their two matrix elements. Only potential faces containing f occur in $[H, E_f]$. If e shares no face with f, E_e commutes with every one of their multiplication and derivative terms. Thus

$$\boxed{\mathcal{N}_{ef} = 0 \text{ exactly, at every positive coupling, when e,f share no face.}} \tag{S14}$$

There is no contradiction between (S4) and (S14): a matrix-valued spectral measure can have nonzero zeroth moment and zero first moment off diagonal. Explicitly, for the spectral resolution of A=H−E on the physical space,

$$d\Sigma_{ef}(\omega) = \langle v_e, \mathbf{1}_{d\omega}(A)v_f\rangle, \quad C_{ef} = \int d\Sigma_{ef}, \quad \mathcal{N}_{ef} = \int \omega\, d\Sigma_{ef}. \tag{S15}$$

The measure matrix is positive semidefinite on every Borel set because its quadratic form is the squared norm of a projected linear combination of the v_e. An individual off-diagonal entry, however, need not be a nonnegative measure; (S15) retains its signs. Smoothness gives the finite moments used here. These identities are the exact relation between the nonlocal covariance contribution and the local energy matrix, not an assumption that one determines the other's support.

Finally $\xi^4 = 1/(256g^{16})$. The physical coefficient in (S4) is therefore $127a_{ef}/(56070144g^{16})$; C has no extra energy factor, whereas $\mathcal{N}$ has one. The scalar energy shift remains in H and cancels only in H−E. The fixed-box coefficient is a concrete next input to the extensive denominator calculation; control of all higher orders and summability at fixed coupling is not inferred from its finite range at order four.

A.5 The complete fourth-order spectral functional and time correlation

The same calculation can resolve the energy dependence, rather than only its zeroth moment. Put $A_\xi = (H - \mathcal{E})/\kappa$, and write

$$F_{ef,\xi}(h) = \langle v_e, h(A_\xi)v_f\rangle.$$

For each fixed box, for distinct links sharing no face, and every bounded continuous real or complex function h which is differentiable at 3,

$$\boxed{\begin{aligned}\lim_{\xi\to 0} \xi^{-4} F_{ef,\xi}(h) = a_{ef}\Big[&-\frac{59}{275184}h(3) - \frac{1}{336}h'(3) \\ &+ \frac{1}{1296}h(9/2) + \frac{3}{132496}h(13/2)\Big].\end{aligned}} \tag{S16}$$

Here the limit is a functional on the specified test functions. In particular the derivative term must not be omitted or represented as an ordinary positive spectral atom. Each original matrix-valued spectral measure is still the exact positive-semidefinite measure (S15).

We prove (S16), including the relation to the actual first interacting spectral subspace. Let P project onto the free physical energy-three space $\mathcal{P} = \operatorname{span}\{W_p\}$, and let

$$R_3 = (I - P)((H_0 - 3)|_{\mathcal{P}^\perp})^{-1}(I - P).$$

The inverse is on the physical space: its constant channel is -1/3. To verify isolation, a nonzero physical spin network has no degree-one support vertex. Its support contains a cycle of length at least four in the bipartite cubic graph, costing at least 3. With fewer than six nontrivial edges it must be one elementary square: a fifth edge would have a new degree-one endpoint, or be a forbidden same-bipartition chord of the four-cycle. The degree-two invariant contractions require equal spins on that square. Only spin 1/2 has energy below 9/2. A six-edge rectangular loop realizes energy 9/2. Thus the first physical free band is exactly P, with the next energy 9/2.

For sufficiently small real ξ, the contour $|z-3| = 3/4$ defines the actual spectral projection P_ξ, and the exact isometry

$$V_\xi = P_\xi P (P P_\xi P|_{\mathcal{P}})^{-1/2} : \mathcal{P} \longrightarrow \operatorname{ran} P_\xi \tag{S17}$$

uses its full Gram factor. Indeed the free contour resolvent has norm at most 4/3. For $|\xi| \leq 3/(64M)$, its Neumann factor is at most 1/8, and integrating the resolvent difference gives $\|P_\xi - P\| \leq 1/7$. The displayed inverse square root therefore exists by its binomial series; idempotence and orthogonality of the real projection show $V_\xi^* V_\xi = I$. Its range is smooth, and the finite-rank resolvent formula and elliptic regularity justify all operator products and fixed Sobolev derivatives below. It is orthogonal to the actual vacuum for a smaller interval, for example $|\xi| \leq 3/(128M)$, by the free gaps and the bounded perturbation estimate $2M|\xi|$.

Let Q be the unchanged open-face signless adjacency matrix, with diagonal the actual face degrees and off-diagonal 1 for shared-edge faces. Direct virtual-channel calculation gives

$$V_\xi^* A_\xi V_\xi = 3I + \xi^2 F + O_L(\xi^4), \qquad F = \frac{7}{15} I - \frac{1}{21} Q. \tag{S18}$$

Here is the coefficient calculation rather than an assumption of (S18). Differentiating (S17) gives $V_0' = R_3 S P$. The second matrix coefficient before vacuum subtraction is $-PSR_3SP$. The vacuum channel gives +1/3 to every entry; a repeated-face channel gives -1/5 to its diagonal; an edge-disjoint pair gives -1/3 to its two diagonals and mutual entries; an adjacent pair gives $-(1/4)/(9/2-3) - (3/4)/(13/2-3) = -8/21$ to that same four-entry pattern. Thus the diagonal is $7/15 - M/3 - d_p/21$, and the off-diagonal is -1/21 exactly for adjacent faces. The actual vacuum coefficient -M/3 from (S5) gives (S18). The central cochain used in Appendix A.3 acts as -I on P; it makes the matrix in (S18) even and accounts for the fourth-order remainder.

The electric vector can be treated with arbitrary fixed real link weights w, retaining its linear dependence on all of them. Define $\Gamma_w = \sum w_e E_e$, $a_p(w) = \sum_{e \in p} w_e/4$, and $v(w) = (\Gamma_w - \langle \Gamma_w \rangle)\psi$. Put

$$y_\xi(w) = V_\xi^* v(w), \qquad \rho_\xi(w) = (I - P_\xi) v(w).$$

These are exact orthogonal components, both orthogonal to the actual vacuum. Coefficient comparison using (S7) and $V_0' = R_3 S P$ gives

$$y_\xi(w) = \xi x(w) + \xi^3 z(w) + O_L(\xi^5), \qquad x(w) = \sum_p a_p(w) W_p,$$

$$\rho_\xi(w) = \xi^2 r(w) + O_{H^k,L}(\xi^3), \tag{S19}$$

where, for $s = a_p(w) + a_q(w)$ and g the shared edge,

$$r(w) = \sum_p \frac{2a_p(w)}{15}(W_p^2 - 1) + \sum_{\{p,q\} \text{ adjacent}} \left[-\frac{2(s + w_g)}{9} Z_{pq,0} + \frac{2(3s + 7w_g)}{273} Z_{pq,1} \right]. \tag{S20}$$

For completeness, Γ_w has eigenvalues $8a_p$ on the repeated face, $3s$ on an edge-disjoint pair, and $3s - 3w_g/2, 3s + w_g/2$ on the adjacent zero and one channels. The shared edge is counted twice as fundamental in 3s; replacing that by its actual spin accounts for precisely the last two corrections. In $\Gamma_w A_2$, the respective coefficients are $a_p/3, s/3, (4s-2w_g)/9, (12s+2w_g)/39$. Subtract the full dressing $R_3 S x(w)$, whose coefficients are $a_p/5, s/3, 2s/3, 2s/7$. The two constant coefficients are both $-\sum_p a_p/3$ and also cancel. This proves (S20) with no omitted channel. The centered projection identity $V_\xi^* \psi_\xi = 0$ is exact. The same central cochain makes $y_\xi(w)$ odd, proving the first expansion (S19).

For a single link e use the original weight $w_g^{(e)} = \delta_{eg}$, and denote the resulting x,z,r by $x_e, z_e, r_e^{\mathrm{vec}}$. This superscript distinguishes the vector from the incidence count r_e. Then $x_e = \sum_{p \ni e} W_p/4$. If e,f share no face, $\langle x_e, x_f\rangle = 0$, and (S18) gives exactly

$$\langle x_e, F x_f\rangle = -a_{ef}/336. \tag{S21}$$

Each ordered adjacent pair contributes $-1/(4\cdot 21\cdot 4)$. In the cross spectral measure of $r_e^{\mathrm{vec}}, r_f^{\mathrm{vec}}$, only the pairs counted by (S3) occur. Repeated-face cross terms vanish. For every contributing pair, e and f are outer links of different faces, $s = 1/4$ for each weight, and $w_g = 0$. Thus their zero-channel coefficients in (S20) are both -1/18, and their one-channel coefficients are both 1/182. Orthogonality and (S6) give the complete free cross measure

$$\langle r_e^{\mathrm{vec}}, h(H_0) r_f^{\mathrm{vec}}\rangle = a_{ef}\left[\frac{1}{1296} h(9/2) + \frac{3}{132496} h(13/2)\right]. \tag{S22}$$

No edge-disjoint channel survives in (S20).

As $A_\xi - H_0 = -\xi S - e(\xi) I$ tends to zero in operator norm, its resolvents converge in norm. For bounded continuous h, this implies convergence of the cross expectations on ρ_ξ/ξ^2 to (S22): first use continuous functions vanishing at infinity and norm resolvent convergence, then approximate h on a compact interval and control the discarded tail by the uniformly bounded second spectral moments. Those moments follow from the H^2 convergence in (S19). Polarization gives the cross statement from the four corresponding diagonal statements. Hence (S22) is the entire fourth-order out-of-band contribution.

The fourth coefficient of the band mass is obtained without knowing the third vacuum coefficient: subtract (S22) at h=1 from (S4). It is

$$\langle x_e, z_f\rangle + \langle z_e, x_f\rangle = a_{ef}\left(\frac{127}{219024} - \frac{1}{1296} - \frac{3}{132496}\right) = -\frac{59}{275184} a_{ef}. \tag{S23}$$

Differentiability of h at 3 and spectral calculus of the finite matrix (S18) give $h(V_\xi^* A_\xi V_\xi) = h(3) I + \xi^2 h'(3) F + o_L(\xi^2)$. Combine this with (S19), (S21), (S23), and the exact orthogonal spectral decomposition into the band and its complement. This proves (S16).

In particular the complete dimensionful ground-state time correlation, with physical $\kappa > 0$ and physical time $t \geq 0$ fixed, is

$$\boxed{\begin{aligned}\langle v_e, e^{-t(H-\mathcal{E})} v_f\rangle = a_{ef}\xi^4 \Bigg[&-\frac{59}{275184} e^{-3\kappa t} + \frac{\kappa t}{336} e^{-3\kappa t} \\ &+ \frac{1}{1296} e^{-9\kappa t/2} + \frac{3}{132496} e^{-13\kappa t/2}\Bigg] + O_{L,t,\kappa}(\xi^6).\end{aligned}} \tag{S24}$$

Here $h(q) = e^{-\kappa t q}$ is used on the nonnegative excitation spectrum, with any bounded continuous extension below zero, so it belongs to the test class of (S16). The bounded potential perturbation gives $e^{-\kappa t K_\xi}$ its norm-convergent Duhamel series: each order is bounded by $(2M|\xi|\kappa t)^n/n!$. The excitation semigroup retains the additional analytic scalar $e^{\kappa t e(\xi)}$. The vacuum and vectors are analytic in fixed Sobolev norms, and the central cochain makes the whole expression even. These facts justify the even remainder in (S24), not only a pointwise little-o limit. For varying physical κ, the same proof uses the explicit parameter $\tau = \kappa t$ and keeps its dependence; no estimate uniform in τ or L is implied.

At t=0 the bracket is 127/219024. Its derivative at t=0 is exactly zero, because

$$3\left(-\frac{59}{275184}\right)-\frac{1}{336}+\frac{9}{2\cdot 1296}+\frac{39}{2\cdot 132496}=0. \tag{S25}$$

This agrees with the exact identity (S14). It exhibits which separated-link spectral contributions cancel in the first moment: the actual band-mass correction, the actual band-energy derivative, and both higher spin channels. The term proportional to t is essential to that cancellation. Equations (S16)–(S25) describe the fixed-box fourth coefficient of the original interacting correlations; they do not exchange a volume limit with the coupling expansion or assert a continuum spectrum.

A.6 Primary context

The original Hamiltonian framework is J. Kogut and L. Susskind, *Hamiltonian formulation of Wilson's lattice gauge theories*, Physical Review D **11** (1975), 395–408, `https://doi.org/10.1103/PhysRevD.11.395`. Full conventions appear in (S1). For historical linked-cluster calculations in the same spatial dimension, see A. C. Irving, T. E. Preece and C. J. Hamer, *Cluster expansion approach to non-abelian lattice gauge theory in (3+1)D (I). SU(2)*, Nuclear Physics B **270** (1986), 536–552, `https://doi.org/10.1016/0550-3213(86)90567-5`. This note derives (S4) directly, with no historical novelty assertion or imported coefficient.

B The primary arithmetic object and the topology actually specified

The starting point is Connes–Consani–Moscovici, *Zeta zeros and prolate wave operators: Semilocal adelic operators*, Section 3.6, proof-PDF pages 17–19, especially Lemma 3.3, equations (3.16)–(3.22), and Proposition 3.6 [5]. The preceding arithmetic map is

$$\mathcal{E}f(x)=x^{1/2}\sum_{n\geq 1}f(nx),\qquad \mathcal{F}_\mu g(s)=\int_0^\infty g(x)x^{-is}\,\frac{dx}{x}.$$

The input consists of even Schwartz functions with $f(0)=\widehat{f}(0)=0$. The paper's two Hermite families are

$$\psi_\ell^+=h_{4\ell}-\frac{h_{4\ell}(0)}{h_0(0)}h_0,\qquad \psi_\ell^-=-h_{4\ell+2}+\frac{h_{4\ell+2}(0)}{h_2(0)}h_2\quad(\ell\geq 1).$$

These signs and the minus sign in the Mellin exponent are retained. The completed function is

$$\Xi(s)=\xi(\tfrac{1}{2}+is),\qquad \xi(z)=\tfrac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z),\qquad \Xi(s)=8H_0(2s). \tag{157}$$

The paper proves

$$\mathcal{F}_\mu\mathcal{E}\psi_\ell^\pm=R_\ell^\pm\Xi,\qquad P_\ell^\pm=-\tfrac{1}{2}(\tfrac{1}{4}+s^2)R_\ell^\pm. \tag{158}$$

It also states immediately before Proposition 3.6 that these give all polynomial multiples of Ξ. Here is the precise algebra behind the last assertion. The top summands of (3.19) and (3.20) are nonzero; their degrees are 2ℓ and $2\ell+1$. Division by $-\frac{1}{2}(\frac{1}{4}+s^2)$ gives degrees $2\ell-2$ and $2\ell-1$, respectively. The even and odd parities remain. Thus the ordered family $R_1^+,R_1^-,R_2^+,R_2^-,\ldots$ has one polynomial of each degree, with a nonzero leading coefficient. Subtracting the appropriate multiple of the polynomial of largest degree, and iterating down to degree zero, expresses every polynomial as a finite linear combination. Linear independence follows by the largest-degree coefficient of a finite relation. Consequently

$$N_{\mathrm{ar}}=\overline{\operatorname{span}\{\mathcal{F}_\mu\mathcal{E}\psi_\ell^\pm\}}^{\,\tau}=\overline{\mathbb{C}[s]\Xi}^{\,\tau},\qquad Q_{\mathrm{ar}}=(\mathcal{H}_{\leq 1},\tau)/N_{\mathrm{ar}}. \tag{159}$$

Here τ denotes precisely the source's topology. No equality of this closed space with a principal algebraic ideal is being asserted. Multiplication $Mf(s) = sf(s)$ is continuous in the topological ring; it preserves the dense generating subspace, and hence N_{ar}. Its induced operator S is the operator of Proposition 3.6. We use the source's spectral assertion as an explicitly imported theorem:

$$\mathrm{Spec}(S) = \{\lambda \in \mathbb{C} : \zeta(\tfrac{1}{2} + i\lambda) = 0 \text{ at a nontrivial zero}\}. \tag{160}$$

Spectrum here means failure of a continuous two-sided inverse. No proof of the entire source spectral theorem is claimed as a new result of this manuscript.

There is a material source limitation. Page 19 calls $\mathcal{H}_{\leq 1}$ the Hadamard topological ring of entire functions of order at most one; it does not state seminorms or a convergence criterion. The cited Connes–Marcolli Proposition 2.24, printed pages 379–381, proves the polynomial-image assertion and explicitly discusses different function-space realizations [6]. It does not supply that missing specification. We therefore retain τ in every source quotient. Below, all assertions depending on another topology name and define that topology. The comparison with the source is an exact quotient diagram, rather than an identification by terminology. The bounded edition audit gives the same limitation in the primary arXiv v1 source, lines 683–704 (Proposition 3.9), and in v2, `mainc2m24fine.tex`, lines 668–689 (Proposition 3.6): each states the order class, polynomial trace image, and spectral set, without defining its topology or algebraic multiplicities. These exact edition locators supplement the author-proof locator in [5]; no claim of byte identity with a publisher's final subscription text is required.

C A globally defined heat map on the original underlying functions

For an entire function f, let $M_f(r) = \max_{|s|\leq r} |f(s)|$ and define its order by

$$\limsup_{r\to\infty} \frac{\log\log\max(e, M_f(r))}{\log r}.$$

The zero function is included. For $1 < p < 2$, $a > 0$ set

$$\|f\|_{p,a} = \sup_{s\in\mathbb{C}} |f(s)|e^{-a|s|^p}.$$

The underlying set $\mathcal{H}_{\leq 1}$ is exactly the set where all these seminorms are finite. Indeed order at most one gives, for $1 < q < p$, $M_f(r) \leq \exp(r^q)$ for sufficiently large r; the expression $r^q - ar^p$ is bounded above. Conversely finiteness gives order at most p for every $p > 1$. Write γ for this explicitly defined growth topology. It is enough to take $p = 1 + 1/j$, $j \geq 2$, and $a = 1/k$, $k \geq 1$: given $p > 1, a > 0$, select one of these $p_j < p$ and use the boundedness of $|s|^{p_j}/k - a|s|^p$. This proves equivalence of the countable family with all displayed seminorms. A sequence Cauchy in these seminorms converges uniformly on compact sets to an entire function. Taking pointwise limits in each weighted Cauchy inequality gives convergence in that seminorm and finiteness of its norm. Thus $(\mathcal{H}_{\leq 1}, \gamma)$ is a complete metrizable locally convex space.

Lemma C.1 (Derivative bounds and the heat group)**.** *Put $D = \partial_s$. Differentiation and multiplication by s map the original underlying set $\mathcal{H}_{\leq 1}$ into itself. For $m \geq 1$,*

$$\|D^m f\|_{p,a} \leq m! \left(\frac{epa}{m}\right)^{m/p} \|f\|_{p,a/2^{p-1}}. \tag{161}$$

For every $t \in \mathbb{C}$, the series

$$U_t f = \sum_{k=0}^{\infty} \frac{(-t/4)^k}{k!} D^{2k} f \tag{162}$$

converges in γ, locally uniformly in t; U_t is a continuous automorphism of $(\mathcal{H}_{\le 1}, \gamma)$ and $U_t U_u = U_{t+u}$, $U_t^{-1} = U_{-t}$.

Proof. Cauchy's circle formula on a circle of radius R about s gives

$$|D^m f(s)| \le m! R^{-m} \max_{|z-s|=R} |f(z)|.$$

For $b = a/2^{p-1}$ the inequality $|s+z|^p \le 2^{p-1}(|s|^p + |z|^p)$ bounds the last expression, after multiplication by $e^{-a|s|^p}$, by $m! R^{-m} e^{aR^p} \|f\|_{p,b}$. Choosing $R = (m/(ap))^{1/p}$ proves (161). For multiplication use

$$\|Mf\|_{p,a} \le \sup_{r \ge 0} r e^{-ar^p/2} \|f\|_{p,a/2}.$$

The kth heat summand is at most

$$\frac{(|t|/4)^k}{k!} (2k)! \left(\frac{epa}{2k}\right)^{2k/p} \|f\|_{p,b}.$$

Since $(2k)!/k! \le (2k)^k$, its scalar factor is at most $[C_{p,a}|t|k^{1-2/p}]^k$, whose kth root tends to zero. This proves summability, continuity, and locally uniform convergence of every time-differentiated series by the same estimate with the additional polynomial factors in k. Commutation with D follows from continuity of D. In the double composition series group terms with $n = k + \ell$. The sum of the absolute scalar coefficients at $D^{2n} f$ is $((|t| + |u|)/4)^n/n!$; the preceding estimate justifies regrouping. The binomial theorem gives U_{t+u}. The value at zero is the identity, proving the inverse. □

Lemma C.2 (Polynomial approximation)**.** *The Taylor polynomials of any $f \in \mathcal{H}_{\le 1}$ converge to f in γ. Multiplication of two elements is continuous in γ.*

Proof. If $f = \sum c_n s^n$, Cauchy's coefficient bound with parameter $b < a$ gives, for $n \ge 1$,

$$|c_n| \le \|f\|_{p,b} (epb/n)^{n/p}.$$

Also $\|s^n\|_{p,a} = (n/(epa))^{n/p}$. Therefore $\|c_n s^n\|_{p,a} \le \|f\|_{p,b}(b/a)^{n/p}$. The tail is bounded by a convergent geometric series. Finally $\|fg\|_{p,a} \le \|f\|_{p,a/2}\|g\|_{p,a/2}$. □

Retain the actual real heat family, rather than substitute a polynomial:

$$\Phi(u) = \sum_{n \ge 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}},$$

$$Z_t(s) = 8H_t(2s) = 8 \int_0^\infty e^{tu^2} \Phi(u) \cos(2su)\, du, \qquad \partial_t Z_t = -\tfrac{1}{4} D^2 Z_t, Z_0 = \Xi. \tag{163}$$

These are the Rodgers–Tao conventions, Section 1 [7]. For clarity the analytic exchanges in this transport can be proved directly. Since $n^2 e^{4u} \ge (n^2 + e^{4u})/2$, the absolutely convergent sums $\sum n^j e^{-\pi n^2/2}$ give constants $C, c > 0$ such that $|\Phi(u)| \le C e^{9u} e^{-ce^{4u}}$ for $u \ge 0$. For $|t| \le T$ and $|s| \le R$, every derivative of the integrand is bounded by a polynomial in u times $C \exp(Tu^2 + (2R+9)u - ce^{4u})$, which is integrable. The exponential series in t can therefore be integrated absolutely. Because $D^{2k} \cos(2su) = (-1)^k (2u)^{2k} \cos(2su)$, it gives exactly $Z_t = U_t \Xi$ with the coefficient $(-t/4)^k/k!$. In particular Z_t lies in the original order-at-most-one set for every complex t, by Lemma C.1; it is never identically zero because U_t is invertible and $\Xi(0) \ne 0$.

D Transport of the actual closed arithmetic image

Define τ_t on the same underlying vector space $\mathcal{H}_{\le 1}$ by declaring $U_t : (\mathcal{H}_{\le 1}, \tau) \longrightarrow (\mathcal{H}_{\le 1}, \tau_t)$ a homeomorphism. This definition specifies every open set: V is τ_t-open exactly when $U_{-t}V$ is τ-open. It uses the globally defined bijection just proved and does not require unproved continuity in τ. Set

$$N_t = U_t N_{\mathrm{ar}} = \overline{\operatorname{span}\{U_t(R_\ell^{\pm}\Xi)\}}^{\tau_t}, \qquad Q_t = (\mathcal{H}_{\le 1}, \tau_t)/N_t. \tag{164}$$

The closure equality follows because homeomorphisms carry closure to closure. In particular N_t is closed. The quotient map

$$\mathcal{U}_t : Q_{\mathrm{ar}} \longrightarrow Q_t, \qquad [f] \longmapsto [U_t f] \tag{165}$$

is well defined, bijective, and continuous, with continuous inverse $[g] \mapsto [U_{-t}g]$. Indeed changing f by $n \in N_{\mathrm{ar}}$ changes $U_t f$ by $U_t n \in N_t$, and the inverse gives necessity; the quotient universal property proves both continuity statements.

Theorem D.1 (Exact arithmetic heat intertwiner)**.** *On the entire-function set,*

$$B_t = U_t M U_{-t} = M - \tfrac{t}{2} D. \tag{166}$$

This operator preserves N_t, is continuous in τ_t, and induces S_t on Q_t. It satisfies $S_t \mathcal{U}_t = \mathcal{U}_t S$ and

$$\operatorname{Spec}(S_t) = \operatorname{Spec}(S) = \{\lambda : \zeta(\tfrac{1}{2} + i\lambda) = 0 \ \textit{nontrivially}\}. \tag{167}$$

The time-dependent trace functions are exactly $U_t(R_\ell^{\pm}\Xi) = R_\ell^{\pm}(B_t) Z_t$.

Proof. The product rule gives $D^{2k} M f = M D^{2k} f + 2k D^{2k-1} f$. Insert this identity in the convergent heat series to obtain $U_t M = (M - (t/2)D) U_t$, proving (166). If $n \in N_{\mathrm{ar}}$, then $B_t U_t n = U_t M n \in N_t$. Continuity is conjugation of the continuous M by the defining homeomorphism. The induced identity follows on representatives. If T is a continuous inverse of $S - \lambda$, then $\mathcal{U}_t T \mathcal{U}_t^{-1}$ is a continuous inverse of $S_t - \lambda$. Conjugating in the opposite direction proves the converse, and hence the spectrum equality using (160). Apply the polynomial identity $U_t p(M) = p(B_t) U_t$ to Ξ for the last assertion. $\square$

The heat map also transports the full multiplication, with every cross term retained. For $f, g \in \mathcal{H}_{\le 1}$ define

$$f \star_t g = U_t\big((U_{-t}f)(U_{-t}g)\big) = \sum_{n=0}^{\infty} \frac{(-t/2)^n}{n!} (D^n f)(D^n g). \tag{168}$$

We prove the series identity, including convergence. For $F_t = U_t f$, $G_t = U_t g$, the series

$$J_t = \sum_{n=0}^{\infty} \frac{(-t/2)^n}{n!} (D^n F_t)(D^n G_t)$$

converges in γ, uniformly on compact time sets: applying (161) with output norm $(p, a/2)$ to both factors bounds its nth term by

$$K \left[\frac{T}{2} \left(\frac{epa}{2} \right)^{2/p} n^{1-2/p} \right]^n \qquad (|t| \le T),$$

where K bounds the product of the two input norms $(p, a/2^p)$, and is finite by the heat estimate. The root test applies. Cauchy's formula in time permits termwise time derivatives; continuity of D permits spatial derivatives. Differentiating the coefficient contributes

$-\frac{1}{2}\sum(-t/2)^n(D^{n+1}F_t)(D^{n+1}G_t)/n!$. Differentiating the factors contributes the two pure second-derivative terms with coefficient $-1/4$. The complete product rule therefore gives $\partial_t J_t = -D^2 J_t/4$ and $J_0 = fg$. Its entire time Taylor coefficients are uniquely $(-1/4)^k D^{2k}(fg)/k!$, so $J_t = U_t(fg)$. Replacing the inputs by $U_{-t}f, U_{-t}g$ proves (168). Conjugation of ordinary multiplication proves associativity, commutativity, unit 1, and continuity for τ_t, even when continuity of the series in the source topology is unknown. The identity $s \star_t f = sf - tf'/2$ recovers exactly B_t. No ideal property of N_{ar} beyond its stated polynomial invariance is required for this identity or for the quotient in Theorem D.1.

The transported actual trace image has no common point-evaluation zero at any nonzero time. Suppose λ were such a common zero. It would annihilate each $U_t(s^n\Xi)$. For every $a \in \mathbb{C}$, the Taylor series of e^{as} converges in γ by Lemma C.2. Multiplication, U_t, and evaluation are continuous in γ, so

$$0 = \sum_{n=0}^{\infty} \frac{a^n}{n!} U_t(s^n\Xi)(\lambda) = e^{a\lambda - ta^2/4} Z_t(\lambda - ta/2).$$

The last equality follows from the just-proved product formula and the entire Taylor formula for Z_t. At $t \neq 0$ varying a forces Z_t to vanish everywhere, a contradiction. This argument uses only the polynomial generators contained in N_t; it assumes no source-topology convergence of the exponential series. Thus the source arithmetic spectrum in (167) is carried by the explicitly transported operator and relation, even though the transported relation has no common ordinary evaluation zero.

The first two nonconstant polynomial generators make the extra terms visible without changing any polynomial coefficient:

$$\begin{aligned} U_t(s\Xi) &= sZ_t - \tfrac{t}{2}Z_t', \\ U_t(s^2\Xi) &= s^2 Z_t - tsZ_t' - \tfrac{t}{2}Z_t + \tfrac{t^2}{4}Z_t''. \end{aligned} \tag{169}$$

For $L = \partial_t + \frac{1}{4}D^2$, the complete product rule is

$$L(AP) = A(P_t + \tfrac{1}{4}P_{ss}) + (A_t + \tfrac{1}{4}A_{ss})P + \tfrac{1}{2}A_s P_s. \tag{170}$$

Thus $L(R(s)Z_t) = \frac{1}{4}R''Z_t + \frac{1}{2}R'Z_t'$. The family $R(B_t)Z_t$ solves the homogeneous heat equation, whereas the displayed expression is the full defect of replacing it by $R(s)Z_t$. For a local factorization $Z_t = AP$ with $A \neq 0$, the same formula retains the analytic unit A and its derivatives. At a simple zero $\lambda(t)$, differentiating $Z_t(\lambda(t)) = 0$ gives

$$\lambda'(t) = \frac{1}{4}\frac{Z_{ss}}{Z_s}(t, \lambda(t)) = \frac{1}{4}\frac{P_{ss}}{P_s}(t, \lambda(t)) + \frac{1}{2}\frac{A_s}{A}(t, \lambda(t)). \tag{171}$$

This statement is on the open locus $Z_s \neq 0$, on which the analytic implicit-function theorem supplies the local curve; it claims no chosen zero is simple. At a multiple zero the division is undefined, and the full equation (170) remains the equation to use.

E Derivative domains, multivalued operators, and jet extensions

Every derivative has the full underlying domain $\mathcal{H}_{\leq 1}$ by Lemma C.1. Continuity in the unnamed source topology is not assumed. The following gives exact source-quotient operators and their domains without imposing it. For $k \geq 1$ put

$$K_k = \{n \in N_{\mathrm{ar}} : D^k n \in N_{\mathrm{ar}}\}, \qquad W_k = (D^k N_{\mathrm{ar}} + N_{\mathrm{ar}})/N_{\mathrm{ar}} \subseteq Q_{\mathrm{ar}}.$$

The derivative correspondence is the actual linear relation

$$\mathfrak{D}_k = \{([f],[D^k f]) : f \in \mathcal{H}_{\le 1}\} \subseteq Q_{\mathrm{ar}} \times Q_{\mathrm{ar}}. \tag{172}$$

For fixed $[f]$, its output fibre is precisely $[D^k f] + W_k$: all representatives are $f+n$ with $n \in N_{\mathrm{ar}}$, and linearity gives all and only the stated outputs. The kernel of its presenting map $f \mapsto ([f],[D^k f])$ is exactly K_k. The relation has domain all of Q_{ar}; its multivalued part at zero is exactly W_k. Thus it is the graph of an operator exactly when $D^k N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$. This is a calculated equivalence, with an explicit obstruction space, rather than an assumption of invariance.

There is no nonempty domain of quotient classes on which the same representative formula is unambiguous if $W_k \neq 0$ and every representative of a class is admitted: addition of the same offending n gives two different outputs for every class. A chosen linear lift is a different map. More precisely any vector subspace $A \subseteq \mathcal{H}_{\le 1}$ supplies a single-valued map on its quotient image $q(A)$ exactly when $D^k(A \cap N_{\mathrm{ar}}) \subseteq N_{\mathrm{ar}}$; the identical difference-of-representatives argument proves both directions. The first heat generator is $-D^2/4$, so its ambiguity space is $-W_2/4$, with the coefficient and sign unchanged.

E.1 Exact algebraic spectral defects of the source relation

These calculations keep the actual N_{ar} and require only the already proved $MN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$. For every $\lambda \in \mathbb{C}$, division by $x = s - \lambda$ gives the algebraic identities

$$\begin{aligned} \operatorname{coker}_{\mathrm{alg}}(S-\lambda) &= \mathcal{H}_{\le 1}/(N_{\mathrm{ar}} + x\mathcal{H}_{\le 1}), \\ \ker(S-\lambda) &\simeq (N_{\mathrm{ar}} \cap x\mathcal{H}_{\le 1})/(xN_{\mathrm{ar}}), \quad [n] \longmapsto [n/x]. \end{aligned} \tag{173}$$

Division belongs to the original order class: if $f(\lambda) = 0$, the filled quotient is entire, is bounded on a fixed disk, and outside $|s| \ge 2|\lambda| + 1$ its modulus is at most $2|f(s)|$. Thus $x\mathcal{H}_{\le 1}$ is exactly the kernel of evaluation at λ. The cokernel identity follows from the image of the induced multiplication. For the second identity, changing n by xn_0 changes n/x by $n_0 \in N_{\mathrm{ar}}$; its image is in the stated kernel. Conversely any kernel representative f has $xf \in N_{\mathrm{ar}} \cap x\mathcal{H}_{\le 1}$. Finally $[n/x] = 0$ exactly when $n \in xN_{\mathrm{ar}}$. This proves every asserted kernel and image.

The cokernel has only two possibilities. When every $n \in N_{\mathrm{ar}}$ vanishes at λ, it is one-dimensional, with its exact identification given by evaluation. Otherwise retain any $h_\lambda \in N_{\mathrm{ar}}$ with $h_\lambda(\lambda) \neq 0$. Then

$$\begin{aligned} A_\lambda f(s) &= \frac{f(s) - f(\lambda)h_\lambda(s)/h_\lambda(\lambda)}{s-\lambda}, \\ f &= \frac{f(\lambda)}{h_\lambda(\lambda)} h_\lambda + (s-\lambda)A_\lambda f \end{aligned} \tag{174}$$

proves surjectivity of $S - \lambda$, and its exact remaining kernel is

$$\ker(S-\lambda) \simeq N_{\mathrm{ar}}/((s-\lambda)N_{\mathrm{ar}} \oplus \mathbb{C}h_\lambda), \qquad n \longmapsto [A_\lambda n]. \tag{175}$$

The map is onto because a kernel class $[g]$ is the image of $n = (s-\lambda)g \in N_{\mathrm{ar}}$. Its kernel consists exactly of $n = (s-\lambda)n_0 + ch_\lambda$, $n_0 \in N_{\mathrm{ar}}$, by (174). The sum is direct by evaluation. In this second case algebraic invertibility is therefore equivalent to the exact identity $N_{\mathrm{ar}} = (s-\lambda)N_{\mathrm{ar}} \oplus \mathbb{C}h_\lambda$. On that locus the representative formula $[f] \mapsto [A_\lambda f]$ is independent of representatives, since $A_\lambda N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$; multiplication in (174) proves the right inverse, and $A_\lambda((s-\lambda)f) = f$ proves the left inverse. These are algebraic statements; continuity of this inverse in τ has not been inserted.

The first derivative and the heat generator have exactly the same frozen invariance question for the source subspace:

$$DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \quad \Longleftrightarrow \quad D^2 N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \quad \Longleftrightarrow \quad -\tfrac{1}{4}D^2 N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}. \tag{176}$$

Indeed the product rule is $D^2(sn) - sD^2n = 2Dn$. Since $sn \in N_{\mathrm{ar}}$, second-derivative invariance puts both terms on the left in N_{ar}, proving first-derivative invariance. Iteration proves the reverse implication; the retained nonzero scalar $-1/4$ proves the last equivalence. If these equivalent inclusions hold, all $D^j\Xi$ belong to N_{ar}. At each λ at least one is nonzero: otherwise the Taylor series would make Ξ identically zero. Thus every $S-\lambda$ is then algebraically surjective by (174). Its prescribed topological spectral points would be represented entirely by the kernel (175) or a noncontinuous algebraic inverse. This computes the precise alternatives without assigning either one to the source's stated spectral set.

There is also a full multiplicity consequence of algebraic derivative descent. Writing $d[f] = [Df]$ on its exact descent locus, the product rule gives $[d, S] = I$. For $0 \neq v \in \ker(S-\lambda)$,

$$(S-\lambda)d^k v = -kd^{k-1}v, \qquad (S-\lambda)^k d^k v = (-1)^k k! v \quad (k \geq 0), \tag{177}$$

with zero right side in the first identity at $k = 0$. To prove induction, commute the next d using $(S-\lambda)d = d(S-\lambda) - I$; the coefficient $-k$ becomes $-(k+1)$. Iteration proves the second identity. A finite linear dependence among $v, dv, \ldots, d^n v$, with largest nonzero coefficient at index j, becomes that coefficient times $(-1)^j j! v$ after applying $(S-\lambda)^j$, a contradiction. Thus every such eigenvector generates an infinite independent generalized eigenvector sequence. No finite source multiplicity assertion has been assumed.

Finally even a single forward nonzero heat inclusion has a calculated consequence. Fix $t \neq 0$ and $c = t/2$, and define

$$\begin{gathered} A_0 = U_t, \qquad A_{k+1} = MA_k - A_kM, \qquad A_k = P_k(D)U_t, \\ P_0(X) = 1, \qquad P_{k+1}(X) = cXP_k(X) - P_k'(X). \end{gathered} \tag{178}$$

The identity follows by induction from $[M, U_t] = cDU_t$ and $[M, D^j] = -jD^{j-1}$: commuting M through $P_k(D)U_t$ contributes $-P_k'(D)U_t + cP_k(D)DU_t$. The degree of P_k is exactly k, with leading coefficient $c^k \neq 0$. Under $U_tN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$, induction with $MN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ gives $A_kN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$; subtracting the lower-degree terms of P_k then gives exactly

$$D^kU_tN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \quad (k \geq 0). \tag{179}$$

In particular every derivative of the nonzero $Z_t = U_t\Xi$ lies in N_{ar}. At every point one derivative is nonzero by the entire Taylor theorem, so every $S-\lambda$ is algebraically surjective in this case too. This proof uses only the one forward inclusion. For the stronger equality $U_tN_{\mathrm{ar}} = N_{\mathrm{ar}}$, it also proves $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$, by writing each $n \in N_{\mathrm{ar}}$ as U_tm in (179). None of these exact implications presumes the inclusion or equality for the actual source closure. The explicit obstruction spaces above remain the calculated source question.

Even when W_k is not closed, the smallest source-closed enlargement of the target relation subspace is explicit:

$$[f] \longmapsto [D^kf] : Q_{\mathrm{ar}} \longrightarrow \mathcal{H}_{\leq 1}/\overline{N_{\mathrm{ar}} + D^kN_{\mathrm{ar}}}^{\tau}. \tag{180}$$

It is well defined algebraically. Equip its source with the quotient of the graph topology $f \mapsto (f, D^kf)$ in $(\mathcal{H}_{\leq 1}, \tau)^2$ to obtain a continuous map. This specifies a topology change explicitly, without claiming it is the original one. The lost part of the original target Q_{ar} is exactly $\overline{N_{\mathrm{ar}} + D^kN_{\mathrm{ar}}}^{\tau}/N_{\mathrm{ar}}$.

Here is an extension which retains correlated derivatives rather than quotienting them away. For $j \geq 1$ define the bijection

$$T_j(f_0, \ldots, f_j) = (f_0, f_1 - Df_0, \ldots, f_j - D^jf_0).$$

Its inverse is $(f_0, g_1, \ldots, g_j) \mapsto (f_0, g_1 + Df_0, \ldots, g_j + D^jf_0)$. Give $V_j = \mathcal{H}_{\leq 1}^{j+1}$ the topology transported by T_j from $(\mathcal{H}_{\leq 1}, \tau) \times (\mathcal{H}_{\leq 1}, \gamma)^j$. The residual factors are explicitly given the proved Hausdorff growth topology; the source factor keeps its original topology. For $J_j f = (f, Df, \ldots, D^j f)$, $T_j(J_j N_{\mathrm{ar}}) = N_{\mathrm{ar}} \oplus 0^j$, so $J_j N_{\mathrm{ar}}$ is closed. Define

$$X_j = V_j / J_j N_{\mathrm{ar}}. \tag{181}$$

Then the map

$$[f_0, \ldots, f_j] \longmapsto ([f_0], f_1 - Df_0, \ldots, f_j - D^j f_0)$$

is a homeomorphism $X_j \simeq Q_{\mathrm{ar}} \oplus (\mathcal{H}_{\leq 1}, \gamma)^j$; independence, the inverse, and continuity follow directly from T_j and the quotient definition. In particular there is the split exact sequence

$$0 \longrightarrow \mathcal{H}_{\leq 1}^j \longrightarrow X_j \xrightarrow{[f_0,\ldots,f_j] \mapsto [f_0]} Q_{\mathrm{ar}} \longrightarrow 0, \qquad [f] \longmapsto [J_j f]. \tag{182}$$

Its relation on a source generator retains every product term:

$$D^k(R\Xi) = \sum_{h=0}^{k} \binom{k}{h} R^{(h)} \Xi^{(k-h)} \quad (0 \leq k \leq j). \tag{183}$$

The binomial formula follows by induction from the ordinary product rule, including the two edge terms at each step.

The original spectral coordinate acts on V_j by

$$(\mathcal{M}_j f)_k = s f_k + k f_{k-1} \quad (0 \leq k \leq j), \qquad f_{-1} = 0.$$

The identity $D^k(sf) = sD^k f + kD^{k-1} f$ proves $\mathcal{M}_j J_j f = J_j(sf)$; it therefore preserves the exact relation subspace. In the T_j coordinates it acts by S on Q_{ar} and by $g_k \mapsto sg_k + kg_{k-1}$ on the residual terms ($g_0 = 0$). This also proves continuity in the stated topology. Notice that (182) adds a free residual module: the spectrum of the entire extension is not claimed to equal the source spectrum. The arithmetic quotient and its split inclusion are explicitly preserved.

F A fully computed analytic realization and its exact comparison

We now compute the closures and the failure of descent in specified topologies. This does not identify them with τ. Let κ be the topology of uniform convergence on compact sets on the same set $\mathcal{H}_{\leq 1}$, and define

$$N_\gamma = \overline{\mathbb{C}[s]\Xi}^{\gamma}, \qquad N_\kappa = \overline{\mathbb{C}[s]\Xi}^{\kappa}.$$

Evaluations of all derivatives are continuous in both topologies by Cauchy's formula. The growth topology is finer than κ, so $N_\gamma \subseteq N_\kappa$. If λ is a zero of Ξ with its actual multiplicity m_λ, every element of either closed subspace has vanishing derivatives of orders $0, \ldots, m_\lambda - 1$ at λ.

Lemma F.1 (Entire division with its growth and topology retained)**.** *Let $g \in \mathcal{H}_{\leq 1}$ with $g(0) \neq 0$. If $f \in \mathcal{H}_{\leq 1}$ and $h = f/g$ extends to an entire function, then $h \in \mathcal{H}_{\leq 1}$. For $1 < p < 2$, $a > 0$, put $b = a/(6 \cdot 2^p)$. There is the linear division estimate*

$$\|f/g\|_{p,a} \leq \left(\frac{\max(1, \|g\|_{p,b})}{|g(0)|} \right)^3 \|f\|_{p,b}. \tag{184}$$

Multiplication by g is a topological linear isomorphism from $(\mathcal{H}_{\leq 1}, \gamma)$ onto the closed subspace of functions divisible by g, with its subspace topology. Compact-topology division is also continuous.

Proof. Choose $R > 0$ with no zero of g on $|s| = R$. Factor its finitely many zeros in the disk, retaining multiplicities. The residual is holomorphic and zero-free there, so its log modulus is harmonic. For $|z| < R$ the boundary mean of $\log |Re^{i\theta} - z|$ equals $\log R$: expand $\log(1 - (z/R)e^{-i\theta})$ in its uniformly convergent series and take real parts. The mean-value formula for the residual gives the exact Jensen identity

$$\frac{1}{2\pi} \int_0^{2\pi} \log |g(Re^{i\theta})| \, d\theta = \log |g(0)| + \sum_{|z|<R} m_z \log(R/|z|) \geq \log |g(0)|.$$

There is no zero at zero. With $\log^+ x = \max(0, \log x)$ and $\log^- x = \max(0, -\log x)$, the inequality $\log^+ |h| \leq \log^+ |f| + \log^- |g|$ gives

$$\frac{1}{2\pi} \int_0^{2\pi} \log^+ |h(Re^{i\theta})| \, d\theta \leq \log^+ M_f(R) + \log^+ M_g(R) - \log |g(0)|. \tag{185}$$

The right side is nonnegative since $M_g(R) \geq |g(0)|$. The harmonic Poisson extension of the continuous nonnegative boundary function $\log^+ |h|$ majorizes $\log |h|$ by the maximum principle and is nonnegative; hence it majorizes $\log^+ |h|$ in the disk. For $|s| \leq r < R$ the Poisson kernel is at most $(R + r)/(R - r)$. Take $R \downarrow 2r$ through circles avoiding zeros. Continuity of the two maximum functions gives, for every $r > 0$,

$$\log^+ M_h(r) \leq 3\big(\log^+ M_f(2r) + \log^+ M_g(2r) - \log |g(0)|\big). \tag{186}$$

No pointwise lower bound for g on a large circle has been assumed. For $f \neq 0$, put $F = \|f\|_{p,b} > 0$ and apply this to $f/F, h/F$. Then $\log^+ M_{f/F}(2r) \leq b(2r)^p$, and $\log^+ M_g(2r) \leq \log \max(1, \|g\|_{p,b}) + b(2r)^p$. The coefficient of r^p is $6b2^p = a$, giving (184); continuity fills in $r = 0$. The zero numerator is immediate. All p, a are allowed, so h belongs to the original order class and division is continuous. Multiplication is continuous by Lemma C.2. The requisite vanishing jets at every zero are closed conditions; local Taylor division identifies them with entire divisibility. The estimate proves this closed set is exactly $g\mathcal{H}_{\leq 1}$. For the compact topology choose $R > r$ with no boundary zero and $\delta = \min_{|s|=R} |g(s)| > 0$. The maximum principle for the filled entire quotient gives $M_h(r) \leq \delta^{-1} M_f(R)$, proving compact continuity too. ☐

For general nonzero $g \in \mathcal{H}_{\leq 1}$, choose and retain a point c with $g(c) \neq 0$ and use disks centered at c. The functions and spectral coordinate remain unchanged. The bound $|s + c|^p \leq 2^{p-1}(|s|^p + |c|^p)$ proves continuity of translations by c and $-c$ in γ; compact containment proves it in κ. Thus all conclusions hold with the stated translated estimates. The original center $c = 0$ already applies to Ξ.

Proposition F.2 (The computed closures are the same subspace)**.** *With their definitions unchanged,*

$$\begin{aligned} N_\gamma = N_\kappa &= \Xi \mathcal{H}_{\leq 1} \\ &= \{f \in \mathcal{H}_{\leq 1} : D^h f(\lambda) = 0 \textit{ for every } \Xi(\lambda) = 0,\ 0 \leq h < m_\lambda\}. \end{aligned} \tag{187}$$

This is equality of subsets of the original ring, not an identification of γ, κ, or τ.

Proof. Jet continuity puts both closures inside the displayed vanishing set and gives $N_\gamma \subseteq N_\kappa$. Local Taylor division makes $h = f/\Xi$ entire for any f in that set; the division lemma then puts h in $\mathcal{H}_{\leq 1}$. Its Taylor polynomials p_n converge in γ, and multiplication gives $p_n \Xi \to f$ there. Thus the vanishing set is contained in N_γ, proving all equalities. The algebraic-ideal equality is the conclusion of this argument; it was not substituted for an unevaluated closure. ☐

Theorem F.3 (Compact closure, multiplicities, and coordinate spectrum)**.** *The compact closure is exactly*

$$N_\kappa = \{f \in \mathcal{H}_{\le 1} : f/\Xi \text{ extends to an entire function}\}$$
$$= \{f \in \mathcal{H}_{\le 1} : D^h f(\lambda) = 0 \ (h < m_\lambda),\ \Xi(\lambda) = 0\}. \tag{188}$$

On $Q_\kappa = (\mathcal{H}_{\le 1}, \kappa)/N_\kappa$, multiplication by s has spectrum exactly the zeros of Ξ in the original s coordinate. For a zero λ of multiplicity m, there is a continuous surjection

$$Q_\kappa \longrightarrow \mathbb{C}[\epsilon]/(\epsilon^m), \qquad [f] \longmapsto \sum_{h=0}^{m-1} \frac{D^h f(\lambda)}{h!}\epsilon^h,$$

intertwining multiplication by s with $\lambda + \epsilon$.

Proof. The vanishing conditions are closed and hold on every $p\Xi$, so the closure is contained in the last displayed set. Local Taylor division at each isolated zero identifies that set with entire divisibility by Ξ. For such an f, put $g = f/\Xi \in \mathcal{O}(\mathbb{C})$. Its Taylor polynomials p_n converge uniformly on every compact; $p_n\Xi \to f$ in κ. Every $p_n\Xi$ belongs to the original order class even though no growth assertion about g is needed. This proves (188) with the closure retained. The local-jet map is independent of representatives; the polynomials $1, (s-\lambda), \dots, (s-\lambda)^{m-1}$ prove surjectivity. Leibniz's rule gives the stated action, and the finite jet evaluations are continuous.

If $\Xi(\lambda) \neq 0$, define

$$T_\lambda f(s) = \frac{f(s) - f(\lambda)\Xi(s)/\Xi(\lambda)}{s - \lambda},$$

filling in the removable value. Division by a linear factor preserves the order bound: outside a disk about λ the denominator is bounded below, and inside a disk the filled entire function is bounded. It is continuous in κ by Cauchy's formula on a larger disk applied to the numerator. For $n \in N_\kappa$ the quotient $T_\lambda n/\Xi = (n/\Xi - (n/\Xi)(\lambda))/(s-\lambda)$ is entire; thus it descends. Direct multiplication proves $(S_\kappa - \lambda)[T_\lambda f] = [f]$. Applying T_λ to $(s-\lambda)f$ gives f exactly, proving the other inverse identity. If $\Xi(\lambda) = 0$, evaluation at λ descends to a nonzero continuous functional which annihilates the image of $S_\kappa - \lambda$. Hence that operator is not surjective. This proves the spectrum claim. □

The same coordinate spectrum and the same finite-jet surjections hold for $Q_\gamma = (\mathcal{H}_{\le 1}, \gamma)/\Xi\mathcal{H}_{\le 1}$. Indeed the numerator in $T_\lambda f$ depends continuously on f in γ. Division by $s - \lambda$, on its entire-divisibility domain, is continuous by Lemma F.1 with a nonzero center retained when needed. The already proved two inverse identities therefore give a continuous resolvent in γ as well. The evaluation obstruction at each zero is continuous in that topology too. This proves the claim, without transferring a resolvent through an unproved topology identification.

The ambient topologies κ and γ are in fact distinct. The original functions $f_n(s) = \exp(ns - n^2)$ tend to zero uniformly on each compact set. On the positive real axis the maximum of $nx - x^{3/2}$ is $4n^3/27$, attained at $x = 4n^2/9$. Since $\operatorname{Re} s \le |s|$, that axis attains the full weighted supremum, giving the exact value

$$\|f_n\|_{3/2,1} = \exp(4n^3/27 - n^2) \longrightarrow \infty.$$

Thus equality of the closed arithmetic relation subspaces and of their spectral sets does not identify the ambient topologies. The identity induces a continuous linear bijection $Q_\gamma \to Q_\kappa$; the next theorem proves that its inverse is not continuous.

F.1 The actual difference between the two analytic quotient topologies

Retain the original entire-function set $\mathcal{H}_{\leq 1} = \mathcal{H}_{\leq 1}$, the original coordinate s, and $\Xi(s) = \xi(\frac{1}{2} + is)$ from (157). Let Λ be the set of its distinct zeros and let $m_\lambda = \mathrm{ord}_\lambda \Xi \geq 1$ be each actual multiplicity. Put

$$I = \Xi\mathcal{H}_{\leq 1}, \qquad Q_\kappa = (\mathcal{H}_{\leq 1}, \kappa)/I, \qquad Q_\gamma = (\mathcal{H}_{\leq 1}, \gamma)/I, \qquad \mathcal{P}_\Xi = \prod_{\lambda \in \Lambda} \mathbb{C}^{m_\lambda}.$$

The product has its ordinary product topology. Coordinates in $\mathbb{C}^{m_\lambda}$ are the actual derivatives of orders $0, \ldots, m_\lambda - 1$, with no factorial rescaling. Define

$$J_\Xi f = (f^{(j)}(\lambda))_{\lambda \in \Lambda,\ 0 \leq j < m_\lambda}, \qquad \mathcal{V}_\Xi = J_\Xi(\mathcal{H}_{\leq 1}) \subset \mathcal{P}_\Xi. \tag{189}$$

Every point of $\mathcal{V}_\Xi$ retains the entire input's arithmetic jets. The restriction to this image is essential.

Theorem F.4 (Jet topology and its completion)**.** *The map induced by* (189) *is a topological linear isomorphism*

$$\overline{J}_\Xi : Q_\kappa \xrightarrow{\cong} \mathcal{V}_\Xi$$

onto the subspace topology from $\mathcal{P}_\Xi$. The image is dense and proper. Consequently Q_κ is incomplete, and its Hausdorff completion is canonically $\mathcal{P}_\Xi$ via this map. In contrast Q_γ is a Fréchet space and is complete. The identity on the same underlying quotient vector space induces a continuous linear bijection

$$\iota : Q_\gamma \longrightarrow Q_\kappa \tag{190}$$

whose inverse is not continuous.

Proof. We first construct each finite interpolation map with its constants. For a finite nonempty set $F \subset \Lambda$, write $d_F = \sum_{\lambda \in F} m_\lambda$ and set

$$B_{\lambda,F}(s) = \prod_{\substack{\nu \in F \\ \nu \neq \lambda}} (s - \nu)^{m_\nu}.$$

The empty product is 1. Since $B_{\lambda,F}(\lambda) \neq 0$, the germ $(s-\lambda)^j/(j! B_{\lambda,F}(s))$ is holomorphic at λ. Denote its Taylor polynomial through degree $m_\lambda - 1$, in powers of the unchanged difference $s - \lambda$, by $T_{\lambda, m_\lambda - 1}$ of that germ. Put

$$L_{\lambda,j,F}(s) = B_{\lambda,F}(s) T_{\lambda,m_\lambda-1}\left(\frac{(s-\lambda)^j}{j! B_{\lambda,F}(s)}\right), \qquad 0 \leq j < m_\lambda. \tag{191}$$

This is a polynomial of degree at most $d_F - 1$. For $\nu \in F \setminus \{\lambda\}$ it vanishes to order at least m_ν at ν. At λ its difference from $(s-\lambda)^j/j!$ vanishes to order at least m_λ. Therefore, for all $\nu \in F$ and $0 \leq k < m_\nu$,

$$L^{(k)}_{\lambda,j,F}(\nu) = \begin{cases} 1, & (\nu, k) = (\lambda, j), \\ 0, & \text{otherwise.} \end{cases}$$

It follows that the linear map to polynomials

$$H_F(v)(s) = \sum_{\lambda \in F} \sum_{j=0}^{m_\lambda - 1} v_{\lambda,j} L_{\lambda,j,F}(s) \tag{192}$$

realizes every prescribed finite array of actual derivatives $v = (v_{\lambda,j})$. For each $R > 0$ define

$$M_f(R) = \max_{|s|\le R} |f(s)|, \qquad C_{R,F} = \sum_{\lambda\in F} \sum_{j=0}^{m_\lambda - 1} M_{L_{\lambda,j,F}}(R).$$

The triangle inequality gives the exact finite bound

$$M_{H_F(v)}(R) \le C_{R,F} \max_{\lambda\in F,\, j<m_\lambda} |v_{\lambda,j}|. \tag{193}$$

When F is empty set $H_F = 0$, $C_{R,F} = 0$ and the maximum over that empty set to 0.

The kernel of J_Ξ is precisely I. Indeed $J_\Xi f = 0$ means that f/Ξ has a removable singularity at each zero of Ξ by its local Taylor expansions. Away from those zeros it is already holomorphic. Thus it extends to an entire function, which belongs to the original order class $\mathcal{H}_{\le 1}$ by Lemma F.1; consequently $f \in \Xi\mathcal{H}_{\le 1}$. The converse follows by the product rule at each zero. This proves that $\overline{J}_\Xi$ is a well-defined linear bijection onto $\mathcal{V}_\Xi$, with inverse sending the jet array of any $f \in \mathcal{H}_{\le 1}$ to its class $[f]$. The kernel calculation proves that this inverse is independent of the choice of such f.

The quotient topology on Q_κ is generated by

$$q_R([f]) = \inf_{h\in\mathcal{H}_{\le 1}} M_{f-\Xi h}(R), \qquad R > 0. \tag{194}$$

For completeness, these are seminorms because they are infima of the representatives' seminorms modulo a linear subspace. The identity $\{q_R < \epsilon\} = \pi_\kappa(\{M_f(R) < \epsilon\})$ holds for every $\epsilon > 0$, and the quotient map π_κ is open since the inverse image of the image of an open set U is the union $U + I$ of its translates. The increasing disk seminorms generate the compact-open topology; the displayed identity therefore proves the assertion about the quotient topology. Integer $R \ge 1$ suffice.

Fix $R > 0$. Choose $R' > R$ such that the circle $|s| = R'$ contains no zero of Ξ, and put $F = \{\lambda \in \Lambda : |\lambda| < R'\}$, a finite set. For $f \in \mathcal{H}_{\le 1}$ form $H = H_F((f^{(j)}(\lambda))_{\lambda\in F, j<m_\lambda})$. The function $(f - H)/\Xi$ is holomorphic throughout $|s| < R'$ by the exact finite interpolation just proved. Its Taylor polynomials $p_n(s)$ at $s = 0$ consequently converge uniformly on $|s| \le R$ to $(f - H)/\Xi$. In particular $p_n \in \mathcal{H}_{\le 1}$, and

$$M_{f-\Xi p_n}(R) \le M_H(R) + M_\Xi(R) M_{(f-H)/\Xi - p_n}(R).$$

Taking the infimum over admissible representatives and then the limit gives

$$q_R([f]) \le M_H(R) \le C_{R,F} \max_{\lambda\in F,\, j<m_\lambda} |f^{(j)}(\lambda)|. \tag{195}$$

No entire extension or growth bound for $(f - H)/\Xi$ outside this disk was used. In particular there is no additional division assumption hidden in the approximation. If F is empty, this proof gives $q_R = 0$, which is compatible with the other quotient seminorms separating points.

Conversely, fix a finite set $F \subset \Lambda$, and choose $R > \max_{\lambda\in F} |\lambda|$ and $0 < \delta < R - \max_{\lambda\in F} |\lambda|$. Cauchy's formula on the radius-δ circle about each λ gives, for every $h \in \mathcal{H}_{\le 1}$,

$$|f^{(j)}(\lambda)| = |(f - \Xi h)^{(j)}(\lambda)| \le j!\delta^{-j} M_{f-\Xi h}(R), \qquad \lambda \in F, \quad j < m_\lambda.$$

Here $0! = 1$, so this bound includes the zeroth derivative without altering its value. Taking infima gives

$$\max_{\lambda\in F,\, j<m_\lambda} |f^{(j)}(\lambda)| \le \left(\max_{\lambda\in F,\, j<m_\lambda} j!\delta^{-j} \right) q_R([f]). \tag{196}$$

Equations (195) and (196) prove continuity in both directions, with the exact stated topologies.

We next verify that the zero set needed for incompleteness is indeed infinite and has an enumeration escaping every compact set. The source Hadamard product for the original even Ξ, retaining its nonzero constant $\Xi(0)$, implies that finitely many zeros would make Ξ a polynomial. This is impossible: at the original coordinate $s = -i(2n - \frac{1}{2})$, $n \geq 1$, the defining completion gives exactly

$$\Xi(-i(2n - \tfrac{1}{2})) = \xi(2n) = n(2n-1)(n-1)!\,\pi^{-n}\zeta(2n). \tag{197}$$

For integer $n \geq 2$, $\zeta(2n) > 1$ by its positive Dirichlet series, and the last $\lfloor n/2 \rfloor$ factors of $(n-1)!$ are at least $n/2$. Thus the logarithm of the right hand side is at least

$$\log(n(2n-1)) + \lfloor n/2 \rfloor \log(n/2) - n \log \pi,$$

which exceeds $d \log(2n)$ for every fixed d and all sufficiently large n. Polynomial values of degree d on these points are bounded by a constant times $(2n)^d$, a contradiction. Since Ξ is nonzero and entire, its zeros are isolated, each multiplicity is finite and each closed disk contains finitely many distinct zeros. We may therefore enumerate all distinct zeros as $\Lambda = \{\lambda_1, \lambda_2, \ldots\}$ with $|\lambda_n| \to \infty$, retaining each m_{λ_n}. This argument uses neither reality nor simplicity of any zero.

Every projection of $\mathcal{V}_\Xi$ onto finitely many full zero-jet blocks is onto, by (192). Such finite coordinate cylinders form a base of the product topology. Consequently $\mathcal{V}_\Xi$ is dense in $\mathcal{P}_\Xi$. It is proper: define the particular array $v \in \mathcal{P}_\Xi$ by

$$v_{\lambda_n,0} = \exp(|\lambda_n|^2), \qquad v_{\lambda_n,j} = 0 \quad (1 \leq j < m_{\lambda_n}). \tag{198}$$

For each $f \in \mathcal{H}_{\leq 1}$ there is a finite constant $A_f > 0$ with $M_f(r) \leq A_f \exp(r^{3/2})$ for every $r \geq 0$; this is exactly the order-at-most-one bound, extended over a bounded interval by increasing A_f. Were $J_\Xi f = v$, we would have

$$\exp(|\lambda_n|^2) \leq A_f \exp(|\lambda_n|^{3/2}) \quad \text{for all } n,$$

which contradicts $|\lambda_n| \to \infty$.

The incompleteness can be exhibited by an actual sequence of polynomial classes. Set $F_n = \{\lambda_1, \ldots, \lambda_n\}$, $H_n = H_{F_n}(v|_{F_n})$, and $x_n = [H_n] \in Q_\kappa$. For each fixed zero-jet block, $J_\Xi H_n$ agrees with v from some index onward; hence $J_\Xi H_n \to v$ in the product. Equivalently, for each fixed R, the finite set in (195) lies in every sufficiently large F_n, so that inequality gives

$$q_R(x_n - x_k) = 0 \qquad (n, k \text{ sufficiently large}).$$

Thus (x_n) is Cauchy in Q_κ. If it converged to a class $[f]$, continuity of $\overline{J}_\Xi$ and uniqueness of limits in the product would give $J_\Xi f = v$, already disproved. For example the product is metrized by the complete translation-invariant metric

$$d(v, w) = \sum_{n=1}^{\infty} 2^{-n} \min\left(1, \max_{0 \leq j < m_{\lambda_n}} |v_{\lambda_n,j} - w_{\lambda_n,j}|\right).$$

A Cauchy sequence for this metric is Cauchy in each finite coordinate block, so has a coordinatewise limit; controlling a finite initial sum and the geometric tail proves convergence in the metric. This proves product completeness directly. The isomorphism $\overline{J}_\Xi$, being linear and continuous with a continuous linear inverse, preserves the additive uniformities. Its dense embedding in this complete product therefore gives the claimed Hausdorff completion.

Finally, we prove the completeness assertion for the stronger quotient explicitly. The growth seminorms are

$$\|f\|_{p,a} = \sup_{s \in \mathbb{C}} |f(s)| e^{-a|s|^p}, \qquad 1 < p < 2, \quad a > 0.$$

The countable family $p = 1 + 1/j$, $a = 1/k$ for integers $j \geq 2$ and $k \geq 1$ defines the same topology: for arbitrary $1 < p < 2, a > 0$ choose $p_j < p$, and the finite maximum of $r^{p_j} - ar^p$ controls $\|f\|_{p,a}$ by a constant times $\|f\|_{p_j,1}$. Each countable seminorm belongs to the original family, proving both comparisons. Enumerate this countable family as $(u_\ell)_{\ell\geq 1}$ and put $P_n = \max_{1\leq \ell\leq n} u_\ell$. If (f_n) is Cauchy in all these seminorms, it is Cauchy uniformly on each compact set, and has an entire limit f. In the weighted Cauchy inequality for any fixed seminorm, passage to the pointwise limit in the second index yields the same uniform weighted bound for $f_n - f$. Thus f has all the finite growth seminorms, lies in the original $\mathcal{H}_{\leq 1}$, and $f_n \to f$ in γ. This proves completeness of $(\mathcal{H}_{\leq 1}, \gamma)$.

The ideal I is closed: the proved kernel description is an intersection of kernels of derivative evaluations, and Cauchy's estimate shows that each of these evaluations is continuous for γ. The quotient is therefore Hausdorff. Its topology has the countable increasing seminorm family

$$\widetilde{P}_n([f]) = \inf_{h\in I} P_n(f + h).$$

Let (y_n) be a Cauchy sequence in this quotient. Choose a strictly increasing subsequence of indices n_k so far out that

$$\widetilde{P}_k(y_{n_{k+1}} - y_{n_k}) < 2^{-k-1} \quad (k \geq 1).$$

This is possible by the Cauchy property and by taking n_k past the threshold for the kth seminorm before selecting the next index. Choose a representative $d_k \in \mathcal{H}_{\leq 1}$ of each displayed difference with $P_k(d_k) < 2^{-k}$. For any fixed j, the tail $k \geq j$ satisfies $P_j(d_k) \leq P_k(d_k) < 2^{-k}$. Consequently the partial sums of $\sum_{k\geq 1} d_k$ are Cauchy in γ and converge in $\mathcal{H}_{\leq 1}$. Adding a representative of y_{n_1} gives a function whose quotient class is the limit of (y_{n_k}). The full Cauchy sequence (y_n) then has the same limit: for each quotient neighbourhood choose a smaller additive neighbourhood whose sum with itself is contained in it, and combine the Cauchy tail with a convergent subsequence term. The countable seminorm topology is metrizable, so this proves quotient completeness and the Fréchet assertion.

The identity $(\mathcal{H}_{\leq 1}, \gamma) \to (\mathcal{H}_{\leq 1}, \kappa)$ is continuous: for every R, p, a, one has $M_f(R) \leq e^{aR^p}\|f\|_{p,a}$. It preserves the identical algebraic subspace I, so the quotient universal property proves continuity and bijectivity of (190). If its inverse were continuous, it would be a continuous linear map, hence uniformly continuous for the additive uniformities. The Cauchy sequence (x_n) constructed above would then be Cauchy in the complete Q_γ, would converge there, and by continuity of ι would converge in Q_κ, a contradiction. This proves that its inverse is not continuous. No invariance of metric completeness under arbitrary nonlinear homeomorphisms has been used. □

The original coordinate action remains visible after completion. For $v = J_\Xi f$, the exact product rule gives

$$\big(J_\Xi(sf)\big)_{\lambda,j} = \lambda v_{\lambda,j} + j v_{\lambda,j-1}, \qquad 0 \leq j < m_\lambda, \tag{199}$$

where the second term is 0 when $j = 0$. Each output coordinate depends on finitely many input coordinates, so this extends to a continuous linear operator on the entire product completion. The factors j are retained because these coordinates are actual derivatives. The completion adds precisely the nonrealizable jet arrays, including (198); it does not add an entire function of order at most one representing such an array. Neither analytic quotient topology has been identified with the source topology τ.

F.2 Exact derivative domains inside the analytic relation

Let $I = \Xi\mathcal{H}_{\leq 1} = N_\gamma = N_\kappa$. For $k \geq 1$ the exact representative domain in the relation space is

$$\begin{aligned} I \cap D^{-k}I = \{\Xi h : h \in \mathcal{H}_{\leq 1}, \quad &[z^n](A_\lambda(\lambda + z)h(\lambda + z)) = 0 \\ &\text{for } \max(0, k - m_\lambda) \leq n < k, \text{ at every zero } \lambda\}. \end{aligned} \tag{200}$$

where the original local unit is $A_\lambda(s) = \Xi(s)/(s-\lambda)^{m_\lambda}$. To prove this, write $\Xi h = z^m \sum_{n\geq 0} b_n z^n$ at λ, with $m = m_\lambda$ and $b_n = [z^n](A_\lambda h)$. Its kth derivative is

$$\sum_{m+n\geq k} \frac{(m+n)!}{(m+n-k)!} b_n z^{m+n-k}.$$

Every factorial coefficient is nonzero. Vanishing to order at least m is equivalent to $b_n = 0$ exactly for $\max(0, k-m) \leq n < k$, as stated. The derivative has the original order bound, so the vanishing condition is equivalent to membership in I by (187). This proves both directions globally, including all multiplicities and local unit coefficients.

For $k = 1$, this is simply $h(\lambda) = 0$ at every distinct zero, without changing or discarding its original multiplicity in Ξ. The derivative ambiguity has the exact presentation

$$\mathcal{H}_{\leq 1}/\{h : h(\lambda) = 0 \text{ for every } \Xi(\lambda) = 0\} \xrightarrow{\cong} (DI + I)/I, \qquad [h] \longmapsto [\Xi' h]. \tag{201}$$

Indeed $D(\Xi h) = \Xi h' + \Xi' h$, and the first term lies in I. At each zero of multiplicity m, Ξ' has exactly order $m-1$, so its product with h is divisible by Ξ exactly when $h(\lambda) = 0$ for every zero. This proves the precise kernel, image, and inverse on the image in (201).

For the heat generator ($k = 2$), at any zero of multiplicity at least two the condition is $h(\lambda) = h'(\lambda) = 0$. At a simple zero it is instead

$$A_\lambda(\lambda)h'(\lambda) + A'_\lambda(\lambda)h(\lambda) = 0. \tag{202}$$

It does not require $h(\lambda) = 0$. These follow from the index range in (200): the simple zero has only the coefficient b_1 constrained. In particular the second-derivative relation domain need not be a multiplication module: at a simple zero, a local Ξh obeying (202) and with $h(\lambda) \neq 0$ has $D^2(s\Xi h)(\lambda) = 2(\Xi h)'(\lambda) \neq 0$. This is a local statement with its explicit witness condition; it does not assume the existence of a global simple zeta zero.

There is also an exact calculation of every finite projection of the ambiguity $W_k = (D^k I + I)/I$. In the factorial jet coordinates of any finite set F of distinct zeros, its image is precisely

$$\prod_{\lambda \in F} \epsilon^{\max(m_\lambda - k, 0)} \mathbb{C}[\epsilon]/(\epsilon^{m_\lambda}). \tag{203}$$

The local dimension is $\min(k, m_\lambda)$. The differentiated series above proves containment. For surjectivity, prescribe the allowed output coefficient y_j of z^j, where $\max(m-k, 0) \leq j < m$. Set $b_{j+k-m} = j!y_j/(j+k)!$ and give the other coefficients $b_0, \ldots, b_{k-1}$ arbitrary values, say zero. Finite Taylor division by the nonzero unit A_λ then specifies the first k Taylor coefficients of h at λ. These finitely many specifications can be achieved by one polynomial: put $M_\lambda(s) = \prod_{\nu \in F, \nu \neq \lambda}(s-\nu)^k$; take the Taylor polynomial of degree $k-1$ of the prescribed local polynomial divided by M_λ, and sum the resulting M_λ multiples over F. At its own node each summand agrees with the prescribed polynomial modulo $(s-\lambda)^k$, and at the other nodes it vanishes to that order. This constructs the required $h \in \mathbb{C}[s] \subset \mathcal{H}_{\leq 1}$ and proves (203). No claim about arbitrary infinite jet interpolation has replaced this exact finite image.

Theorem F.5 (Direct differential and heat obstructions)**.** *Neither N_γ nor N_κ is invariant under D or D^2. For either frozen quotient, neither representative formula $[f']$ nor $[-f''/4]$ defines an operator on any nonempty class domain admitting all representatives. For every $t \neq 0$, neither $U_t N_\gamma \subseteq N_\gamma$ nor $U_t N_\kappa \subseteq N_\kappa$ holds. Ordinary multiplication by s preserves neither $U_t N_\gamma$ nor $U_t N_\kappa$.*

Proof. Choose any zero λ of the actual Ξ and let $m \geq 1$ be its multiplicity. The existence of its zeros is part of the classical arithmetic datum in the primary source; no realness or simplicity is used. Write $\Xi = (s-\lambda)^m A$ with $A(\lambda) \neq 0$. Then $\Xi' = (s-\lambda)^{m-1}(mA + (s-\lambda)A')$

does not vanish to order m. Since $\Xi \in N_\gamma \subseteq N_\kappa$, both derivative invariance assertions fail. If $m \geq 2$, Ξ'' has leading coefficient $m(m-1)A(\lambda)$ at order $m-2$, proving the second-derivative failure. If $m = 1$, use the actual polynomial multiple $n = (s-\lambda)\Xi$ instead: $n''(\lambda) = 2\Xi'(\lambda) \neq 0$. This retains the product-rule cross term and covers all multiplicities. The nonempty-domain assertion follows from the explicit fibre computation in (172).

For $a \in \mathbb{C}$, $e^{as}\Xi \in N_\gamma$ by polynomial approximation of e^{as} and continuous multiplication, and hence belongs to N_κ. The conjugation $D(e^{as}f) = e^{as}(D+a)f$ and the absolutely convergent heat and translation series give

$$U_t(e^{as}\Xi)(s) = e^{as - ta^2/4} Z_t(s - ta/2). \tag{204}$$

For completeness, the translation series for f is its Taylor series about s and sums to $f(s-ta/2)$; its local uniform convergence and the derivative estimates justify commutation of $e^{-(ta/2)D}$ with U_t. If $U_t N_\bullet \subseteq N_\bullet$ for either choice, evaluate (204) at λ. The exponential never vanishes, so $Z_t(\lambda - ta/2) = 0$ for every a. When $t \neq 0$ these arguments fill $\mathbb{C}$; this contradicts $Z_t \neq 0$. Finally if M preserved $U_t N_\bullet$, conjugation would imply $(M + (t/2)D)N_\bullet \subseteq N_\bullet$. Since M already preserves $N_\bullet$, this forces $DN_\bullet \subseteq N_\bullet$, contrary to the first part. $\square$

The analytic moving-divisor quotient, which does carry the zeros of the actual Z_t under ordinary s, is

$$I_t = \overline{\mathbb{C}[s]Z_t}^{\kappa} = \{f \in \mathcal{H}_{\leq 1} : f/Z_t \text{ is entire}\}, \qquad A_t = (\mathcal{H}_{\leq 1}, \kappa)/I_t. \tag{205}$$

The division lemma and polynomial approximation further prove

$$I_t = Z_t \mathcal{H}_{\leq 1} = \overline{\mathbb{C}[s]Z_t}^{\gamma} \quad (t \in \mathbb{C}).$$

For a complex t with $Z_t(0) = 0$, use the retained nonzero center c in the division lemma; Z_t is not identically zero. The analytically transported relation has the different exact principal presentation for the proved transported multiplication:

$$U_t(\Xi \mathcal{H}_{\leq 1}) = Z_t \star_t \mathcal{H}_{\leq 1}. \tag{206}$$

Indeed $U_t(\Xi h) = Z_t \star_t U_t h$, and U_t is onto. The map $h \mapsto Z_t \star_t h$ is a continuous linear isomorphism of $(\mathcal{H}_{\leq 1}, \gamma)$ onto this closed relation; its inverse on that relation is $n \mapsto U_t((U_{-t}n)/\Xi)$, continuous by the division and heat lemmas. Thus the two multiplications and their exact relation spaces are connected by a proved isomorphism with a proved inverse. The proof of Theorem F.3 applies verbatim with the nonzero entire order-at-most-one function Z_t: it uses only local Taylor division and compact polynomial approximation. Consequently $\mathrm{Spec}(M \text{ on } A_t) = \{\lambda : Z_t(\lambda) = 0\}$, with the full local multiplicity jets, and with the explicit inverse $[f] \mapsto [(f - f(\lambda)Z_t/Z_t(\lambda))/(s-\lambda)]$ off that set. Equation (171) is the evolution of this original-coordinate spectrum on its simple-zero locus. This spectrum can move; (167) is the spectrum of the different, exactly specified transported operator B_t.

We now give the promised comparison morphisms with every kernel retained. Put $J = N_{\mathrm{ar}} \cap N_\kappa$ and $C = (\mathcal{H}_{\leq 1}, \tau \vee \kappa)/J$, where $\tau \vee \kappa$ is the topology generated by both topologies. Since each summand is closed in its own topology, J is closed in their join. The identity maps give continuous surjections

$$Q_{\mathrm{ar}} \longleftarrow C \longrightarrow Q_\kappa, \qquad \ker(C \to Q_{\mathrm{ar}}) = N_{\mathrm{ar}}/J, \quad \ker(C \to Q_\kappa) = N_\kappa/J. \tag{207}$$

Each assertion follows by taking the same representative f and testing membership in the respective relation subspace; the quotient universal property proves continuity. An algebraic common quotient is $\mathcal{H}_{\leq 1}/(N_{\mathrm{ar}} + N_\kappa)$ with the quotient topology from the join; its Hausdorff version uses $\overline{N_{\rm ar}+N_\kappa}^{\tau\vee\kappa}$. The map from C to this further quotient is continuous. Continuity of maps to it from each of the two separately topologized quotients is not asserted. No closedness of the sum is presumed. This describes the precise information lost in that further quotient.

At time t replace the two subspaces by N_t and I_t and the join by $\tau_t\vee\kappa$. The same construction gives continuous surjections from their intersection quotient to the genuine transported arithmetic quotient Q_t and to A_t, with kernels $N_t/(N_t\cap I_t)$ and $I_t/(N_t\cap I_t)$, respectively. The two generator systems are linked by the explicit formula $R(B_t)Z_t$ versus $R(s)Z_t$ and the defects (169)–(170). These are exact morphisms and closure formulas for the actual source space. Determining that either kernel vanishes for τ is not established by the two inspected primary texts. Accordingly the source-frozen statements $DN_{\rm ar}\subseteq N_{\rm ar}$ and $D^2N_{\rm ar}\subseteq N_{\rm ar}$ have not been decided solely from that unnamed topology. The unqualified noninvariance results above have been proved in the two explicitly specified analytic realizations.

G Arithmetic traces and their exact Mellin obstructions

G.1 The full Schwartz trace and arithmetic principal parts

Retain the source domain

$$\mathcal{S}_0^{\rm ev}=\{\psi\in\mathcal{S}(\mathbb{R}):\psi(-x)=\psi(x),\ \psi(0)=\widehat{\psi}(0)=0\},\qquad \widehat{\psi}(y)=\int_{\mathbb{R}}\psi(x)e^{-2\pi ixy}\,dx.$$

Its full trace map takes values in entire functions:

$$\begin{aligned}F_\psi(s)&=\mathcal{F}_\mu\mathcal{E}\psi(s),\\ F_\psi(i(z-\tfrac12))&=\zeta(z)\mathcal{M}\psi(z),\quad \mathcal{M}\psi(z)=\int_0^\infty\psi(x)x^z\,\frac{dx}{x}.\end{aligned}\tag{208}$$

The Mellin integral is holomorphic for $\operatorname{Re}z>-2$, and the factorization holds there with its removable value at $z=1$. This statement does not impose an unproved order bound on the entire F_ψ for arbitrary Schwartz inputs.

Here is a direct proof with all analytic domains retained. For fixed $x>0$, periodize ψ by the smooth function

$$y\longmapsto\sum_{k\in\mathbb{Z}}\psi(x(k+y)).$$

Its Fourier coefficients are $x^{-1}\widehat{\psi}(\ell/x)$: integrate on $y\in[0,1]$, substitute $v=x(k+y)$, and sum the adjacent intervals, justified by Schwartz decay. These coefficients decrease faster than every power of ℓ, so their Fourier series converges with all derivatives and at $y=0$ gives the Poisson identity. Evenness and the two exact zero conditions yield

$$\sum_{n\geq1}\psi(nx)=x^{-1}\sum_{n\geq1}\widehat{\psi}(n/x),\qquad \mathcal{E}\psi(x)=\mathcal{E}\widehat{\psi}(1/x).$$

For $x\geq1$, a Schwartz bound of arbitrarily high order gives $|\sum\psi(nx)|\leq C_Ax^{-A}\sum n^{-A}$ for $A>1$. The same estimate after termwise differentiation follows by choosing the Schwartz exponent larger than the derivative order plus one. The Poisson identity supplies the corresponding bounds at zero. Thus $\mathcal{E}\psi(e^u)$ decreases faster than $e^{-A|u|}$ for every $A>0$ at both ends. Its Fourier integral and each derivative in s are dominated locally uniformly in s by an integrable function. This proves that F_ψ is entire.

Since ψ is even and $\psi(0)=0$, its Taylor formula gives $\psi(x)=O(x^2)$ at zero. Together with Schwartz decay at infinity this proves the stated holomorphy of $\mathcal{M}\psi$ by dominated differentiation, including all powers of $\log x$. For $\sigma = \operatorname{Re} z > 1$,

$$\sum_{n\geq 1} \int_0^\infty |\psi(nx)| x^\sigma \frac{dx}{x} = \Big(\sum_{n\geq 1} n^{-\sigma}\Big) \int_0^\infty |\psi(y)| y^\sigma \frac{dy}{y} < \infty.$$

Absolute interchange and the original exponent x^{-is} now give (208) on this half-plane. Moreover $\mathcal{M}\psi(1) = \int_0^\infty \psi(x)\,dx = \widehat{\psi}(0)/2 = 0$. This cancels the simple pole of ζ at 1, so the right side is holomorphic for $\operatorname{Re} z > -2$. The identity theorem extends the equality to that entire half-plane, proving the assertion.

Let $\mathcal{Z}$ denote the set of distinct nontrivial zeros ρ of ζ, with their original multiplicities m_ρ. The evenness of Ξ, immediate from the retained cosine integral, gives the exact reflected-coordinate identity

$$s = i(z - \tfrac{1}{2}), \qquad z = \tfrac{1}{2} - is, \qquad \Xi(i(z - \tfrac{1}{2})) = \xi(1 - z) = \xi(z). \tag{209}$$

At each $\rho \in \mathcal{Z}$ the completion factor $z(z-1)\pi^{-z/2}\Gamma(z/2)/2$ is a nonvanishing holomorphic unit. Hence $\lambda_\rho = i(\rho - \frac{1}{2})$ is a zero of Ξ of exactly multiplicity m_ρ, and these are all its zeros. No reflection of the retained spectral operator s has been performed: (209) is the exact Mellin parameter dictionary for that same operator. The factorization (208) proves

$$\begin{gathered} D^j F_\psi(\lambda_\rho) = 0 \quad (\rho \in \mathcal{Z},\ 0 \leq j < m_\rho), \\ F_\psi \in \mathcal{H}_{\leq 1} \implies F_\psi \in \Xi\mathcal{H}_{\leq 1}. \end{gathered} \tag{210}$$

The first assertion follows because $\mathcal{M}\psi$ is holomorphic at every ρ and $s - \lambda_\rho = i(z - \rho)$; the second then follows from the proved division theorem. It is an exact intersection assertion about the full Schwartz trace, without enlarging the order-at-most-one domain or closing it in τ.

For each ρ retain the actual local unit $a_\rho(z) = \zeta(z)/(z-\rho)^{m_\rho}$ and the space

$$\mathcal{P}_\rho^- = \left\{ \sum_{j=1}^{m_\rho} c_{-j}(z-\rho)^{-j} : c_{-j} \in \mathbb{C} \right\}.$$

The principal-part map on the original entire-function set is

$$\mathfrak{P} : \mathcal{H}_{\leq 1} \longrightarrow \prod_{\rho \in \mathcal{Z}} \mathcal{P}_\rho^-, \qquad F \longmapsto \left(\mathrm{PP}_{z=\rho} \frac{F(i(z - \frac{1}{2}))}{\zeta(z)} \right)_\rho. \tag{211}$$

The kernel of this map is exactly $\Xi\mathcal{H}_{\leq 1}$. It induces an injective linear map from the underlying vector space of each Q_γ, Q_κ onto the same specified image. Every finite projection is surjective. The map from Q_κ to its image, with the product subspace topology, is a topological linear isomorphism; the product of all principal-part spaces is its completion. From Q_γ it is a continuous bijection onto that image whose inverse is not continuous.

To prove each claim while retaining the analytic units, write $a_\rho(\rho + w)^{-1} = \sum_{\ell \geq 0} \alpha_{\rho,\ell} w^\ell$. The exact Laurent coefficient at degree $-m_\rho + q$, $0 \leq q < m_\rho$, of the ρ component is

$$c_{-m_\rho + q} = \sum_{j=0}^{q} \frac{i^j}{j!} F^{(j)}(\lambda_\rho) \alpha_{\rho, q-j}. \tag{212}$$

This follows by multiplying the two convergent local Taylor series and the retained factor w^{-m_ρ}. Its finite matrix is triangular with nonzero diagonal entries $i^j/(j!a_\rho(\rho))$. Explicitly, given a prescribed principal part $P_\rho(w)$, take the coefficients b_j through degree $m_\rho - 1$ of $a_\rho(\rho + w) w^{m_\rho} P_\rho(w)$; the required original derivatives are $j! i^{-j} b_j$. Multiplying back by a_ρ^{-1}

recovers the prescribed negative powers, since the discarded error is divisible by w^{m_ρ}. This proves a finite-dimensional linear isomorphism at every zero, including its inverse and every factorial and power of i.

The vanishing of all principal parts is therefore equivalent to the vanishing of all the original multiplicity jets of F. Theorem F.2 identifies that kernel exactly with $\Xi\mathcal{H}_{\le 1}$. At any finite set of zeros, the required derivatives can be realized by the explicit Hermite lift (192); this proves surjectivity of every finite principal-part projection. Each block map and its inverse are continuous because their finite matrices have fixed complex coefficients. Their product is therefore a homeomorphism of the two product spaces: the inverse image of a cylinder fixing finitely many coordinates is again controlled by those finitely many blocks. Applying this product isomorphism to Theorem F.4 proves all stated topology, completion, and comparison assertions for $\mathfrak{P}$.

This map also carries the original spectral action exactly. For $P_\rho = \mathfrak{P}_\rho(F)$ one has

$$\mathfrak{P}_\rho(sF) = \mathrm{PP}_{w=0}((\lambda_\rho + iw)P_\rho(w)). \tag{213}$$

Indeed the full Laurent quotient for sF is the quotient for F multiplied by $i(z - \frac{1}{2}) = \lambda_\rho + iw$. Its holomorphic residual contributes no negative power after this multiplication. Thus the finite block is the retained λ_ρ plus the explicitly specified nilpotent operator $P \mapsto \mathrm{PP}(iwP)$, whose nilpotency index is m_ρ: iteration on w^{-m_ρ} gives a nonzero multiple of w^{-1} at step $m_\rho - 1$ and zero at step m_ρ.

For $I = \Xi\mathcal{H}_{\le 1}$, the derivative ambiguity $W_k = (D^k I + I)/I$ has, under $\mathfrak{P}$, the exact finite image

$$\prod_{\rho\in F} \left\{ \sum_{j=1}^{\min(k,m_\rho)} c_{-j}(z-\rho)^{-j} : c_{-j} \in \mathbb{C} \right\} \quad (F \subset \mathcal{Z} \text{ finite}). \tag{214}$$

The finite image in (203) consists exactly of numerator jets divisible by $w^{\max(m_\rho - k, 0)}$ after the invertible substitution $s - \lambda_\rho = iw$. Multiplication by the nonvanishing unit a_ρ^{-1} preserves this divisibility and is invertible on the truncated Taylor ring. Division by w^{m_ρ} therefore yields exactly all negative powers of order at most $\min(k, m_\rho)$, proving both containment and surjectivity in the displayed finite image. In particular the complete local-unit expression, rather than only a count of vanishing orders, transports the derivative defect to the arithmetic Mellin poles.

G.2 An exact arithmetic inverse for every heated Hermite trace

Let ψ be any finite linear combination of the original $\psi_\ell^\pm$, retaining all its coefficients, and write $F_0 = \mathcal{F}_\mu \mathcal{E}\psi = p\Xi$. For $x > 0$ and $t \in \mathbb{C}$ define

$$\begin{aligned} q(x) &= \sum_{n\ge 1} \psi(nx), & g_t(x) &= e^{t(\log x)^2/4}\mathcal{E}\psi(x), \\ q_t(x) &= x^{-1/2} g_t(x) = e^{t(\log x)^2/4} q(x), & \psi_t(x) &= \sum_{n\ge 1} \mu(n) q_t(nx). \end{aligned} \tag{215}$$

Here $\mu(1) = 1$, $\mu(n) = 0$ when a prime square divides n, and $\mu(n) = (-1)^j$ for a product of j distinct primes. For the inverse ψ_t, the symbol $\mathcal{E}\psi_t$ means the absolutely convergent trace-sum extension $x^{1/2}\sum_{m\ge 1} \psi_t(mx)$ on the positive half-line. Its domain here consists of smooth functions with rapid decrease at infinity; no Schwartz extension through zero is required. The transform and inverse assertions, with their domains, are

$$\mathcal{E}\psi_t = g_t, \qquad \mathcal{F}_\mu g_t = U_t(p\Xi) =: F_t, \qquad \int_0^\infty \psi_t(x) x^z \, \frac{dx}{x} = \frac{F_t(i(z-\frac{1}{2}))}{\zeta(z)} \quad (\operatorname{Re} z > 1). \tag{216}$$

The series for ψ_t converges locally with all derivatives on $(0, \infty)$, and ψ_t and all its derivatives decrease faster than every inverse power at infinity. No behavior at zero has been assumed for this inverse.

We prove all analytic assertions in these formulas. A finite Hermite combination and its Fourier transform are polynomials times the original Gaussian $e^{-\pi x^2}$. Thus at infinity, for every derivative order j, their derivatives are bounded by $C_j(1+x)^{d_j}e^{-\pi x^2}$. For $x \geq 1$, termwise derivatives of $\sum \psi(nx)$ are bounded by a sum of $C_j n^j(1+nx)^{d_j}e^{-\pi n^2 x^2}$; splitting the Gaussian into two equal exponential factors bounds that sum by $C'_j(1+x)^{d_j}e^{-\pi x^2/2}$. This also justifies the differentiation. Both zero conditions give, by the Poisson identity,

$$\mathcal{E}\psi(x) = \mathcal{E}\widehat{\psi}(1/x).$$

Consequently, in $u = \log x$, the functions $\mathcal{E}\psi(e^u)$ and all their u derivatives have bounds of the form $C\exp(C'|u| - ce^{2|u|})$ on the two ends. Multiplication by $e^{tu^2/4}$, and any finite number of its derivatives, preserves integrability against $e^{A|u|}$ for every $A > 0$, locally uniformly for t in compact subsets of $\mathbb{C}$. It follows that $\mathcal{F}_\mu g_t$ is entire in both t, s. Since $D_s^{2k}e^{-isu} = (-1)^k u^{2k}e^{-isu}$, absolute termwise integration of the time exponential gives exactly

$$\int_{\mathbb{R}} e^{tu^2/4}\mathcal{E}\psi(e^u)e^{-isu}\,du = \sum_{k\geq 0}\frac{(-t/4)^k}{k!}D_s^{2k}(p(s)\Xi(s)) = F_t(s).$$

The heat lemma identifies the same series in γ; hence $F_t \in \mathcal{H}_{\leq 1}$ with the original coordinate and coefficient.

The proved estimates imply that q_t and all its x derivatives decrease faster than every power at infinity, and that q_t decreases faster than every power at zero. On a compact interval $x \in [\delta, L] \subset (0, \infty)$, choose $A > j+1$ to bound the jth derivative of the nth inverse summand by $Cn^j(nx)^{-A} \leq C\delta^{-A}n^{j-A}$. The finitely many arguments below one are harmless. This proves local convergence with derivatives. On $x \geq 1$ the same bound, with arbitrarily large A, proves rapid decrease of each derivative. For a fixed $x > 0$ the double sum used below is absolutely bounded by

$$\sum_{m,n\geq 1}|q_t(mnx)| = \sum_{k\geq 1}d(k)|q_t(kx)| < \infty,$$

where $d(k) \leq k$ and rapid decrease permits an exponent greater than two. Unique prime factorization gives $\sum_{n|k}\mu(n) = 1$ for $k = 1$ and $\prod_{p|k}(1-1) = 0$ for $k > 1$. Therefore

$$\sum_{m\geq 1}\psi_t(mx) = \sum_{k\geq 1}\Big(\sum_{n|k}\mu(n)\Big)q_t(kx) = q_t(x).$$

Multiplication by the retained factor $x^{1/2}$ proves the first identity of (216). It also proves uniqueness of this inverse among functions decreasing faster than every power at infinity: for any such v with $\sum_m v(mx) = q_t(x)$, the same absolutely convergent double sum applied to $\sum_n \mu(n)\sum_m v(mnx)$ equals $v(x)$.

For $\sigma = \operatorname{Re} z > 1$, absolute integration gives

$$\sum_{n\geq 1}|\mu(n)|\int_0^\infty |q_t(nx)|x^\sigma\,\frac{dx}{x} \leq \Big(\sum_{n\geq 1}n^{-\sigma}\Big)\int_0^\infty |q_t(y)|y^\sigma\,\frac{dy}{y} < \infty.$$

The divisor identity also gives $(\sum \mu(n)n^{-z})\zeta(z) = 1$ by absolute multiplication of the two Dirichlet series in this half-plane. Finally

$$\int_0^\infty q_t(x)x^z\,\frac{dx}{x} = \int_0^\infty g_t(x)x^{z-1/2}\,\frac{dx}{x} = F_t(i(z-\tfrac{1}{2})),$$

because $-i\,i(z-\frac{1}{2}) = z-\frac{1}{2}$. These equalities prove the last identity of (216), including its absolute domain.

At a nontrivial zero ρ of ζ of multiplicity m, put $\lambda = i(\rho-\frac{1}{2})$ and retain $\zeta(z) = (z-\rho)^m a_\rho(z)$ with $a_\rho(\rho) \neq 0$. The retained cosine integral makes Ξ even, so its definition gives

$$\Xi(i(z-\tfrac{1}{2})) = \xi(1-z) = \xi(z).$$

At a nontrivial zero, the factor $z(z-1)\pi^{-z/2}\Gamma(z/2)/2$ is a nonvanishing holomorphic unit. Thus this reflected coordinate sends the zero at ρ to the zero at λ with exactly the same multiplicity. The meromorphic continuation of the last quotient has pole order

$$\max\{m - \operatorname{ord}_\lambda F_t, 0\}. \tag{217}$$

For the zero seed $F_t = 0$, the quotient is zero and its pole order is zero; the displayed formula uses $\operatorname{ord}_\lambda 0 = +\infty$. Indeed $s-\lambda = i(z-\rho)$, so its numerator has exactly the original order $\operatorname{ord}_\lambda F_t$, with its leading coefficient multiplied by the nonzero power of i. The unit a_ρ is retained in every principal coefficient.

For every fixed $t \neq 0$ and every zero λ of Ξ, there is a polynomial p for which $F_t(\lambda) \neq 0$: otherwise all $U_t(s^n\Xi)$ would vanish there, contrary to the no-common-zero proof in Section D. The exact polynomial-image calculation supplies a finite original Hermite seed for this p. For that seed the unique inverse ψ_t in (215) is unbounded near zero. To prove this last assertion, boundedness near zero together with its proved rapid decrease at infinity would make its Mellin integral holomorphic in $\operatorname{Re} z > 0$: on compact subsets, derivatives in z are dominated by powers of $|\log x| x^{\sigma-1}$ near zero with $\sigma > 0$. Its equality in $\operatorname{Re} z > 1$ would then continue as a holomorphic function through every nontrivial ρ, contradicting the pole of order m just proved at $\rho = \frac{1}{2} - i\lambda$. Thus for each nonzero heat time some actual finite Hermite trace has an explicit arithmetic inverse which fails even boundedness at the origin. This proves a concrete obstruction to Schwartz descent with a unique inverse and its poles; it does not replace the unresolved closure in τ by a different closure.

H The inverse-root cover, the branch locus, and arithmetic fibres

The retained polynomial inverse-fibre slice is $P_{a,0,0}(r) = -2r^2 - 2a$. Its root equation is $a = -r^2$. The arithmetic coordinate is $a = 2s$, so

$$s = -r^2/2. \tag{218}$$

The finite free extension of the actual quotient is

$$\widehat{Q} = \mathbb{C}[r] \otimes_{\mathbb{C}[s]} Q_{\mathrm{ar}} = Q_{\mathrm{ar}} \oplus rQ_{\mathrm{ar}}, \qquad \mathcal{R} = \begin{pmatrix} 0 & -2S \\ 1 & 0 \end{pmatrix}, \qquad -\tfrac{1}{2}\mathcal{R}^2 = \begin{pmatrix} S & 0 \\ 0 & S \end{pmatrix}.$$

The even–odd decomposition of a polynomial in r proves freeness with basis $(1, r)$ and uniqueness coefficient by coefficient. Multiplying $q_0 + rq_1$ by r proves the displayed matrices; the finite direct-sum topology makes every induced map continuous. The involution is $(q_0, q_1) \mapsto (q_0, -q_1)$; the trace is $\operatorname{Tr}(q_0 + rq_1) = 2q_0$, with kernel rQ_{ar}. It is $\mathbb{C}[s]$-linear. It does not provide a $\mathbb{C}[r]$-module quotient with this kernel unless $S = 0$: multiplying rq by r gives $-2Sq$ in the even summand. This proves the exact limitation, rather than suppressing the odd component. The same construction at time t uses S_t and is conjugated by $\operatorname{diag}(\mathcal{U}_t, \mathcal{U}_t)$.

The original s-spectrum is unchanged by the extension: an inverse of $S - \lambda$ duplicates to a diagonal inverse, and conversely projection of an inverse of the doubled matrix to its first component, following inclusion of that component, gives an inverse of $S-\lambda$. Both directions are continuous. Moreover

$$\mathrm{Spec}(\mathcal{R}) = \{\rho \in \mathbb{C} : -\rho^2/2 \in \mathrm{Spec}(S)\}. \tag{219}$$

To prove this without an unsupported spectral-mapping citation, solve $(\mathcal{R}-\rho)(u,v) = (a,b)$. The second row gives $u = b+\rho v$ and the first becomes $-2(S+\rho^2/2)v = a+\rho b$. A continuous inverse of $S+\rho^2/2$ therefore gives a continuous inverse matrix. Conversely the right-hand side $(a,0)$ and the second projection of an inverse matrix supply an inverse of $-2(S+\rho^2/2)$, with both identities checked by the same two rows.

Before any quotient by Ξ, let $\mathcal{O}_s$ be the sheaf of holomorphic functions in s and put $B = \mathcal{O}_s[r]/(r^2+2s)$, identified with $\mathcal{O}_s \oplus r\mathcal{O}_s$. On the punctured base $s \neq 0$, implicit differentiation gives $dr/ds = -1/r = r/(2s)$. Thus the actual lifted derivative is

$$\nabla_s \begin{pmatrix} f_0 \\ f_1 \end{pmatrix} = \begin{pmatrix} f_0' \\ f_1' + f_1/(2s) \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -2s \\ 1 & 0 \end{pmatrix}, \quad [\nabla_s, R] = -R^{-1}. \tag{220}$$

Indeed with $A = \mathrm{diag}(0, 1/(2s))$, $R' + AR - RA = \left(\begin{smallmatrix} 0 & -1 \\ 1/(2s) & 0 \end{smallmatrix}\right) = -R^{-1}$. A second application gives

$$\nabla_s^2 = \mathrm{diag}\left(D^2, D^2 + s^{-1}D - \frac{1}{4s^2}\right). \tag{221}$$

Equivalently, on a function of r the derivative is $-r^{-1}\partial_r$ and the heat generator is $-\frac{1}{4}r^{-2}\partial_r^2 + \frac{1}{4}r^{-3}\partial_r$. Squaring $-r^{-1}\partial_r$ by the product rule proves this with both signs preserved.

The branch extension is exact. The meromorphic connection sends B into $s^{-1}B$ and the connection one-form has a simple pole with residue $\mathrm{diag}(0, 1/2)$. Its maximal domain for holomorphic output at $s=0$ is

$$\mathrm{Dom}_{\mathrm{hol}}\, \nabla_s = \mathcal{O}_s \oplus rs\mathcal{O}_s. \tag{222}$$

In fact the odd coefficient is $f_1' + f_1/(2s)$; its only possible pole has coefficient $f_1(0)/2$, proving necessity and sufficiency. For the square, write $f_1 = \sum_{n\ge 0} a_n s^n$. Its image has coefficient $(n^2 - 1/4)a_n s^{n-2}$. The negative powers vanish exactly when $a_0 = a_1 = 0$, so

$$\mathrm{Dom}_{\mathrm{hol}}\, \nabla_s^2 = \mathcal{O}_s \oplus rs^2\mathcal{O}_s.$$

For k iterates the odd coefficient is $\prod_{h=0}^{k-1}(n+1/2-h)a_n s^{n-k}$, whose prefactor is never zero for integral n. Thus its domain is $\mathcal{O}_s \oplus rs^k\mathcal{O}_s$. No derivation $B \to B$ extending ∂_s exists at the branch: derivation of $r^2+2s = 0$ would give $2r\delta(r) + 2 = 0$, forcing r to be a unit although it vanishes at the branch point. The logarithmic derivative $s\nabla_s$ does extend to all of B, with matrix $sD + \mathrm{diag}(0, 1/2)$.

These are sheaf and representative calculations; they do not assert that ∇_s descends through the frozen arithmetic relation. For a holomorphic local multiplier $R_0(s)$ the relation vector $(R_0\Xi, 0)$ acquires $(R_0'\Xi + R_0\Xi', 0)$, and the odd relation $(0, R_0\Xi)$ acquires $(0, R_0'\Xi + R_0\Xi' + R_0\Xi/(2s))$. The second iterate includes the full $2R_0'\Xi'$ term, as well as the odd $s^{-1}D - 1/(4s^2)$ terms of (221). Jet extension (181), or the moving closed quotient (164), retains these relations exactly.

Finally, the branch point is not an arithmetic spectral point at $t=0$. The source gives $\Xi(0) \neq 0$, so its Proposition 3.6 makes S continuously invertible; then $\mathcal{R}$ is invertible by (219). In the analytic divisor realization the stronger local statement follows from Ξ being a holomorphic unit on a disk about zero: the quotient stalk is zero there. For real t, $Z_t(0) > 0$ directly from the retained kernel:

$$2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u} = \pi n^2 e^{5u}(2\pi n^2 e^{4u} - 3) > 0 \quad (u \ge 0),$$

and the positive integral converges by the bound already proved. Thus the actual real heat divisor never meets the cover branch point $s = 0$. On a simple zero curve its retained lift satisfies $r^2 = -2\lambda$, $r' = -\lambda'/r$; together with (171) these are the original-coordinate and cover equations, including the unit contribution. A nonreal r alone does not specify whether $\lambda = -r^2/2$ is nonreal.

I Exact ingress from compact Weil tests

The accompanying retained Weil-test calculation supplies the concrete space $\mathcal{E} = C_c^\infty(\mathbb{R}; \mathbb{C})$ with coordinate operator $T = -i\partial_u$ and Fourier transform

$$(\mathscr{F} f)(s) = \int_{\mathbb{R}} f(u) e^{-ius}\, du.$$

Here we give the complete heat ingress and its inverse domains; none of the following claims requires identifying the source quotient topology with a test-space topology. Integration by parts gives $\mathscr{F}(Tf) = s\mathscr{F} f$ with no endpoint term. If $\operatorname{supp} f \subset [-L, L]$, then

$$|\mathscr{F} f(s)| \le \|f\|_1 e^{L|\operatorname{Im} s|}, \qquad \|\mathscr{F} f\|_{p,a} \le \|f\|_1 \sup_{r\ge 0} e^{Lr - ar^p} < \infty.$$

Thus the Fourier transform maps the actual test space into the original entire-function set, continuously in γ on every fixed-support smooth-function space and hence in its usual inductive-limit topology.

Define, for all complex t,

$$(G_t f)(u) = e^{tu^2/4} f(u), \qquad U_t \mathscr{F} f = \mathscr{F} G_t f, \qquad T_t = G_t T G_{-t} = T + \frac{it}{2} u. \tag{223}$$

The multiplication G_t is a continuous automorphism on each fixed-support smooth-function space, with inverse G_{-t}. Indeed each derivative of the smooth multiplier is bounded on the fixed compact support, and Leibniz's finite sum bounds every output derivative seminorm by finitely many input ones. The same argument for the inverse proves the assertion. The exponential series and every u-derivative converge uniformly on a fixed compact set, locally uniformly in t; consequently the family is entire in time in that test topology. In the Fourier integral, $D^{2k}\mathscr{F} f = \mathscr{F}((-1)^k u^{2k} f)$. The compact support permits absolute interchange with the exponential series, proving the middle identity with the exact factor $1/4$. Finally differentiation of $e^{-tu^2/4} f$ gives the displayed T_t, and

$$\mathscr{F}(T_t f) = B_t \mathscr{F} f, \qquad D\mathscr{F} f = \mathscr{F}(-iuf).$$

Both equations follow directly from the integral, including their signs. The test-space version of the retained cover is therefore

$$\mathcal{R}_t^{\text{test}} = \begin{pmatrix} 0 & -2T_t \\ 1 & 0 \end{pmatrix}, \qquad (\mathcal{R}_t^{\text{test}})^2 = -2\operatorname{diag}(T_t, T_t).$$

Componentwise Fourier transformation intertwines this matrix with the representative matrix using B_t, and then with the actual quotient matrix using S_t.

To retain every possible arithmetic relation in this ingress, put

$$K = \{f \in \mathcal{E} : \mathscr{F} f \in N_{\text{ar}}\}, \qquad K_t = \{f \in \mathcal{E} : \mathscr{F} f \in N_t\}. \tag{224}$$

Then $K_t = G_t K$: by (223), $\mathscr{F} f \in U_t N_{\text{ar}}$ exactly when $\mathscr{F}(G_{-t} f) \in N_{\text{ar}}$. Thus there are injective linear maps onto their explicitly specified images

$$\mathcal{E}/K \longrightarrow Q_{\text{ar}}, \quad [f] \longmapsto [\mathscr{F} f], \qquad \mathcal{E}/K_t \longrightarrow Q_t, \quad [f] \longmapsto [\mathscr{F} f],$$

and the square formed by G_t and $\mathcal{U}_t$ commutes. The kernel statements follow exactly from (224); no claim that $K = 0$ in the unnamed source topology is inserted. To make the ingress continuous without that identification, give $\mathcal{E}$ the join η of its usual test topology with the inverse-image topology of τ under $\mathscr{F}$. Then K is closed and the first quotient map is continuous. Give the time-t test space the transported topology $(G_t)_*\eta$; the second map and the commutative square are continuous by (223). This specifies the exact topologies rather than claiming an unproved topological embedding of a quotient image equipped with its subspace topology.

There is a fully explicit inverse-domain calculation in the same test coordinates. For $\lambda \in \mathbb{C}$, $k \geq 1$, the image is

$$(T_t - \lambda)^k \mathcal{E} = \left\{ g \in \mathcal{E} : \int_{\mathbb{R}} v^j e^{-tv^2/4 - i\lambda v} g(v)\, dv = 0 \quad (0 \leq j < k) \right\}. \tag{225}$$

On that precise image the inverse is

$$[(T_t - \lambda)^{-k} g](u) = \frac{i^k e^{tu^2/4 + i\lambda u}}{(k-1)!} \int_{-\infty}^{u} (u-v)^{k-1} e^{-tv^2/4 - i\lambda v} g(v)\, dv. \tag{226}$$

To prove both directions, write $f = e^{tu^2/4 + i\lambda u} h$. Direct differentiation gives $(T_t - \lambda) f = e^{tu^2/4 + i\lambda u}(-ih')$. Iteration reduces the inverse equation to a kth primitive, giving (226) because $(-i)^k i^k = 1$. For g supported in $[a, b]$, that formula vanishes below a. Above b, expansion of $(u - v)^{k-1}$ shows it vanishes identically exactly when all k moments in (225) vanish; the binomial coefficients and the nonzero exponential do not change that equivalence. Its smoothness follows by repeated differentiation of the integral; smoothness of g with compact support gives matching derivatives at both endpoints. Necessity for any compact solution also follows by integrating $h^{(k)}$ against $1, v, \ldots, v^{k-1}$ and integrating by parts k times. Uniqueness follows since a compactly supported solution of $h^{(k)} = 0$ is a polynomial of degree at most $k - 1$ which vanishes on a ray, and hence is zero. This proves the exact image, inverse, kernel zero, and preservation of the original support interval.

In particular the inverse of $\mathcal{R}_t^{\mathrm{test}}$ is

$$(g_0, g_1) \longmapsto (g_1, -\tfrac{1}{2} T_t^{-1} g_0), \qquad \mathrm{Dom}(\mathcal{R}_t^{\mathrm{test}})^{-1} = \left\{ (g_0, g_1) : \int e^{-tv^2/4} g_0(v)\, dv = 0 \right\}.$$

Substitution into its two rows proves both inverse identities. Thus the branch-locus obstruction in the concrete test realization is a retained Gaussian-weighted zero-moment condition. The exact arithmetic quotient instead has invertible S_t and root operator, as proved above. Their ingress map and its kernels explain precisely how these statements coexist; one operator's inverse domain is not silently assigned to the other realization.

J Proof scope and reproducibility

The new source-topology assertions proved here are the complete transported closure, homeomorphic quotient, conjugated spectral operator, source-domain differential correspondence, exact coordinate kernels and cokernels, correlated jet extension, finite free cover, compact-test heat ingress with its weighted-moment inverse domains, and exact comparison diagrams. The compact and growth relation subspaces both equal $\Xi\mathcal{H}_{\leq 1}$ by the complete division proof. Their quotient topologies are computed: the compact quotient has its exact jet-image topology and product completion, whereas the growth quotient is complete and the comparison inverse is not continuous. All derivative relation domains and finite ambiguity projections retain the full zero multiplicities and analytic units. The arithmetic principal-part realization has the exact kernel $\Xi\mathcal{H}_{\leq 1}$, every finite image, the same product completion, and explicit original-coordinate spectral blocks. The full Schwartz trace factorization and its order-class intersection are proved without assigning every Schwartz trace to that order class. Both explicit realizations have proved frozen differential and heat noninvariance and the original-coordinate divisor spectrum. Each heated finite Hermite trace also has its proved unique Möbius inverse, precise Mellin domain and full nontrivial-zero pole orders, with actual failures of boundedness at the origin for suitable original seeds at every nonzero time. The missing identification of the paper's unnamed topology is recorded explicitly and is not replaced by a conditional arithmetic theorem. Nothing in these results establishes an off-critical zero, a negative Weil value, or an arithmetic endpoint by analogy with the inverse-fibre collision. The exact transport through that collision cover and the derivative information it requires have been calculated above.

The companion checker uses rational polynomial arithmetic to verify the heat conjugation on finite polynomial inputs, the retained product terms, the matrix commutator, the iterated branch-domain coefficients, the original-coordinate relation, and sample finite jet multiplication, the multiplicity-sensitive derivative domains, the source heat commutator recurrence, exact Möbius divisor sums, and Mellin principal-part coordinate inverses. These are reproducible checks of the displayed identities, not a substitute for their written proofs. Source hashes and exact reading locators are in the companion provenance receipt; no numerical zero search or Lean run is part of this work.

Using the released Navier–Stokes construction as the arithmetic input

The precise imported witness and its unchanged viscosity

We now use the particular field constructed in the released manuscript [8], not merely an arbitrary member of the input class of Section 16. The complete 166-page source proof is retained in this repository as `sources/ns_public/navier_stokes_166.pdf`; its SHA-256 is recorded in the source provenance. The source existence and all-order residual construction are imported from that manuscript. Our independent reading of its witness construction and closing proof is recorded in `logbook/ns_witness_source_read.md`; it is not a claim of independent verification of every source lemma. The formulas below specify the imported object and prove its arithmetic image.

Use a star to retain the source's viscosity-one fields, rather than overwriting the physical viscosity $\nu > 0$. In source equation (9.21), page 115, the actual summed Cartesian vector potential, azimuthal field, and pressure are denoted by $A_*, B_*e_\theta, p_{*,\mathrm{loc}}$. These are the cutoff-summed corrected fields of that equation, including all stages and the finite initialization. Source equation (10.4), page 118, constructs

$$u_* = \mathrm{curl}(c_*A_*) + c_*B_*e_\theta, \qquad p_* = c_*p_{*,\mathrm{loc}}, \qquad c_* = \chi_x\chi_t.$$

Here the source spatial cutoff is smooth in (r^2, z), supported in $r < r_0,\ |z| < z_0$, and equals one on $r \le r_0/2,\ |z| \le z_0/2$. Its time cutoff is zero for $1 - t \ge \tau_0$ and one for $0 \le 1 - t \le \tau_0/2$, where $0 < \tau_0 < 1/2$. Retain the actual source choices of these parameters. The source force before time one is exactly

$$f_* = \partial_t u_* + (u_* \cdot \nabla)u_* - \Delta u_* + \nabla p_*.$$

Thus every cutoff derivative belongs to the force. Its smooth extension through time one is source (10.11), page 120, with all its prescribed Taylor coefficients and cutoff sequence. The source proves that its support is contained in $K_* \times [0, 2]$, where $K_* = \mathrm{supp}\,\chi_x$, and it vanishes near time zero. The pressure is the displayed compactly supported source pressure; no divergence-free condition is imposed on f_*.

For the physical parameter $\nu > 0$, take precisely source (10.22):

$$\begin{aligned} y &= x/\sqrt{\nu}, & x &= \sqrt{\nu}\, y,\\ u_\nu(x,t) &= \sqrt{\nu}\, u_*(x/\sqrt{\nu}, t), & p_\nu(x,t) &= \nu p_*(x/\sqrt{\nu}, t),\\ f_\nu(x,t) &= \sqrt{\nu}\, f_*(x/\sqrt{\nu}, t), & K_\nu &= \sqrt{\nu}\, K_*. \end{aligned}$$

The inverse multiplies the velocity and force by $\nu^{-1/2}$, the pressure by ν^{-1}, and substitutes $x = \sqrt{\nu}\, y$; physical time is unchanged. The chain rule gives

$$\partial_t u_\nu + (u_\nu \cdot \nabla_x)u_\nu - \nu\Delta_x u_\nu + \nabla_x p_\nu = \sqrt{\nu}\,[\partial_t u_* + (u_* \cdot \nabla_y)u_* - \Delta_y u_* + \nabla_y p_*](y,t) = f_\nu.$$

Also $\mathrm{div}_x\, u_\nu = \mathrm{div}_y\, u_* = 0$, $u_\nu(0) = 0$, and u_ν, p_ν have support in K_ν for every $t < 1$. These are exactly the inputs of Section 16 with $T = 1$, with all coordinate and parameter maps specified.

Let

$$F_\nu(t) = \int_0^t \|f_\nu(s)\|_{L^2(\mathbb{R}^3)}\, ds, \qquad E_\nu = F_\nu(1) = \nu^{5/4} F_*(1) < \infty.$$

The last exponent follows from the factor $\sqrt{\nu}$ in f_ν and $dx = \nu^{3/2}dy$. For completeness the energy estimate used below is proved directly for this input. Compact support and incompressibility give, by integrating the equation against u_ν,

$$\frac{1}{2}\frac{d}{dt}\|u_\nu(t)\|_2^2 + \nu\|\nabla u_\nu(t)\|_2^2 = \langle f_\nu(t), u_\nu(t)\rangle.$$

The convection integral is the integral of $\operatorname{div}(u_\nu|u_\nu|^2/2)$, and the pressure integral is minus the integral of $p_\nu \operatorname{div} u_\nu$; both vanish. Divide by $(\|u_\nu\|_2^2+\delta^2)^{1/2}$, use Cauchy–Schwarz, integrate from zero, and let $\delta \downarrow 0$. This proves $\|u_\nu(t)\|_2 \le F_\nu(t)$. Substitution back into the integrated identity proves, with its full dissipation coefficient,

$$\|u_\nu(t)\|_2^2 + 2\nu \int_0^t \|\nabla u_\nu(s)\|_2^2 ds \le 2\int_0^t F_\nu'(s)F_\nu(s)ds = F_\nu(t)^2.$$

The source's actual growing component in two arithmetic channels

The source fixes $0<h<1/100$, $X_{\text{in}}>0$, and $e_0>0$ through its construction. Its closing proof, (10.20)–(10.21), page 124, gives the actual corrected swirl along

$$x_{*,\tau} = (\sqrt{2X_{\text{in}}\tau},0,0), \qquad t=1-\tau, \qquad (u_*)_2(x_{*,\tau},1-\tau) = \tau^{-1/2-h}(e_0+R_*(\tau)).$$

The original remainder is retained as the function $R_*(\tau)=\tau^{1/2+h}(u_*)_2(x_{*,\tau},1-\tau)-e_0$. The source estimate supplies constants $C_*>0,\tau_*>0$ with $|R_*(\tau)|\le C_*\tau^{2h}$ for $0<\tau<\tau_*$. At these points the cylindrical angle is zero, so $e_\theta=(0,1,0)$ and the displayed Cartesian component is precisely the source swirl. The annular and higher-order corrections have already been accounted for in this remainder. The parameter h is the chosen source parameter; no claim that an arbitrary prescribed h works is needed or made.

Define

$$r_{\nu,\tau}=\sqrt{2\nu X_{\text{in}}\tau}, \qquad x_{\nu,\tau}=(r_{\nu,\tau},0,0).$$

The exact viscosity map proves

$$(u_\nu)_2(x_{\nu,\tau},1-\tau) = \sqrt{\nu}\,\tau^{-1/2-h}(e_0+R_*(\tau)). \tag{227}$$

Use the actual material flow $X_{\nu,t}$ of u_ν proved to exist on each interval before one in Section 16. Put

$$X_{\nu,t}(q)=\sqrt{\nu}\,X_{*,t}(q/\sqrt{\nu}).$$

This is an equality of the actual flows: its right side equals q at zero, and its time derivative is $\sqrt{\nu}\,u_*(t,X_{*,t}(q/\sqrt{\nu})) = u_\nu(t,\sqrt{\nu}\,X_{*,t}(q/\sqrt{\nu}))$. The previously proved uniqueness gives the equality, with inverse $X_{\nu,t}^{-1}(x)=\sqrt{\nu}\,X_{*,t}^{-1}(x/\sqrt{\nu})$. Define the labels

$$q_{\nu,\tau}=X_{\nu,1-\tau}^{-1}(x_{\nu,\tau}).$$

This is the exact inverse between the Eulerian sampling points and their material labels. It is a family of labels depending on τ; the source path has not been asserted to be a single fluid trajectory.

Retain $S=(F_1/2,F_2,F_3)$ and $J=DS$ in the physical Cartesian coordinates x, not the viscosity-one coordinate y. Direct substitution into the original polynomials gives

$$S(r,0,0)=(0,0,2r), \qquad J(r,0,0)=\begin{pmatrix}0&0&1/2\\0&1&3r\\2&-3r^2&-r^3\end{pmatrix}.$$

Indeed the first row comes from differentiating $F_1/2=((1+xy)^3w+y^2(1+xy)(4+3xy))/2$; the second comes from $F_2=y+3x(1+xy)^2w+3xy^2(4+3xy)$; the third comes from $F_3=2x-3x^2y-x^3w$, all evaluated at $(x,y,w)=(r,0,0)$. Every entry, including the off-diagonal powers of r, is retained. Writing $v=J(r,0,0)u$ therefore gives the exact component inverse

$$u_3=2v_1, \qquad u_2=v_2-6rv_1,$$
$$u_1=\tfrac{1}{2}v_3+\tfrac{3}{2}r^2v_2-8r^3v_1.$$

Substituting these expressions into all three rows proves both directions of this matrix inverse.

Fix actual arithmetic time zero. Take $H = Z_0 = \Xi$ or $H = K_0$ with the complete four-term coefficient from the preceding proof, and write

$$d_H^2 = \int_{\mathbb{R}} |H'(s)|^2 ds > 0, \qquad \Psi_{\nu,j}(t,q,\zeta) = H(S_j(X_{\nu,t}(q)) + \zeta), \qquad D_{\nu,j} = \partial_t \Psi_{\nu,j}.$$

The exact positive constants are those of (152) at $\theta = 0$, including 32π and, for K_0, the entire factor $|m|^2$. At the particular pair $(t,q) = (1-\tau, q_{\nu,\tau})$, the first two shifts are both zero. Consequently define the actual integral

$$I_{\nu,j}(\tau) = \frac{1}{d_H^2} \int_{\mathbb{R}} \overline{H'(s)} D_{\nu,j}(1-\tau, q_{\nu,\tau}, s) ds, \qquad j = 1, 2.$$

The chain rule gives $D_{\nu,j} = v_j H'(s)$ there. It follows, without an asymptotic replacement of the profiles, that

$$I_{\nu,2}(\tau) - 6r_{\nu,\tau} I_{\nu,1}(\tau) = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)). \tag{228}$$

There is also a whole-profile identity, stronger than its projection:

$$\begin{aligned} &D_{\nu,2}(1-\tau, q_{\nu,\tau}, s) - 6r_{\nu,\tau} D_{\nu,1}(1-\tau, q_{\nu,\tau}, s) \\ &\quad = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)) H'(s). \end{aligned} \tag{229}$$

Its squared $L^2(ds)$ norm is exactly $\nu\tau^{-1-2h}(e_0 + R_*(\tau))^2 d_H^2$. Thus the source's actual component and remainder, not just an abstract possibility of growth, have been carried into the original arithmetic profile. The complex conjugation and integration above retain the complex values of K_0 when that channel is used.

Choose any $0 < \tau_{**} \le \tau_*$ such that $C_* \tau_{**}^{2h} \le e_0/2$ and the source cutoffs equal one on the path for $\tau < \tau_{**}$. Cauchy–Schwarz on the two component coefficients $(-6r_{\nu,\tau}, 1)$ yields

$$\left(\sum_{j=1}^{3} \|D_{\nu,j}(1-\tau, q_{\nu,\tau}, \cdot)\|_2^2 \right)^{1/2} \ge \frac{d_H \sqrt{\nu}\, e_0}{2\sqrt{1 + 72\nu X_{\mathrm{in}} \tau}}\, \tau^{-1/2-h}. \tag{230}$$

This lower bound proves divergence in the spatial-supremum, spectral-L^2 norm for the imported constructed field. It is not a matching upper asymptotic for its global supremum. All forcing, pressure, and material diffusion terms remain those of (149) and (150), with the explicit (u_ν, p_ν, f_ν) just given.

A terminal arithmetic profile with a bounded energy norm

The same constructed input gives a terminal value in a precise function space. This assertion can be proved from its energy identity without controlling its unbounded spatial supremum. Let $M_\nu = \max_{K_\nu} \|J\|_{\mathrm{op}}$ and put

$$\mathcal{H}_\nu = L^2(K_\nu \times \mathbb{R}, dq\, ds; \mathbb{C}^3), \qquad B_{\nu,j}(t,q,s) = \Psi_{\nu,j}(t,q,s) - H(S_j(q) + s).$$

Subtracting the displayed initial value defines an affine coordinate with the explicit inverse $B \mapsto B + (H(S_j(q)+s))_j$. It does not discard the initial profile. Its domain and reference element are specified; outside K_ν the difference is identically zero. The full profile on K_ν is also square integrable because real translation gives norm $|K_\nu|^{1/2}\|H\|_2$ in each channel.

The already proved exact identity gives

$$\|D_\nu(t)\|_{\mathcal{H}_\nu}^2 = d_H^2 \int_{K_\nu} |J(x) u_\nu(t,x)|^2 dx \le d_H^2 M_\nu^2 E_\nu^2.$$

The identity follows by integrating the real spectral translation and then using $\det D_q X_{\nu,t} = 1$; no spatial coordinate is removed. On a closed time interval before one the functions are smooth and the Schwartz bounds justify their Hilbert-space derivatives. Integrating $\partial_t B_\nu = D_\nu$ there and using the triangle inequality for the integral yields, for $0 \le a < b < 1$,

$$\|B_\nu(b) - B_\nu(a)\|_{\mathcal{H}_\nu} \le d_H M_\nu E_\nu(b-a). \tag{231}$$

Completeness constructs a unique terminal element $B_\nu(1)$ and the exact bound $\|B_\nu(1) - B_\nu(t)\| \le d_H M_\nu E_\nu(1-t)$. This constructs a Lipschitz extension on $[0,1]$ in the specified Hilbert space, while (230) proves divergence in the stronger spatial-supremum norm of its time derivative.

We also identify the terminal element through an actual map on labels. Volume preservation gives

$$\int_{K_\nu}\int_0^1 |u_\nu(t, X_{\nu,t}(q))|dt\,dq = \int_0^1 \|u_\nu(t)\|_{L^1(K_\nu)}dt \le |K_\nu|^{1/2}E_\nu.$$

The nonnegative integral is finite, so for almost every q the full trajectory has finite total length before one. Its integral equation therefore constructs the limit $X_{\nu,1}(q) \in K_\nu$ as $t \uparrow 1$. Set its value arbitrarily in K_ν on the null exceptional set. Also the integral triangle inequality and volume preservation give

$$\|X_{\nu,b} - X_{\nu,a}\|_{L^2(K_\nu)} \le \int_a^b \|u_\nu(t)\|_2 dt \le E_\nu(b-a),$$

and the same estimate with $b = 1$ follows by the constructed limit and Fatou's lemma. For every continuous g on compact K_ν, dominated convergence gives

$$\int_{K_\nu} g(X_{\nu,1}(q))dq = \lim_{t\uparrow 1}\int_{K_\nu} g(X_{\nu,t}(q))dq = \int_{K_\nu} g(x)dx.$$

This identifies the pushforward of Lebesgue measure exactly. To see that continuous tests suffice, for each relatively closed set C use the continuous decreasing approximations $g_n(x) = \max(0, 1-n\,\mathrm{dist}(x, C))$ to its indicator, then monotone or dominated convergence; complements give open sets and the uniqueness of finite measures on their generated Borel sigma algebra gives the stated pushforward.

For almost every q the pointwise terminal entire fibre is

$$\Psi_{\nu,j}(1, q, \zeta) = H(S_j(X_{\nu,1}(q)) + \zeta).$$

It agrees with the Hilbert-space limit: real translations are continuous in $L^2(ds)$, as follows from $\|H(\cdot + a) - H(\cdot + b)\|_2 \le d_H|a-b|$ by integrating H'. The shifts converge almost everywhere, remain bounded on K_ν, and their translated-profile squared differences are bounded by $4\|H\|_2^2$. Dominated convergence proves the identification in $\mathcal{H}_\nu$. Replacing H by $H^{(n)}$ in this proof proves the same assertions for every pure spectral derivative, with the retained constant $\|H^{(n+1)}\|_2 M_\nu E_\nu$.

The terminal label map has a further exact operator correspondence. On $L^2(K_\nu)$ define $V_t g = g \circ X_{\nu,t}$, including $t = 1$. For $t < 1$ its inverse is composition with $X_{\nu,t}^{-1}$, so it is unitary. The just-proved measure identity makes V_1 an isometry. For continuous g, dominated convergence gives $V_t g \to V_1 g$ in L^2. Continuous functions are dense for Lebesgue measure restricted to a compact set: approximate measurable-set indicators by continuous distance cutoffs between compact subsets and open supersets using finite-measure regularity, then approximate simple functions. The isometric bounds extend the convergence to every $g \in L^2(K_\nu)$. Thus $V_t \to V_1$ strongly. Its Hilbert adjoint satisfies $V_1^* V_1 = I$ by preservation of inner products, and $V_1 V_1^*$ is the orthogonal projection onto its closed range: it is self-adjoint, idempotent, acts as identity on that range, and annihilates its orthogonal complement. These are the proved terminal inverse and range statements. No pointwise inverse of $X_{\nu,1}$, or surjectivity of V_1, has been supplied by this argument.

The actual zero image, including the terminal fibres

For every time before one and almost every terminal label, the shift $a = S_j(X_{\nu,t}(q))$ is real. The full coordinate map is $\zeta \mapsto a + \zeta$ with inverse $s \mapsto s - a$. If $H(s) = (s - z_0)^m A_0(s)$ with $A_0(z_0) \neq 0$, the actual image is

$$H(a + \zeta) = (\zeta - (z_0 - a))^m A_0(a + \zeta).$$

This retains the whole analytic unit and all its derivatives. In particular its zero has imaginary part $\operatorname{Im} z_0$ at all these times, even for the source input whose arithmetic time-derivative combination obeys (229). With $H = \Xi$ the precise completed-zeta argument is $\rho = 1/2 + i(a + \zeta)$, and $\operatorname{Re} \rho = 1/2 - \operatorname{Im} \zeta$. Thus the actual source witness produces the proved derivative singularity and terminal profile above; this explicit map preserves zero heights. A zero-height change is not obtained by substituting a faster-growing velocity into this same real-translation map.